



CNC Machine Tool

Groover M.P., Automation, Production Systems, and Computer-Integrated Manufacturing: Chapters 6 and 7



The student knows:

- Main components and functionalities of CNC machine tools;
- How to classify machine tools;
- Main characteristics of turning centers;
- Main characteristics of machining centers.





AUTOMATION AND MANUFACTURING

- The process of enabling machines to follow a predetermined sequence of operations with little or no human labor and using specialized equipment and devices that perform and control manufacturing processes.
- Main applications: manufacturing processes, material handling and movement, inspection, assembly, packaging.



AUTOMATION GOALS

Automation has the following primary goals:

- Integrate devices, machines and systems
- Improve productivity
- Improve quality
- Reduce human involvement
- Improve personnel safety
- Optimize use of floor space



- 1947** John T. Parsons defined an accurate computer (still punch-card based) method to interpolate the curves describing helicopter blades.
- 1948** Contract Air Corps - Parsons Corporation to make the innovative and challenging tapered wings for military aircraft. The subcontractors IBM and MIT (Massachusetts Institute of Technology) took care of the servomechanisms.
- 1951** Contract Parsons Corporation - Servomechanisms Lab of the MIT.
- 1952** Demonstration of the first prototype numerical control machine tool at MIT.
- 1952** The Electronic Industries Association defines **NC** as *a system in which actions are controlled by direct insertion of numerical data at some point. The system must automatically interpret at least some portion of this data.*

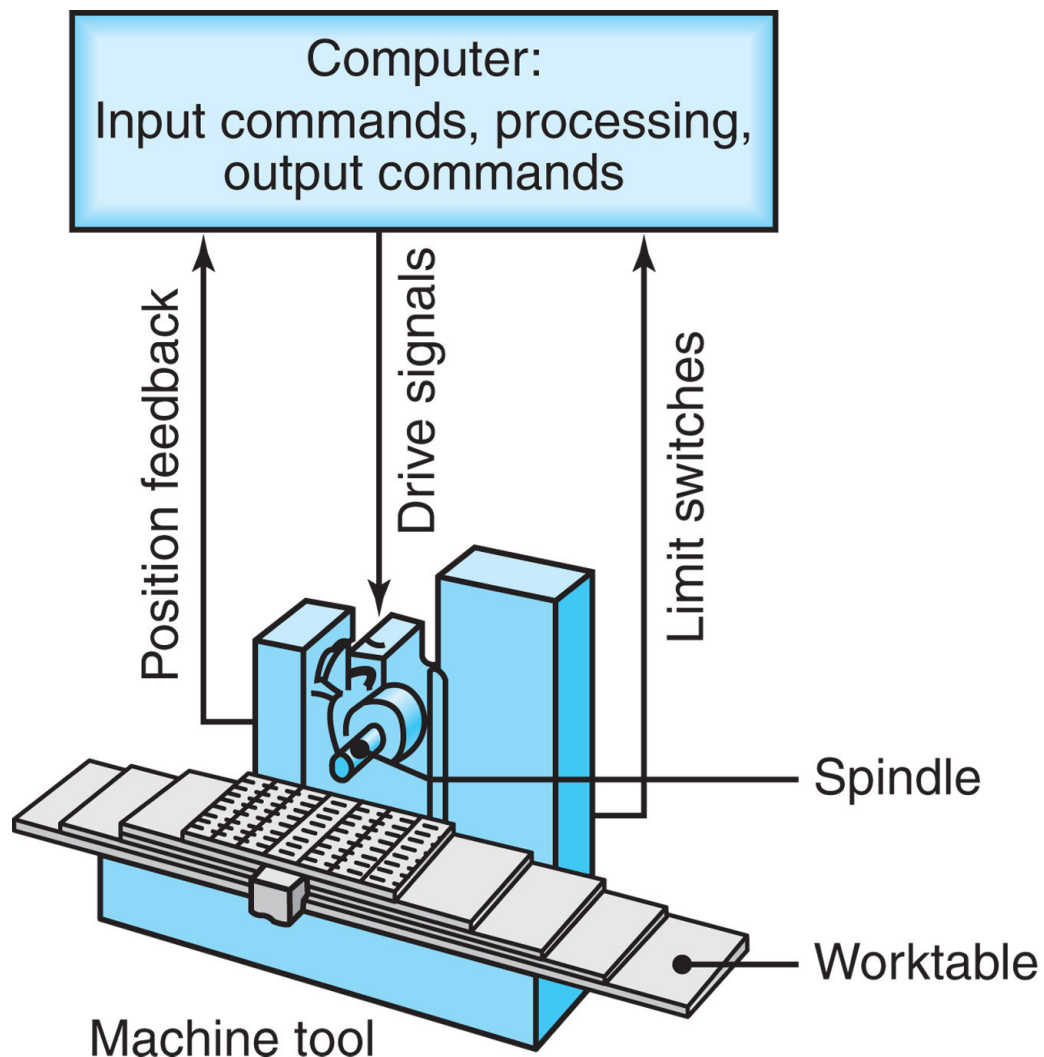


Numerical Control

Automatic control of a process performed by a device that makes use of numerical data introduced while the operation is in progress (ISO 2806:1994)

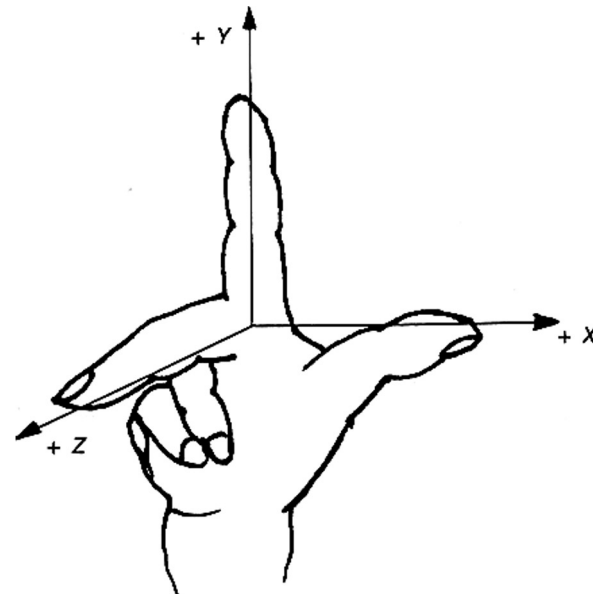
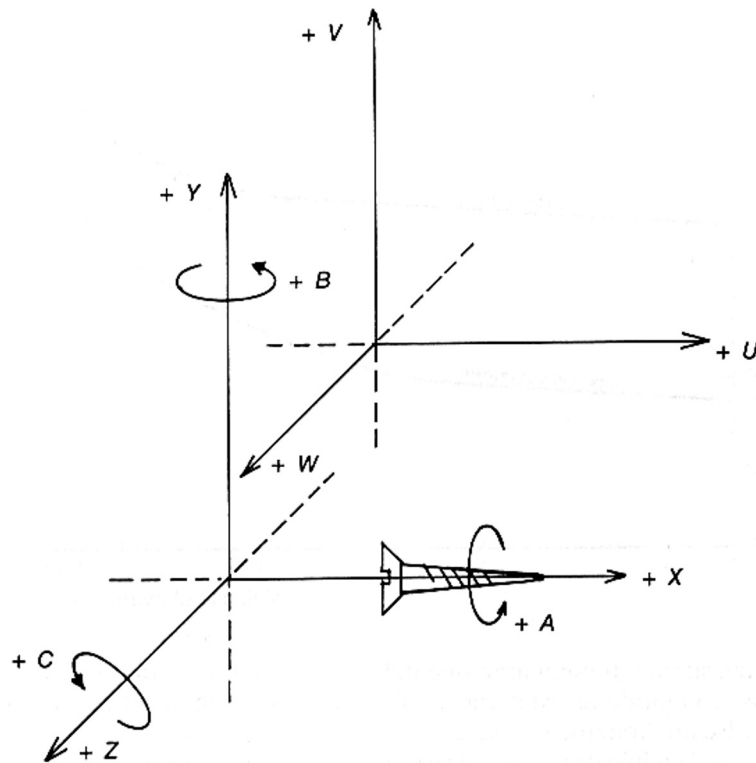
Computer Numerical Control

Realization of NC using a computer to control the machine functions (ISO 2806:1994)





Principal direction in which a part of a machine can move in a linear or rotary mode (ISO 2806:1994).





MACHINE AXIS: CONVENTION

- $Z \Rightarrow$ spindle rotation axis

Positive direction with the tool moving away from the workpiece

- $X \Rightarrow$ parallel to the working table

Positive direction related to Z and Y positive direction

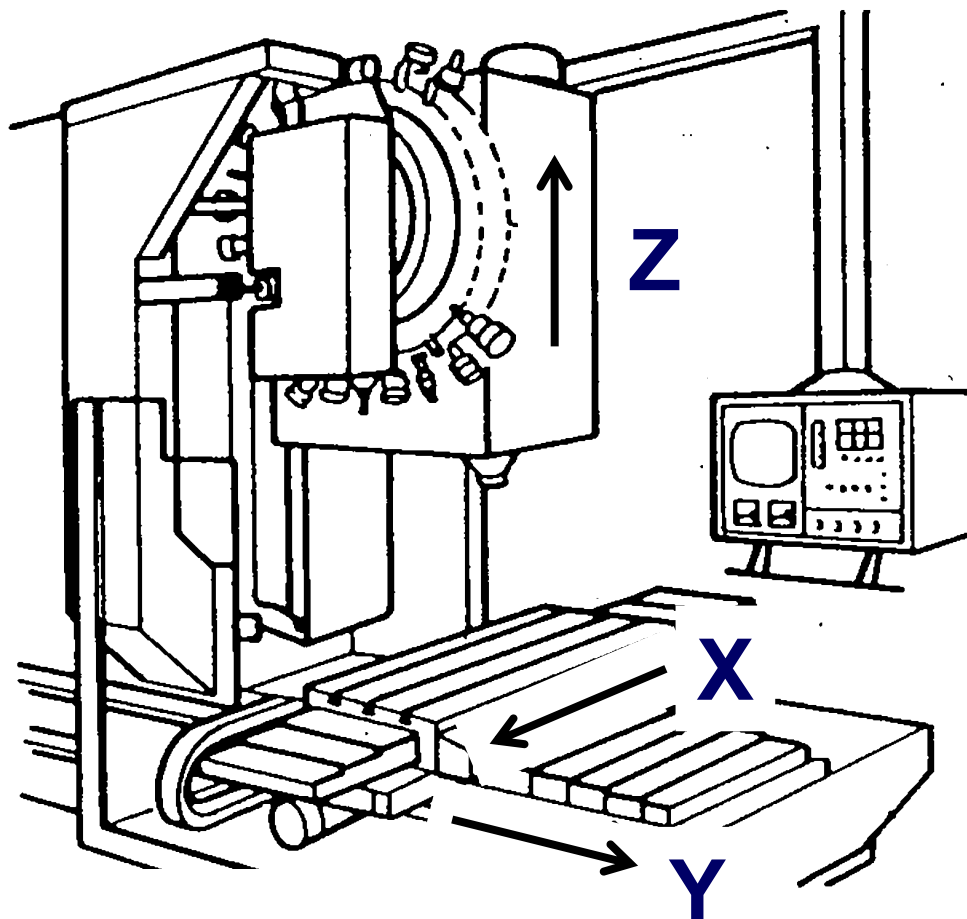
- $Y \Rightarrow$ normal to the ZX plane

Positive direction with the tool moving away from the ZX plane

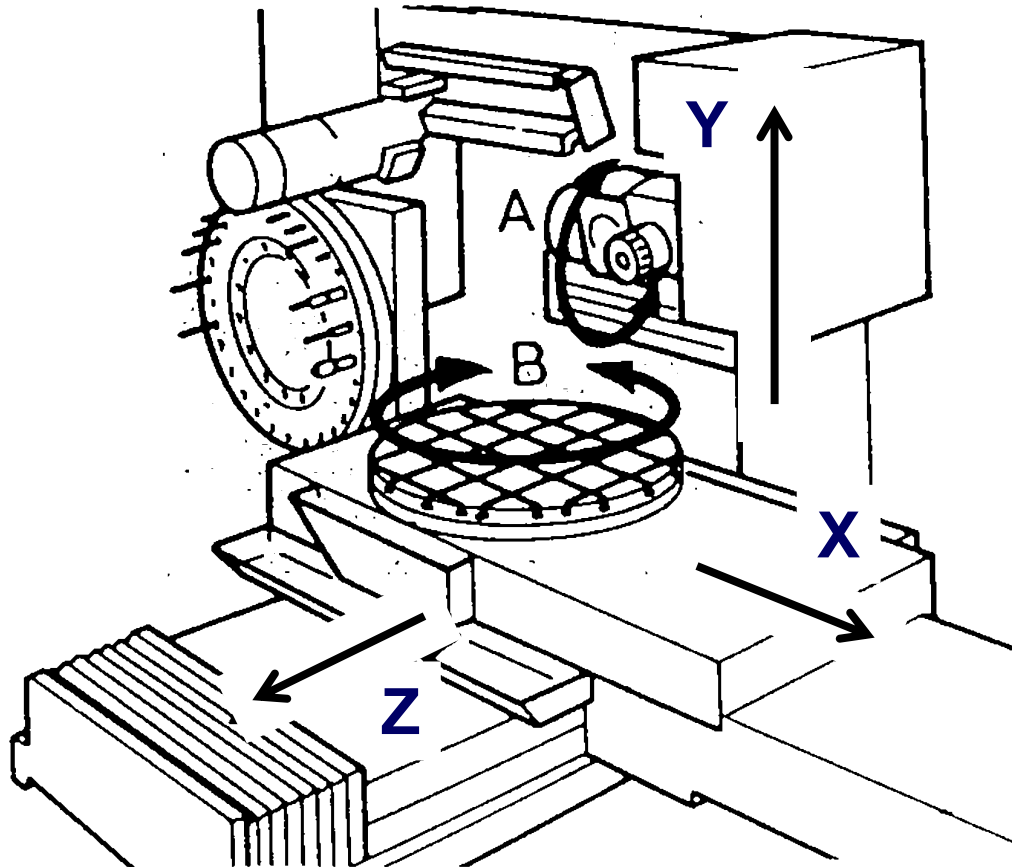


MACHINE AXES: EXAMPLE

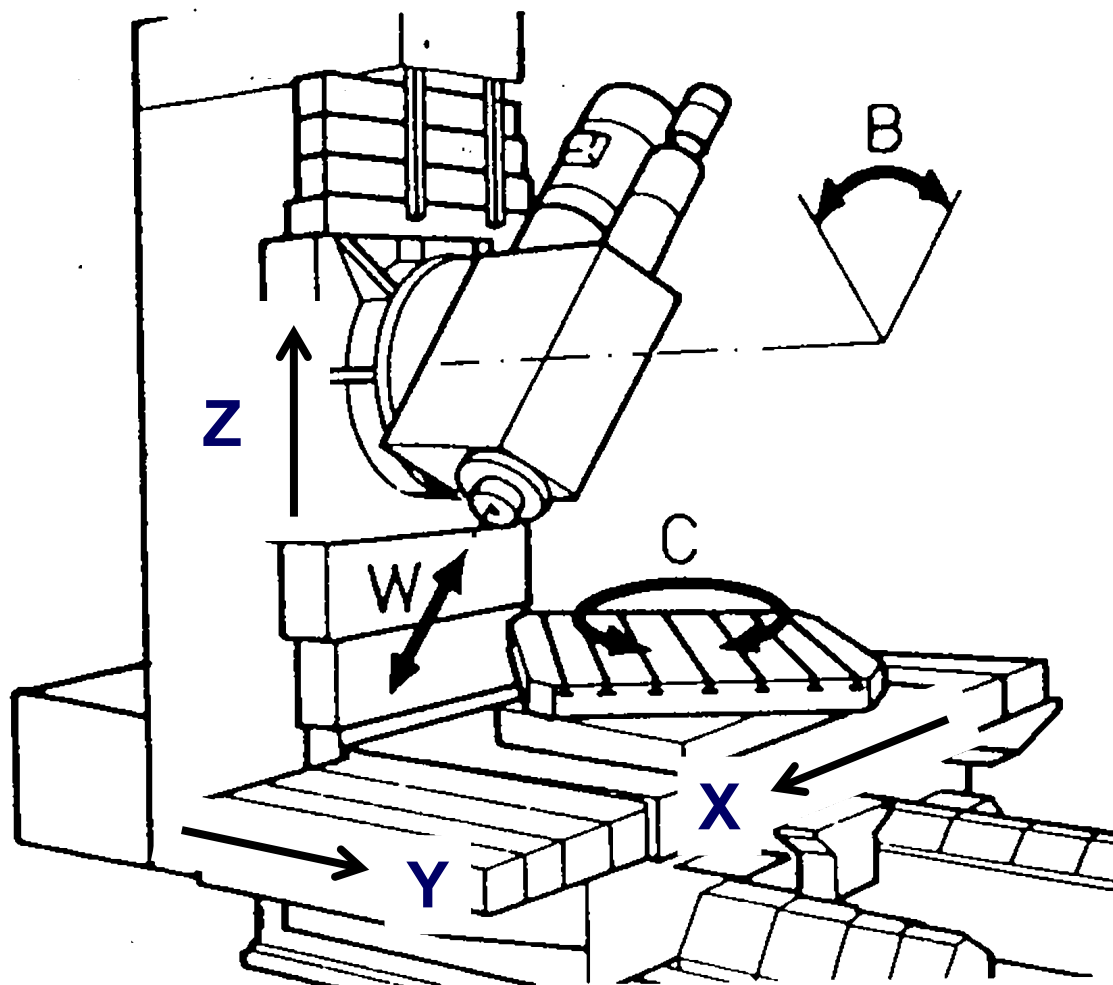
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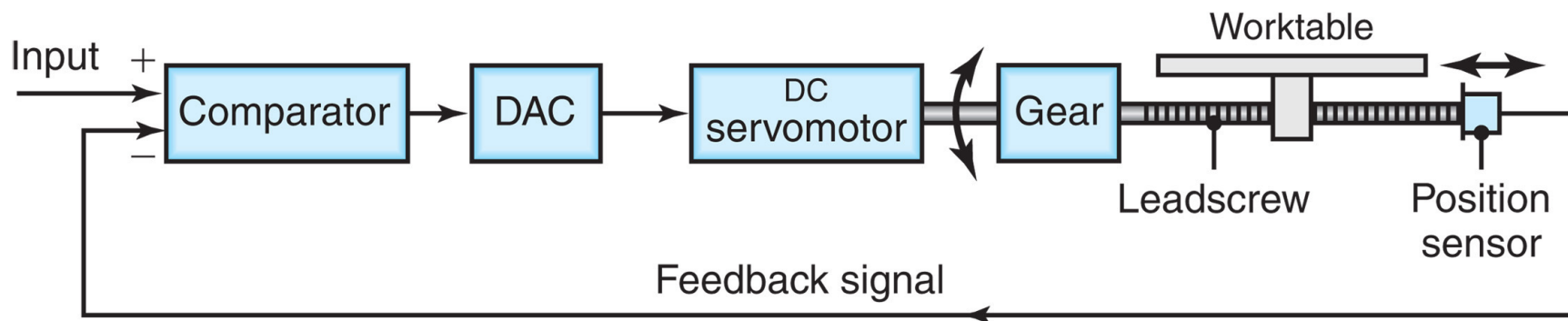
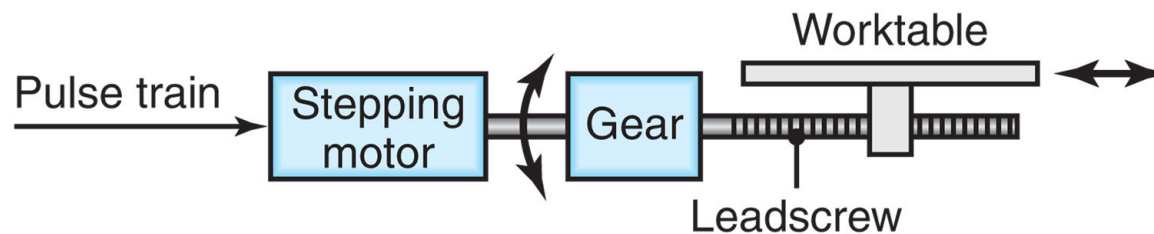
3-axis machining center

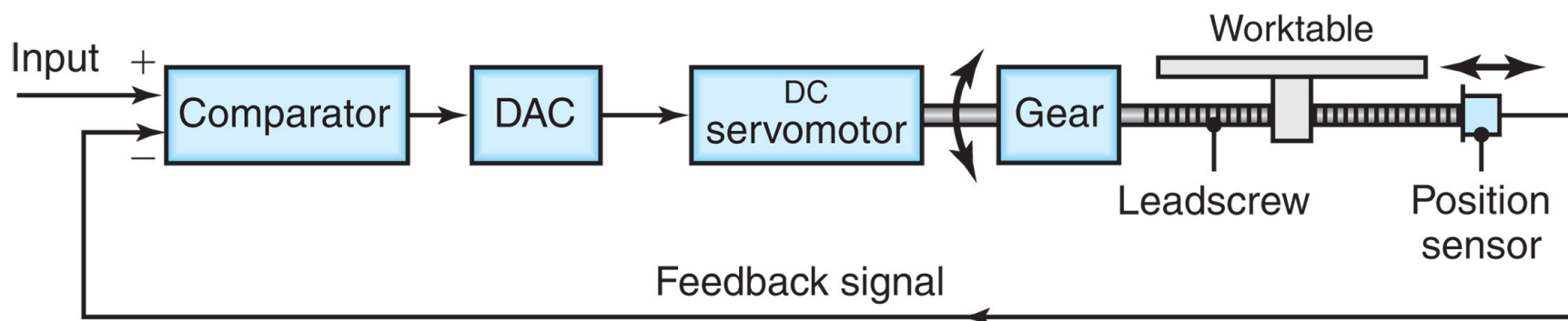
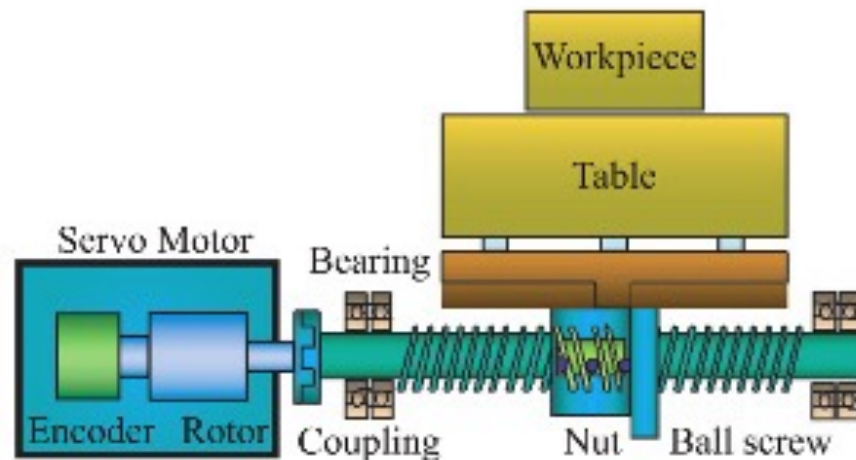
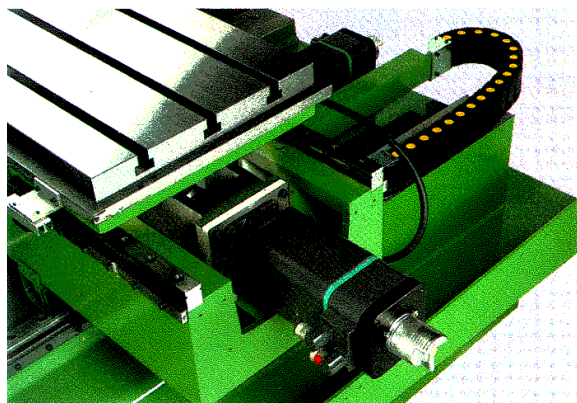


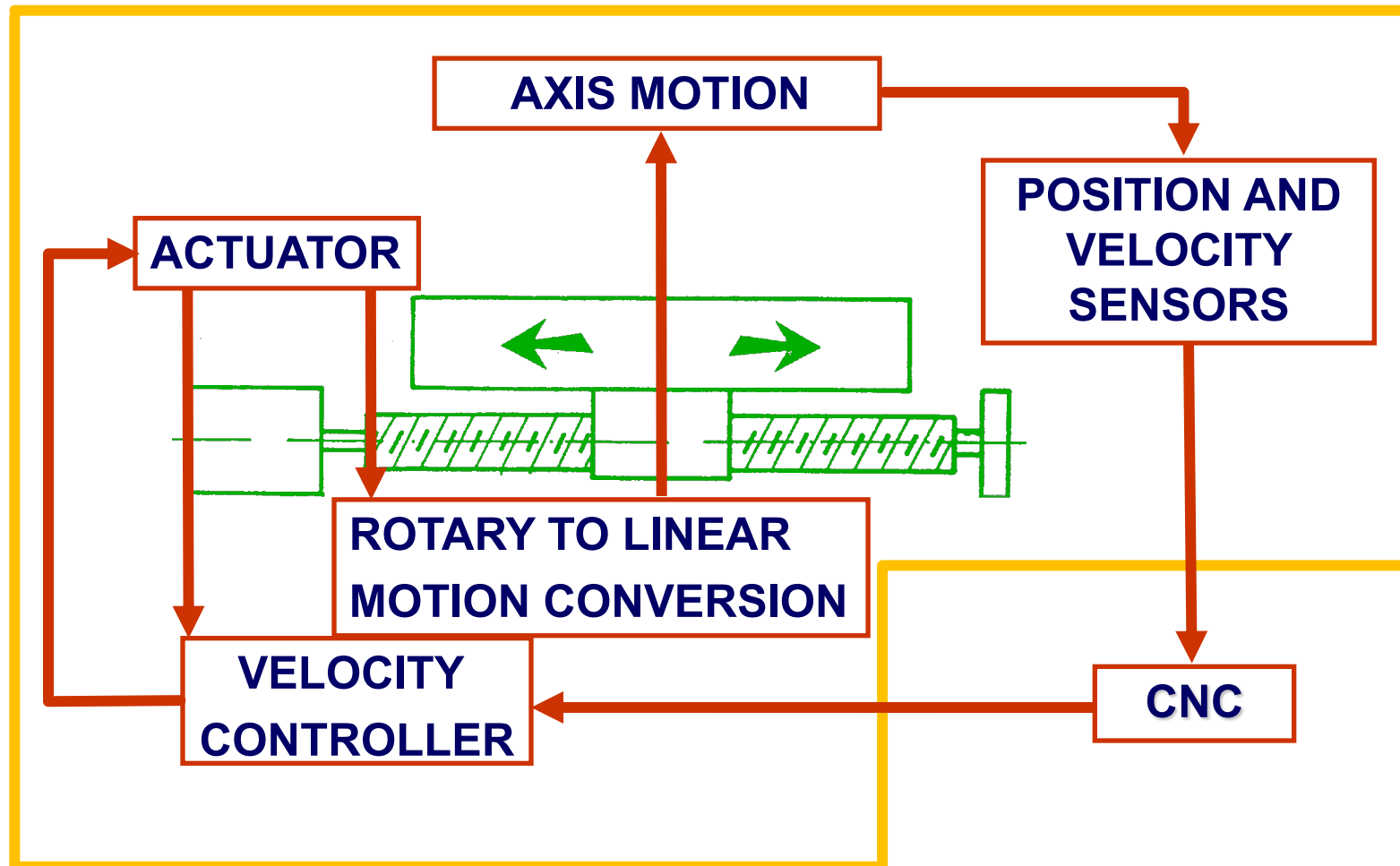
5-axis machining center

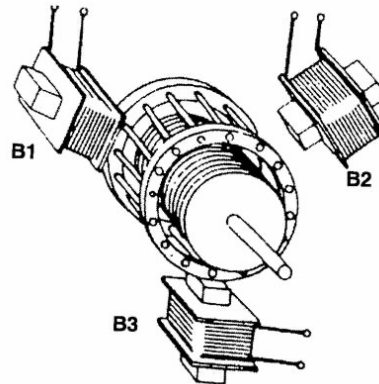
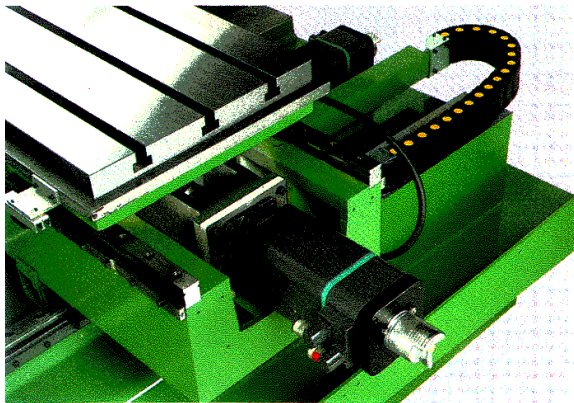


6-axis machining center









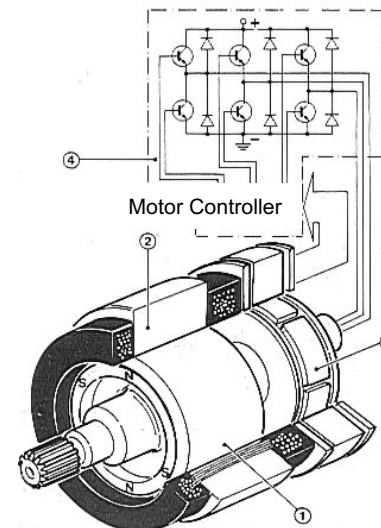
AC motor is an electric motor driven by an alternating current (AC).

Two basic parts:

1. an outside stator having coils supplied with alternating current to produce a rotating magnetic field,
2. an inside rotor attached to the output shaft producing a second rotating magnetic field.

DC motor generates torque directly from DC power supplied to the motor. Brushes and springs carry the electric current from the commutator to the spinning wire windings of the rotor inside the motor.

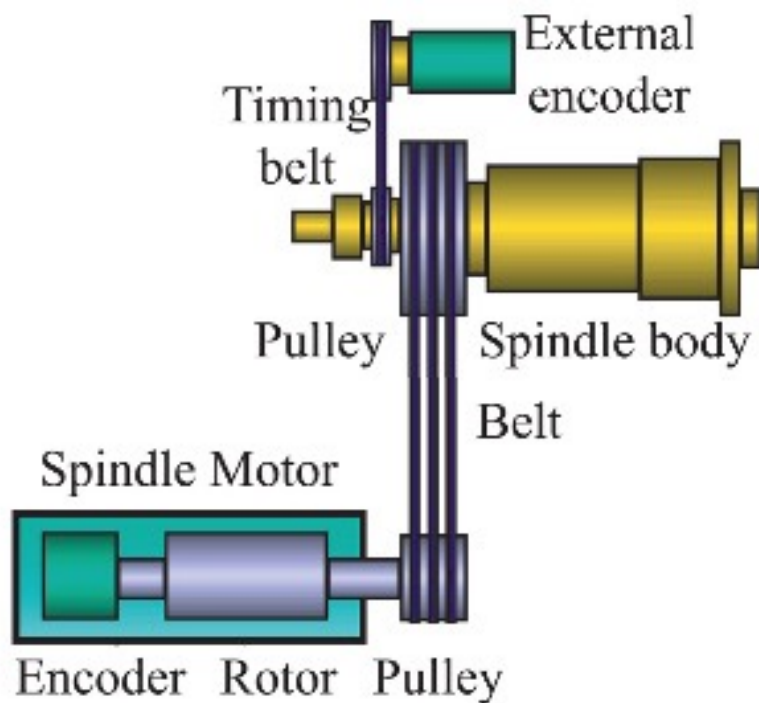
Brushless DC motors use a rotating permanent magnet in the rotor, and stationary electrical magnets on the motor housing. A motor controller converts DC to AC. This design is simpler than that of brushed motors because it eliminates the complication of transferring power from outside the motor to the spinning rotor.



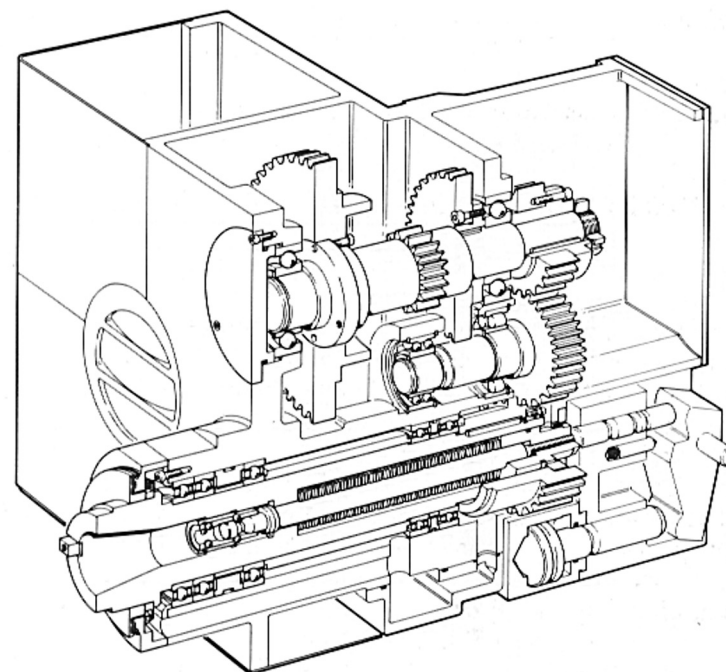


ACTUATORS: ELECTRIC MOTORS

Motor	Advantages	Disadvantages	Use
AC Motor	Robustness, low cost, high reliability	High initial current, not proportional torque–current	Spindle
DC Motor	Easy speed and torque control, high torque at low speed	High maintenance cost	Axis and Spindle
Brushless DC Motor	High and constant torque, high acceleration and speed, high reliability	High cost	Axis and Spindle
Stepper Motor	Open-loop, low cost, high reliability, high torque at low speeds and a simple, rugged construction that operates in almost any environment.	Limited Power (< 5 kW), the resonance effect at low speeds and decreasing torque with increasing speed	Axis



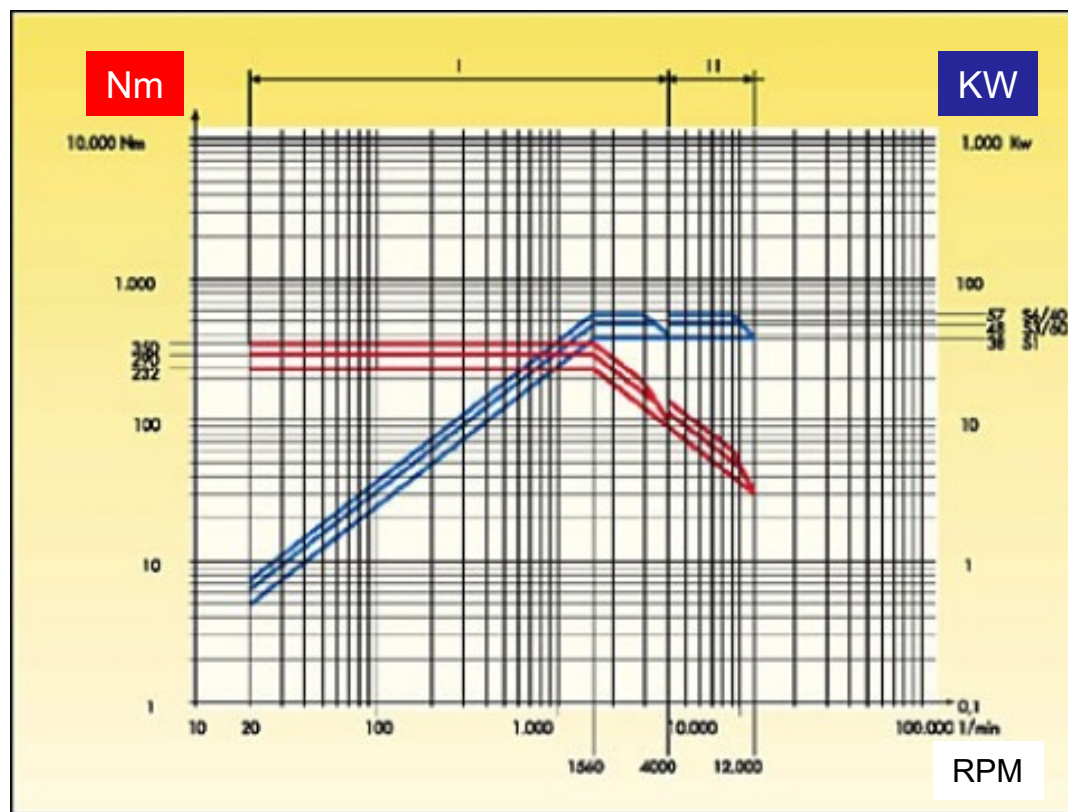
(b) spindle driving mechanism

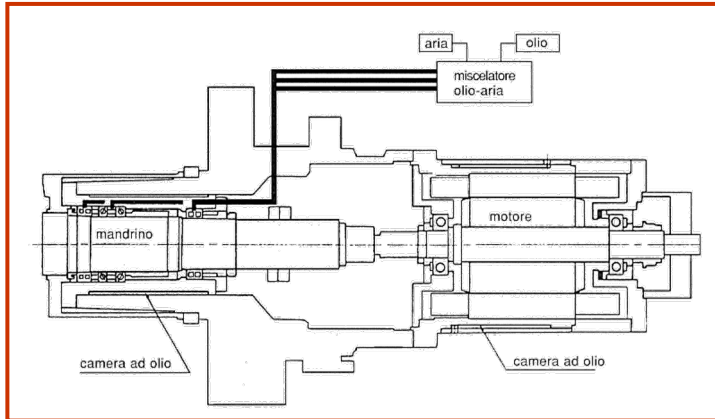




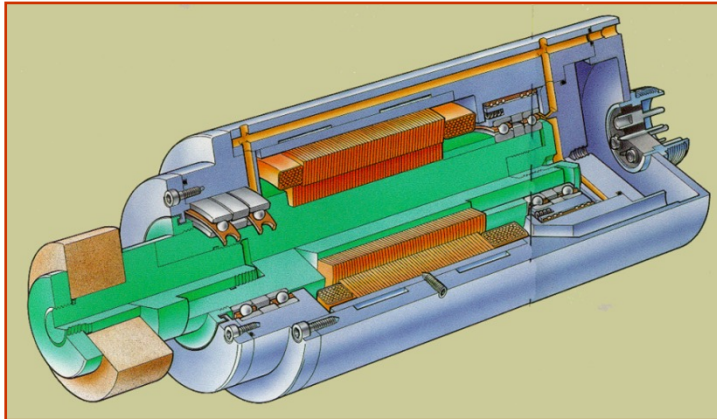
Characteristics

- Power
- Torque
- Speed
- Stiffness





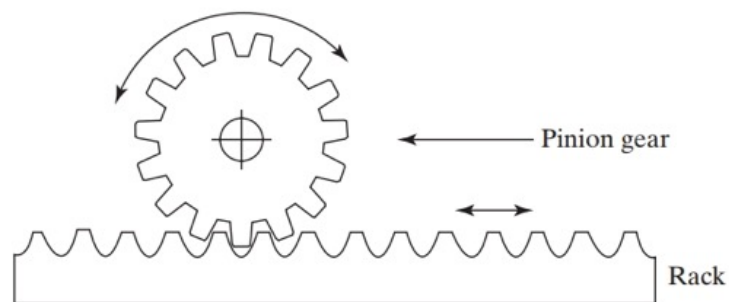
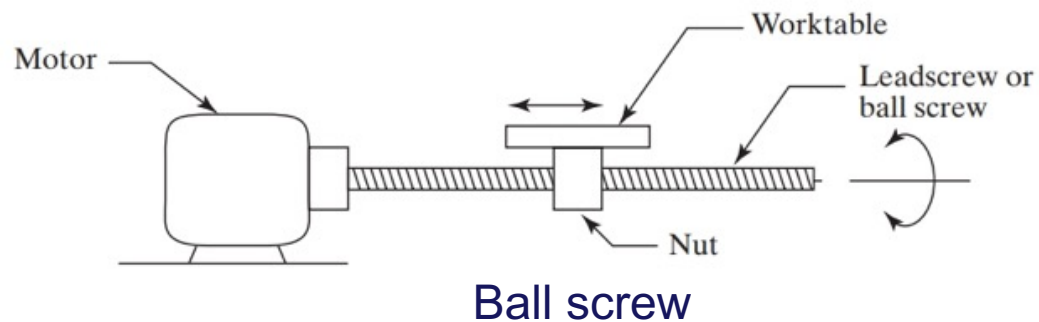
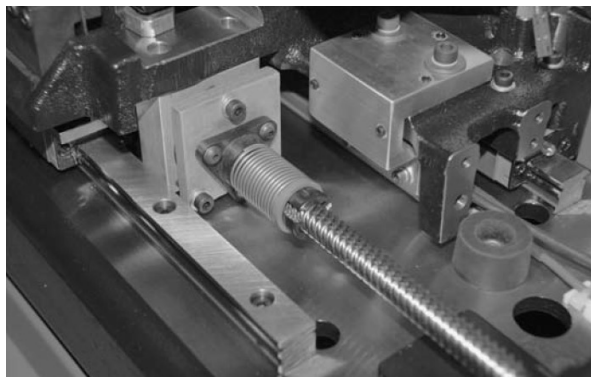
- Direct coupling spindle/motor
- Compact and simple structure (no transmission mechanism)
- Simpler mechanics
- High accuracy (no radial forces)
- High stiffness of the spindle head
- High specific power and rotational speed



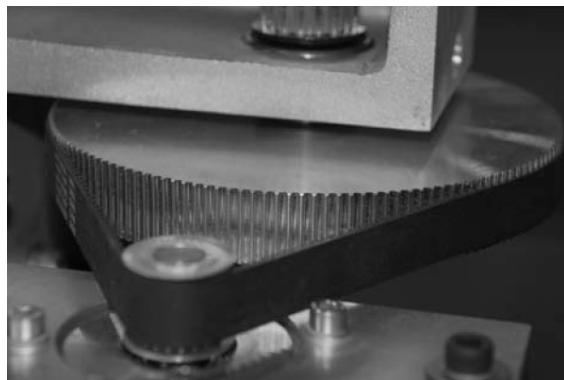


ROTARY-TO-LINEAR MOTION CONVERSION

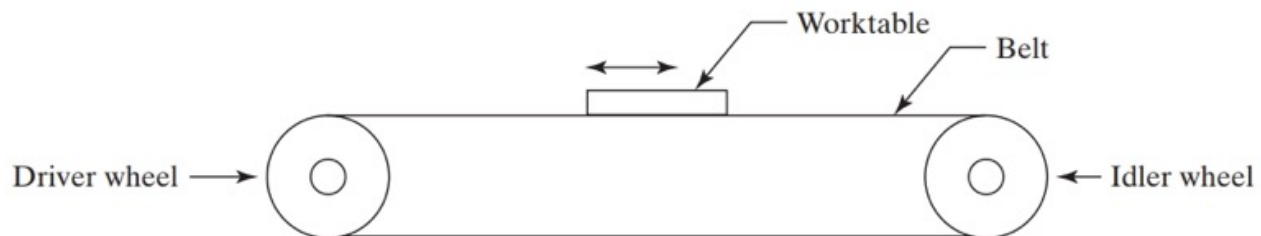
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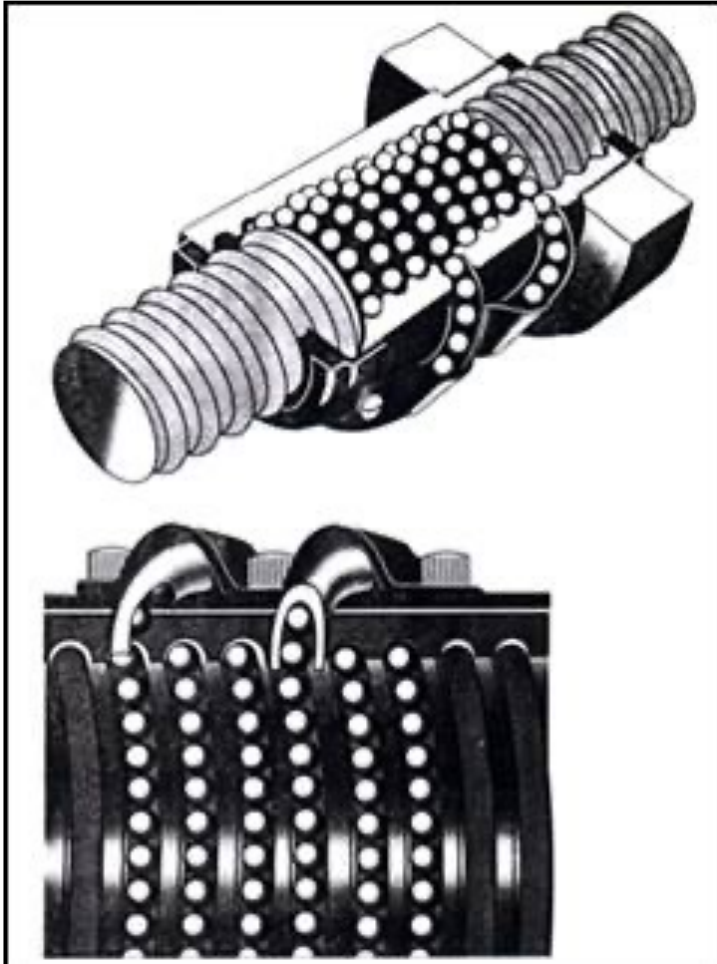
Rack and pinion



Pulley System



ROTARY-TO-LINEAR MOTION CONVERSION



Ball Screw

- Friction reduction
- High efficiency
- Heat dissipation
- No backlash with preload
- High accuracy, the minimum displacement (Basic Length Unit) depends on the lead of the ball screw

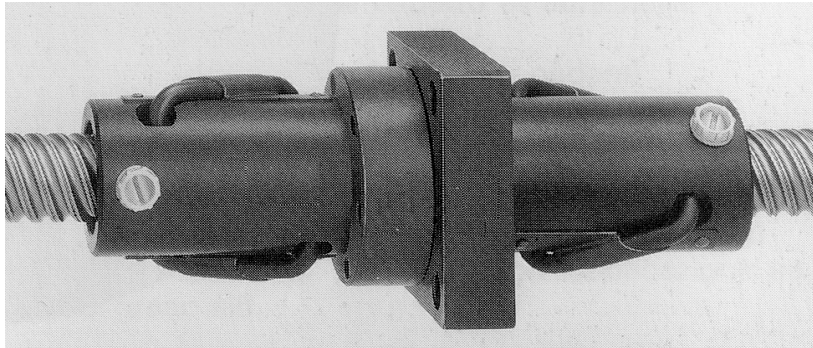


BALL SCREW

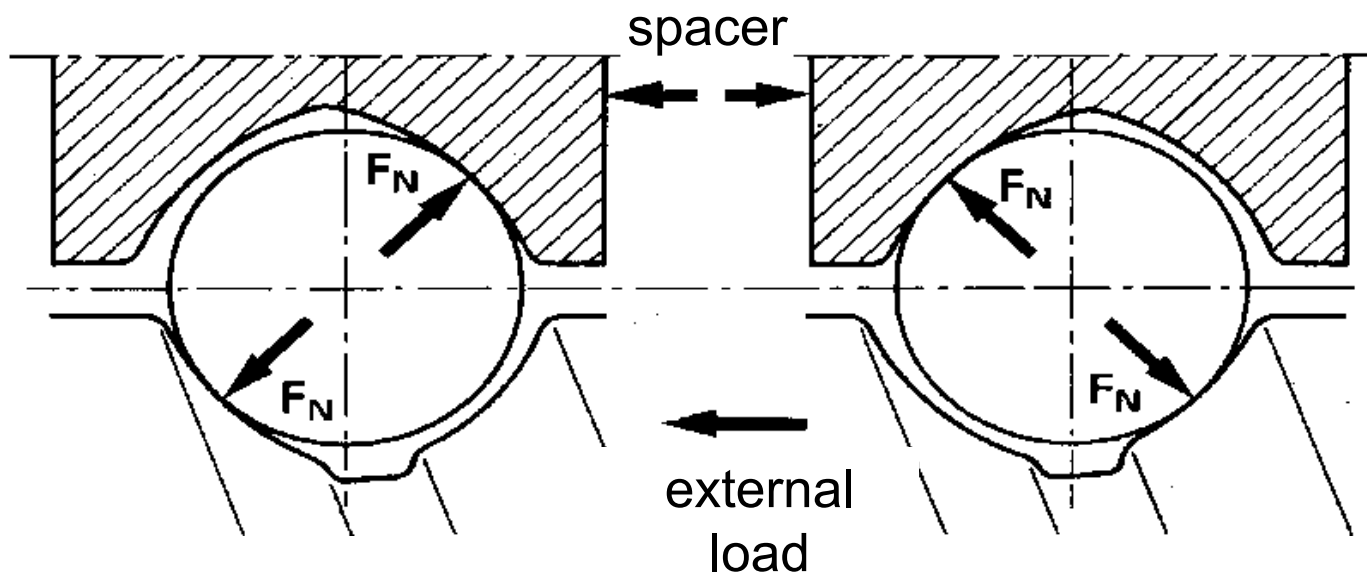
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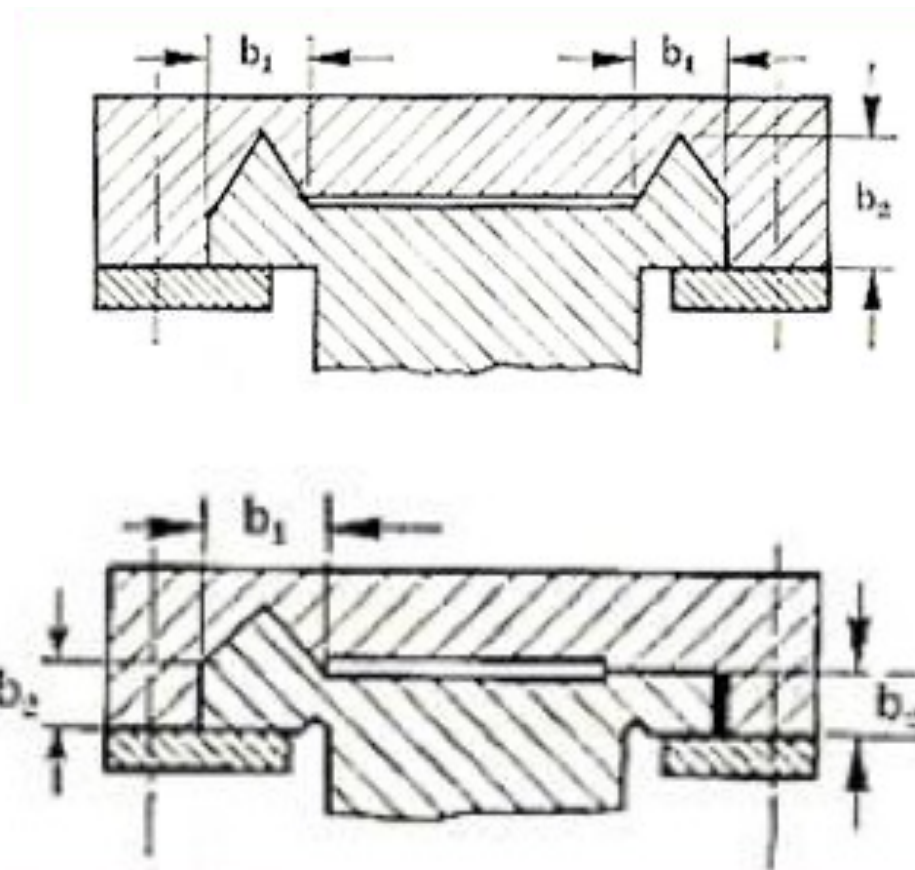
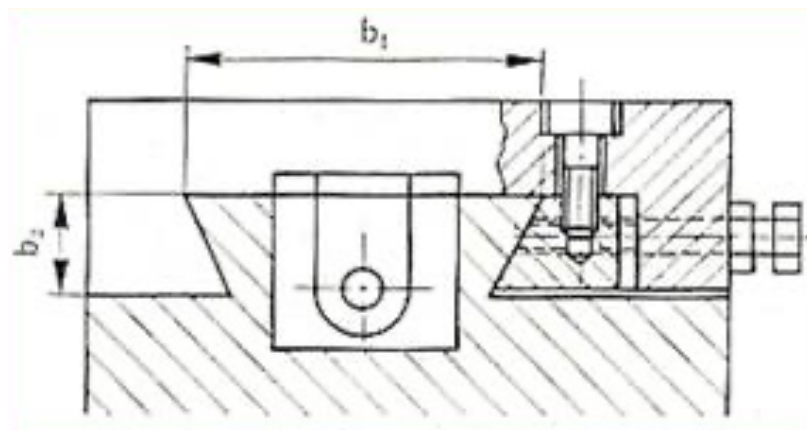
The **preload** eliminates axial and radial backlash.





Sliding guideways

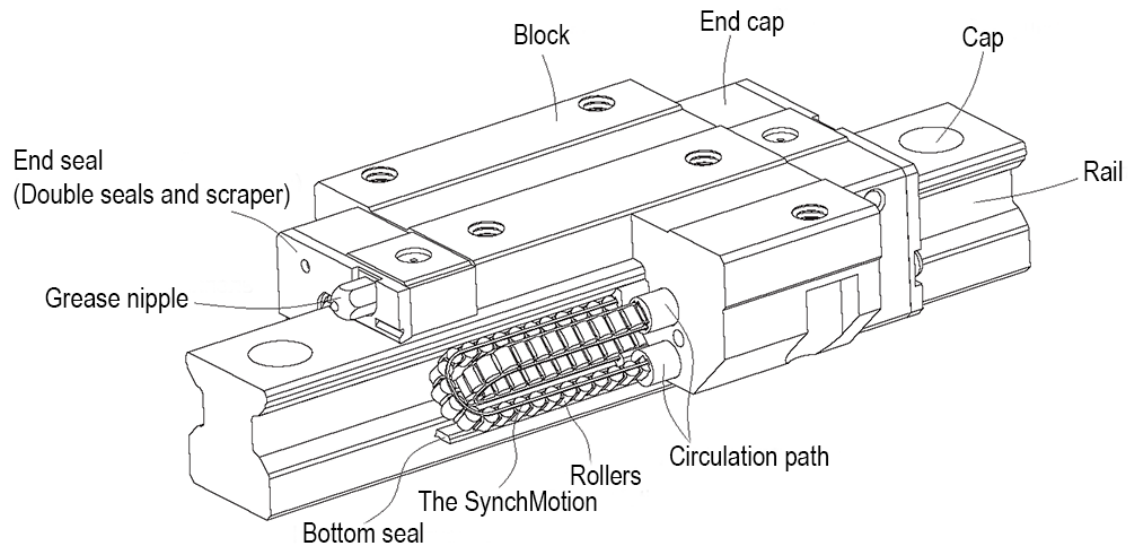
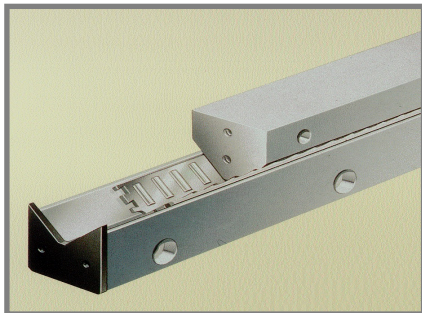
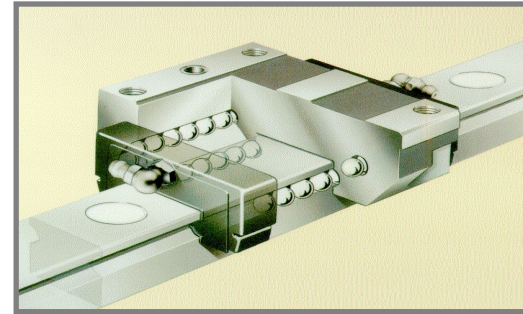
- Integral from casting
- Patch from joining





Rolling guideways

- Low friction $0.0005 < \mu < 0.005$;
- High accuracy
- High load
- No stick-slip
- Low wearing
- Low transmission force
- Low maintenance cost

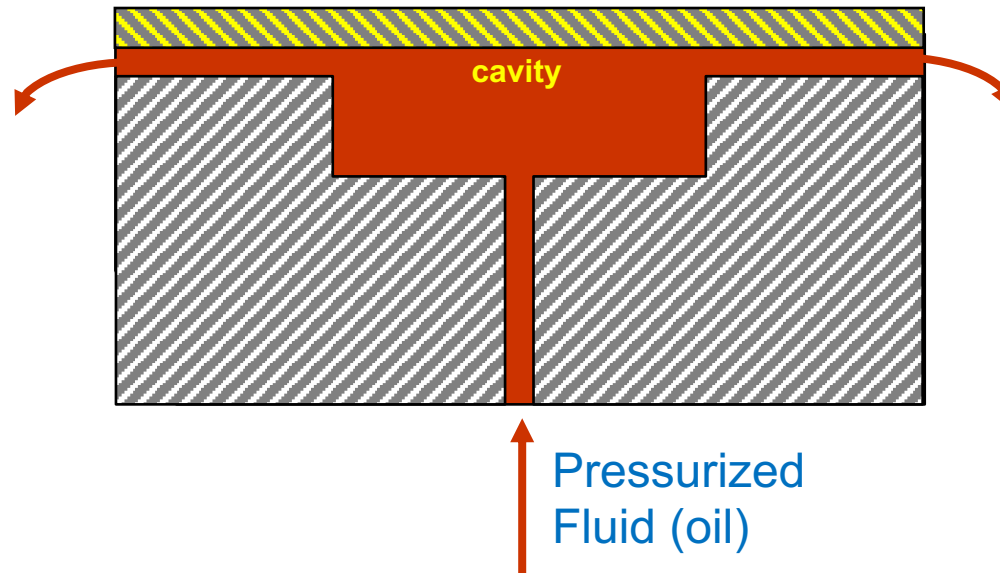


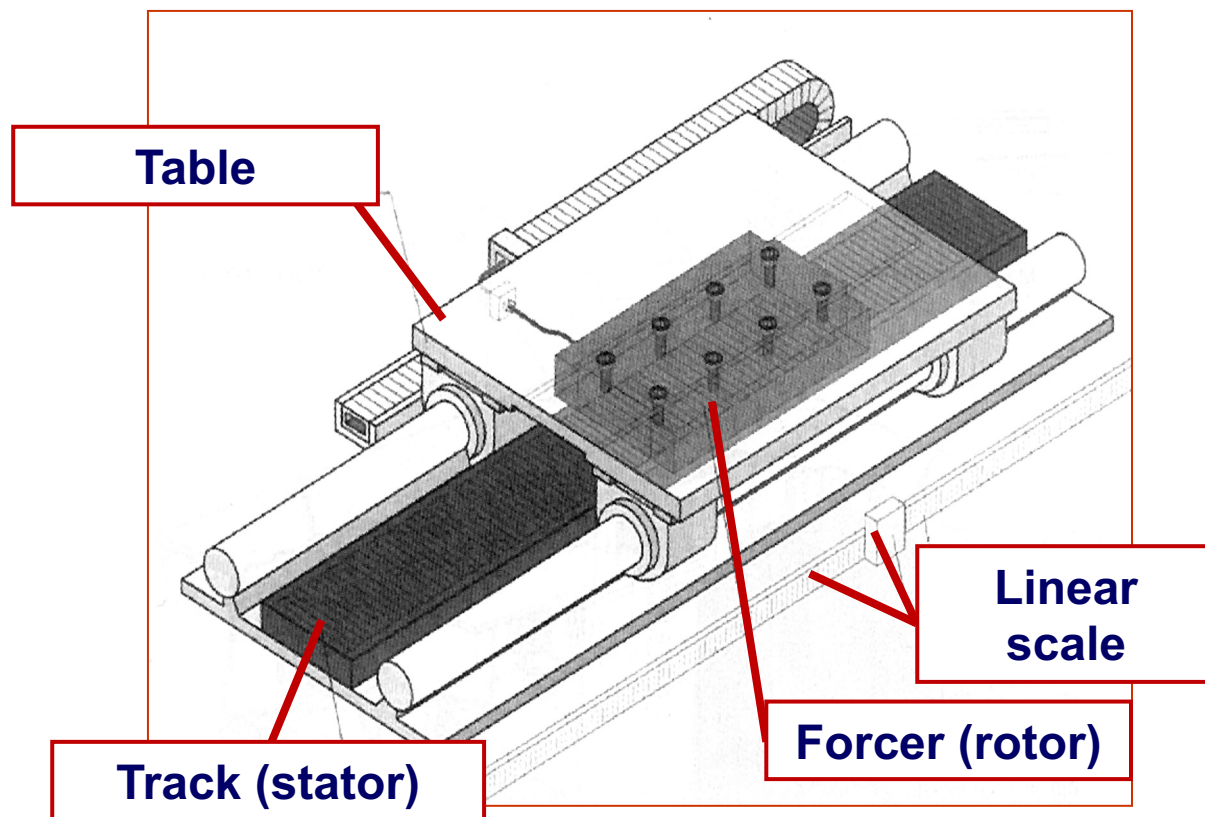


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Hydrostatic guideways





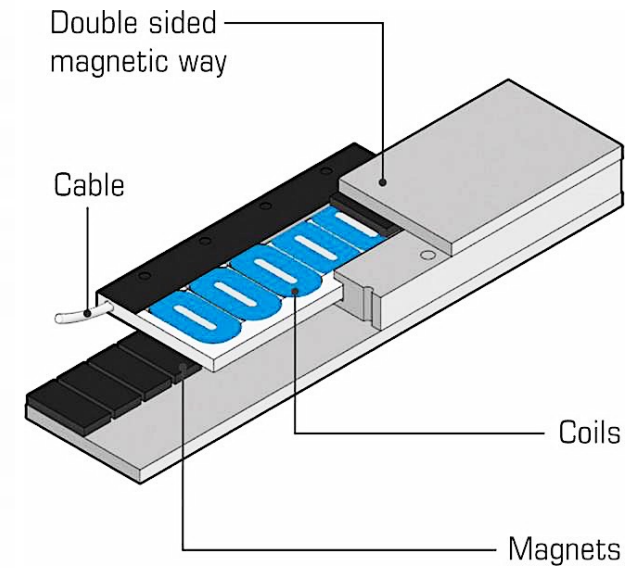


Running on air bearings:
Linear motor axis ELGL

Linear motor axis DNCE-LAS



FESTO



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Advantages

- Rapid feeds (120 m/min)
- High acceleration (20 m/s²)
- No transmission chain
- Low inertia
- High accuracy
- Low maintenance
- High length of axis

Drawbacks

- High cost
- High voltage
- Heating
- Protection from chips
- Complex electronic control
- Problems with magnetic fields



Since a sensor is a device that converts a physical variable of interest (e.g., force, temperature) into a more convenient physical form (e.g., electrical voltage) for purpose of measuring the variable, they are also called transducers.

Two types:

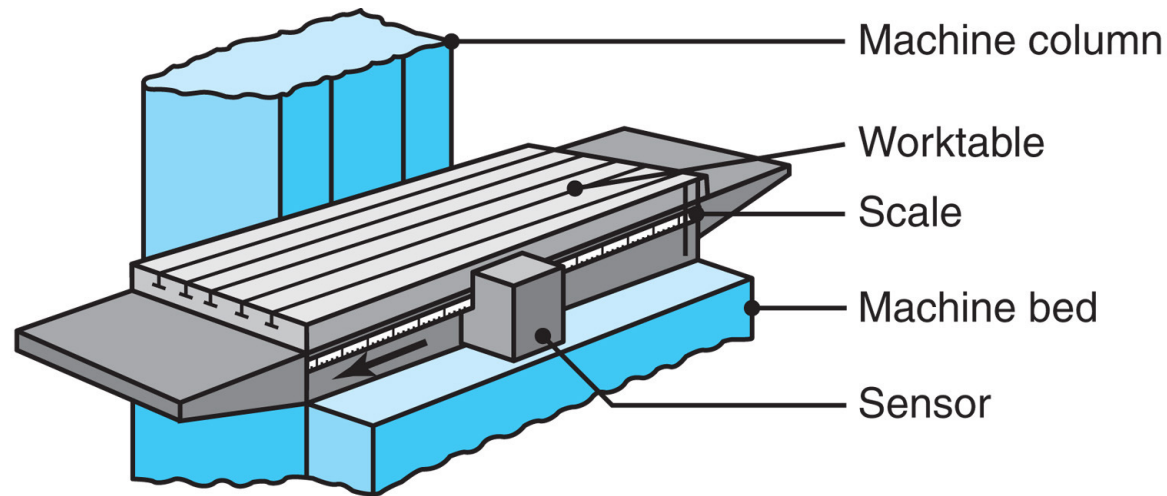
- Position/Displacement Transducers
- Velocity Transducers



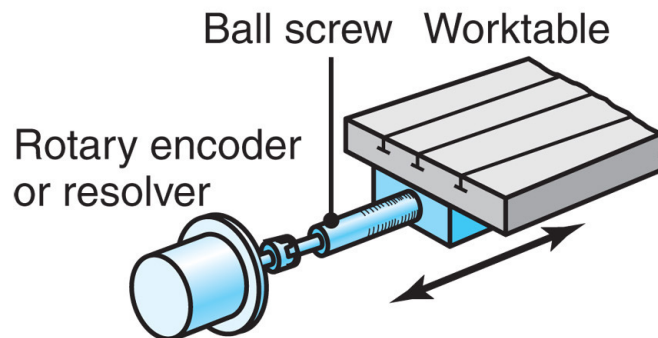
- Object of the measure: position, velocity
- Measurement method: direct, indirect
- Type of signal: analog, digital
- Type of measurement: absolute, incremental



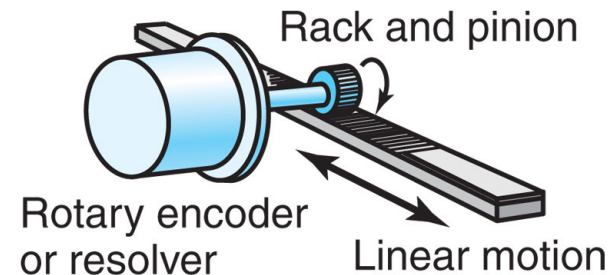
SENSORS: MEASUREMENT METHOD



(a)



(b)



(c)

(a) Direct measurement of the linear displacement of a machine-tool worktable;
(b) and (c) indirect measurement methods.



SENSORS: TYPE OF SIGNAL

- Analog:** The measure is transformed in a proportional electric current that can continuously vary as a function of the measured variable.
- Digital:** The measure is transformed in electrical pulses. Each pulse increases of one step the counter unit.

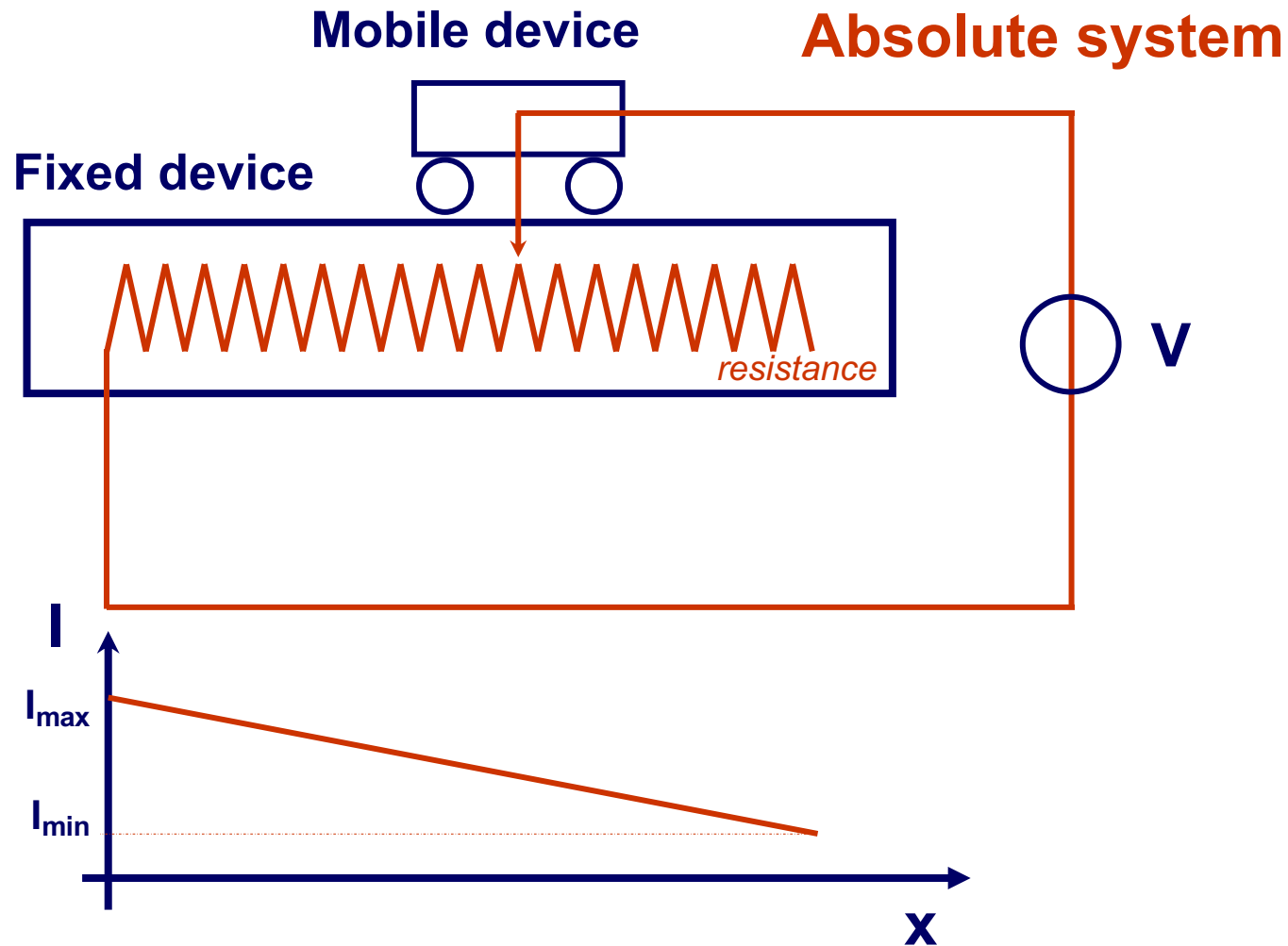


SENSORS: TYPE OF MEASUREMENT

- Absolute:** Different measures at different times have the same reference system.
- Incremental:** A measure always refer to the previous one.
- Cyclic absolute:** Mixed.

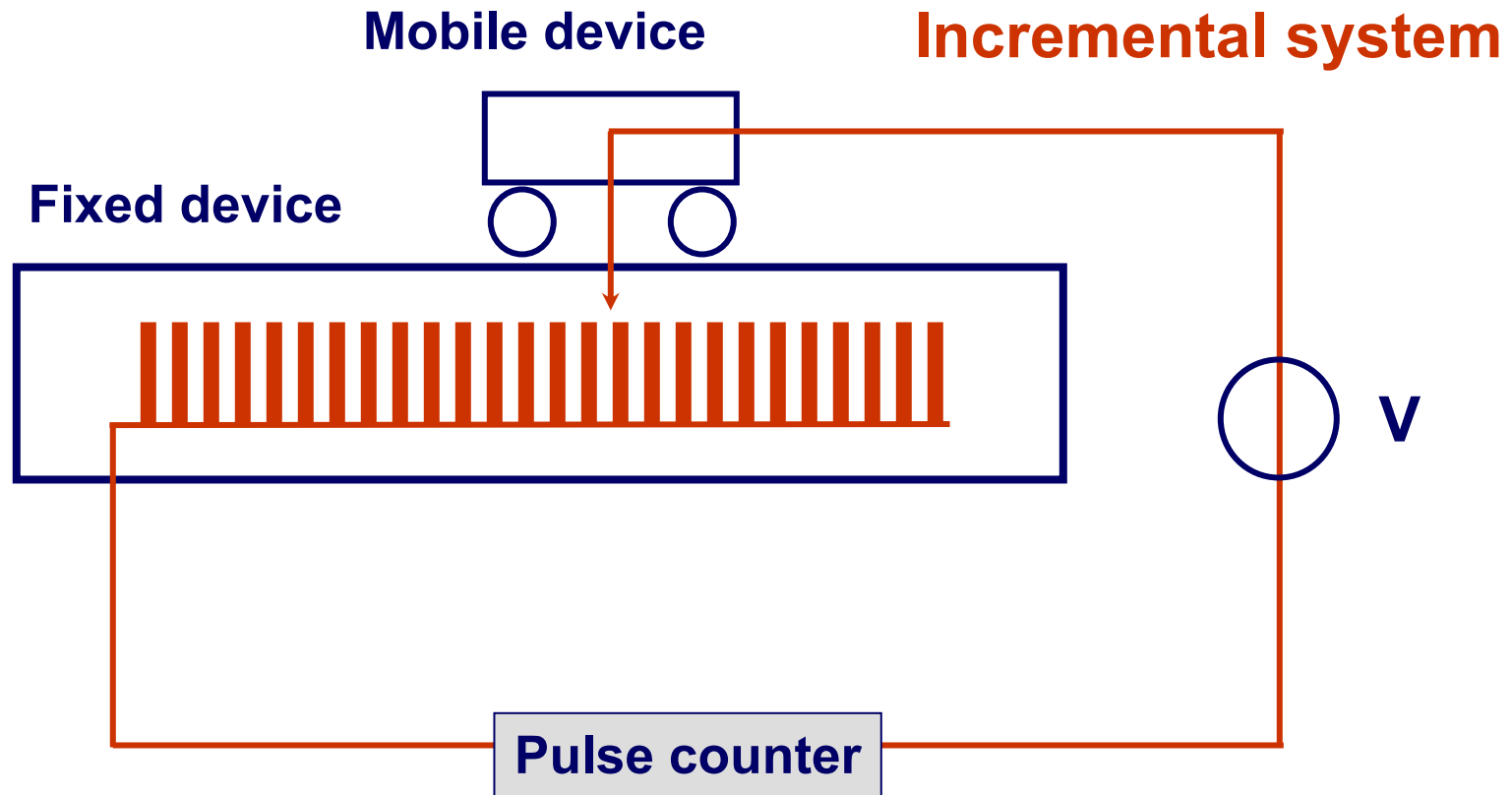


SENSORS: TYPE OF MEASUREMENT



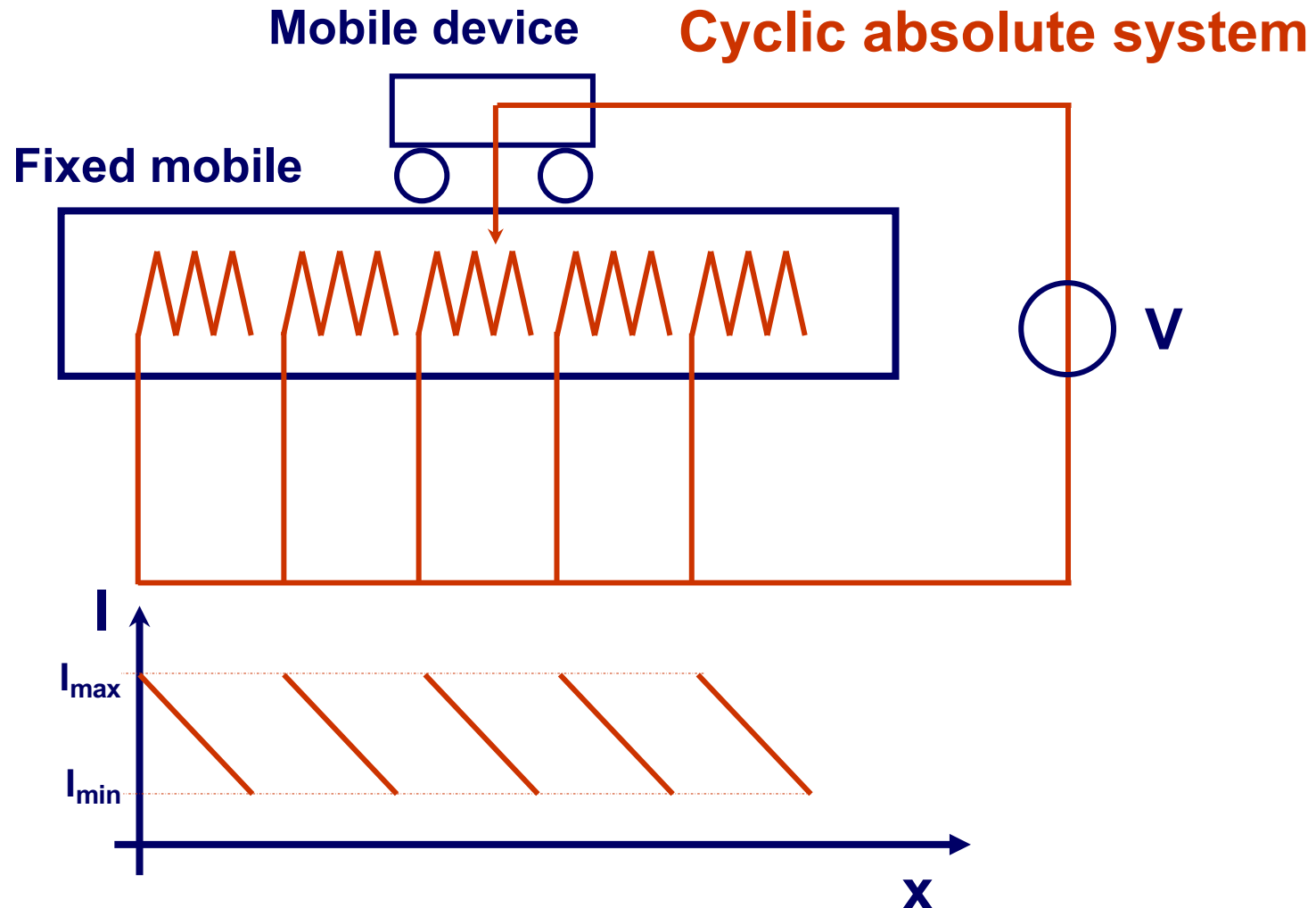


SENSORS: TYPE OF MEASUREMENT



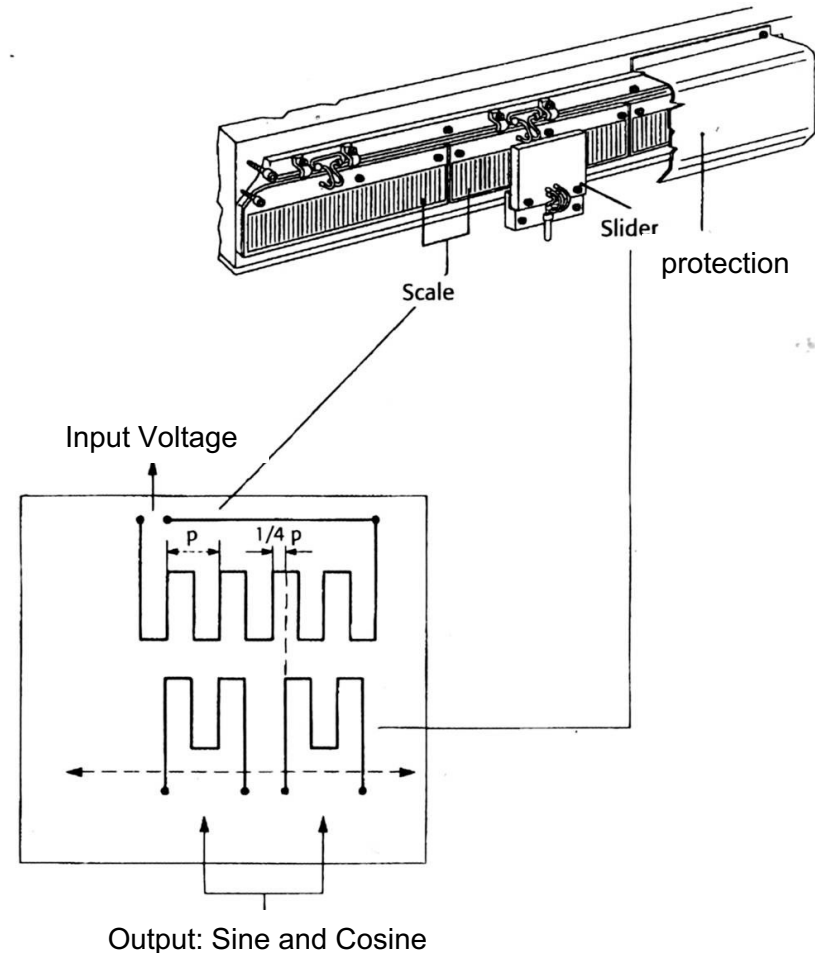


SENSORS: TYPE OF MEASUREMENT



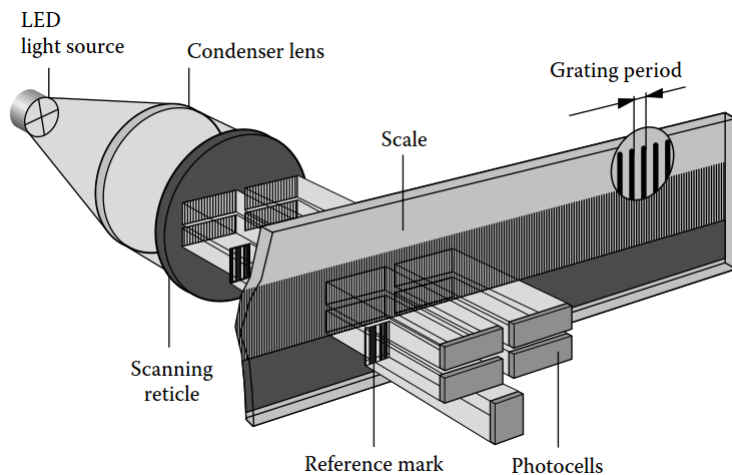
DISPLACEMENT TRANSDUCERS: INDUCTOSYN

- Linear and angular movements
- Precision circuit patterns are etched in copper bonded to the surface of inductosyn elements.
- Inductive coupling between windings on a rotary or linear pair of elements
- The inductosyn elements are attached to the fixed and moveable machine parts with the elements
- Direct, analog, cyclic absolute

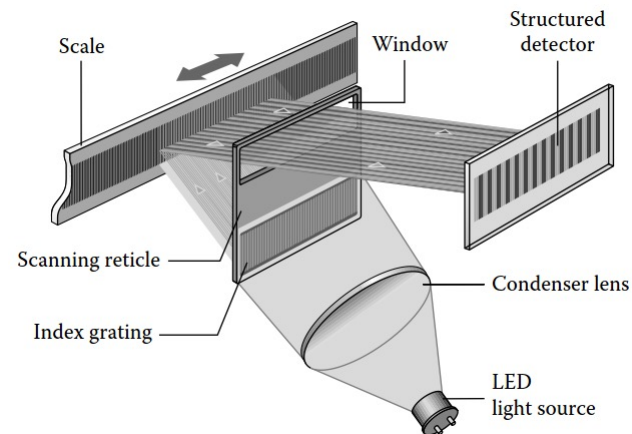




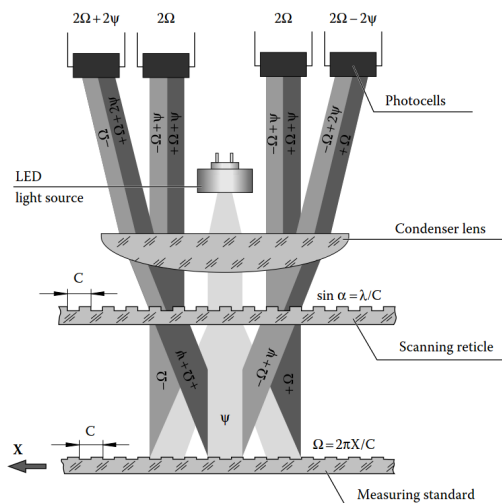
DISPLACEMENT TRANSDUCERS: LINEAR SCALES



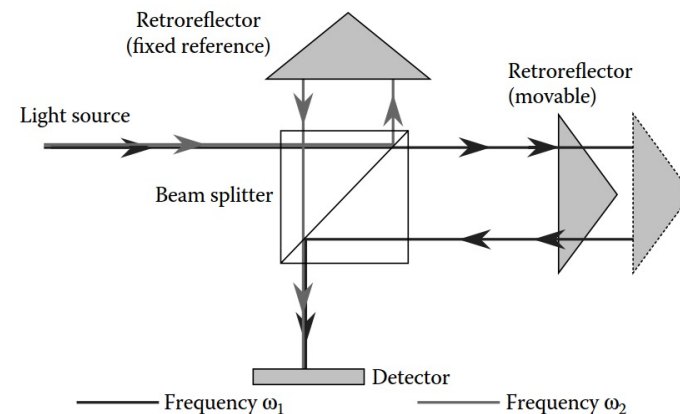
Transmission scale



Reflection scale



Interferential scale

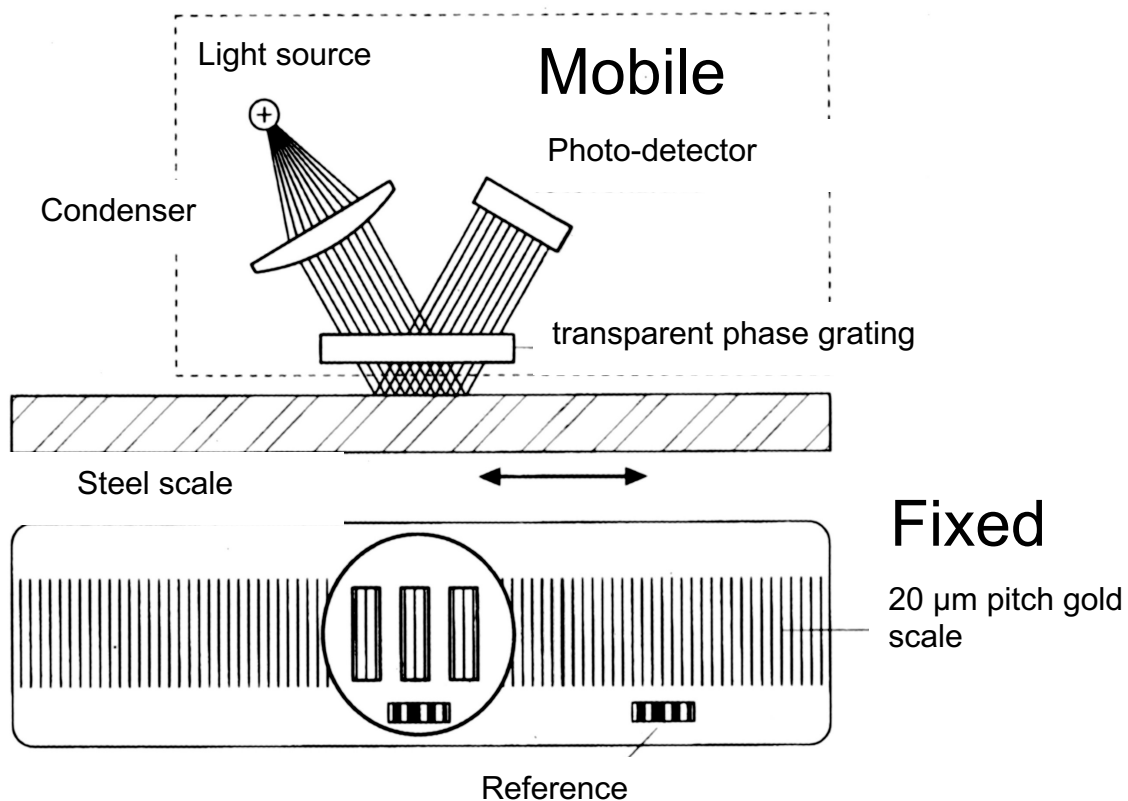
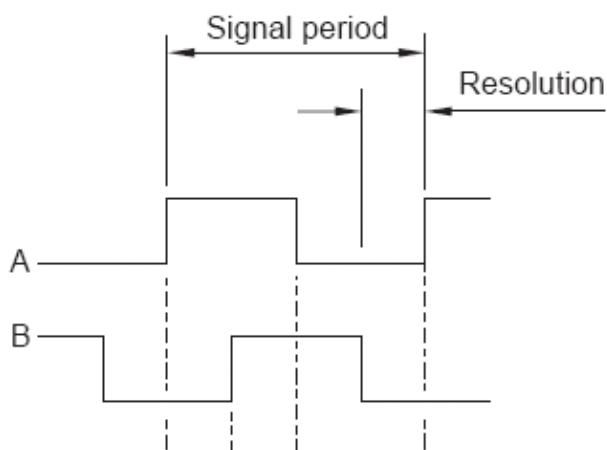


Laser interferometer scale



DISPLACEMENT TRANSDUCERS: REFLECTION SCALES

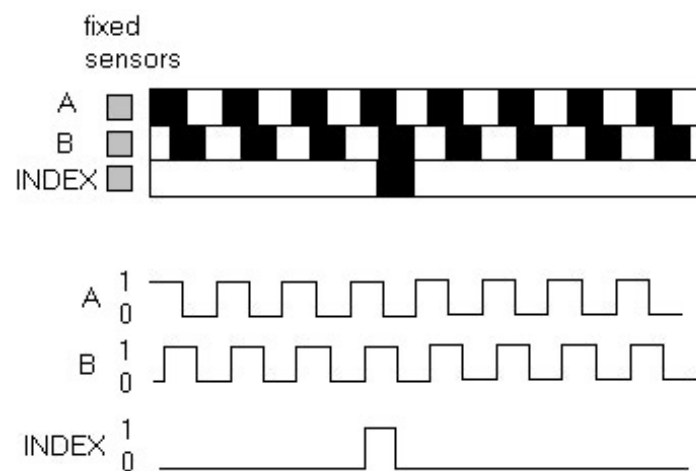
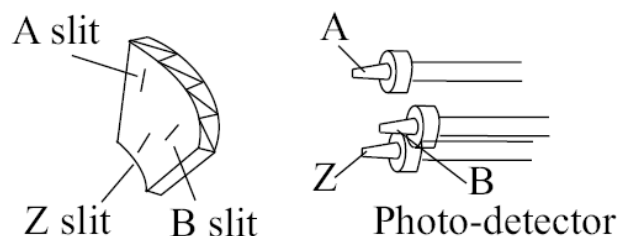
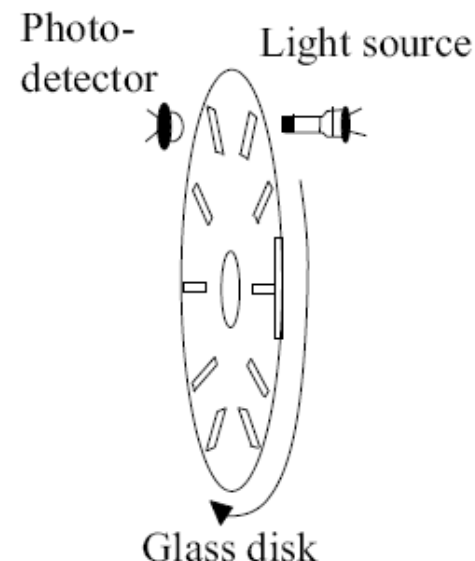
- Direct, digital, incremental or absolute
- Two output signals



Incremental Encoder

Converts the angular position of a shaft to a digital code.

- Direct or indirect, digital, incremental.
- A glass/plastic disk with narrow openings (slits) rotates between a light source and a photo-detector.
- The output signal is a square wave.
- The number of slots defines the resolution.



Absolute Encoder

- The slits are arranged along concentric circles
- A slit provides a binary bit
- The rotation position data is output in binary or decimal form
- The noise is not accumulated



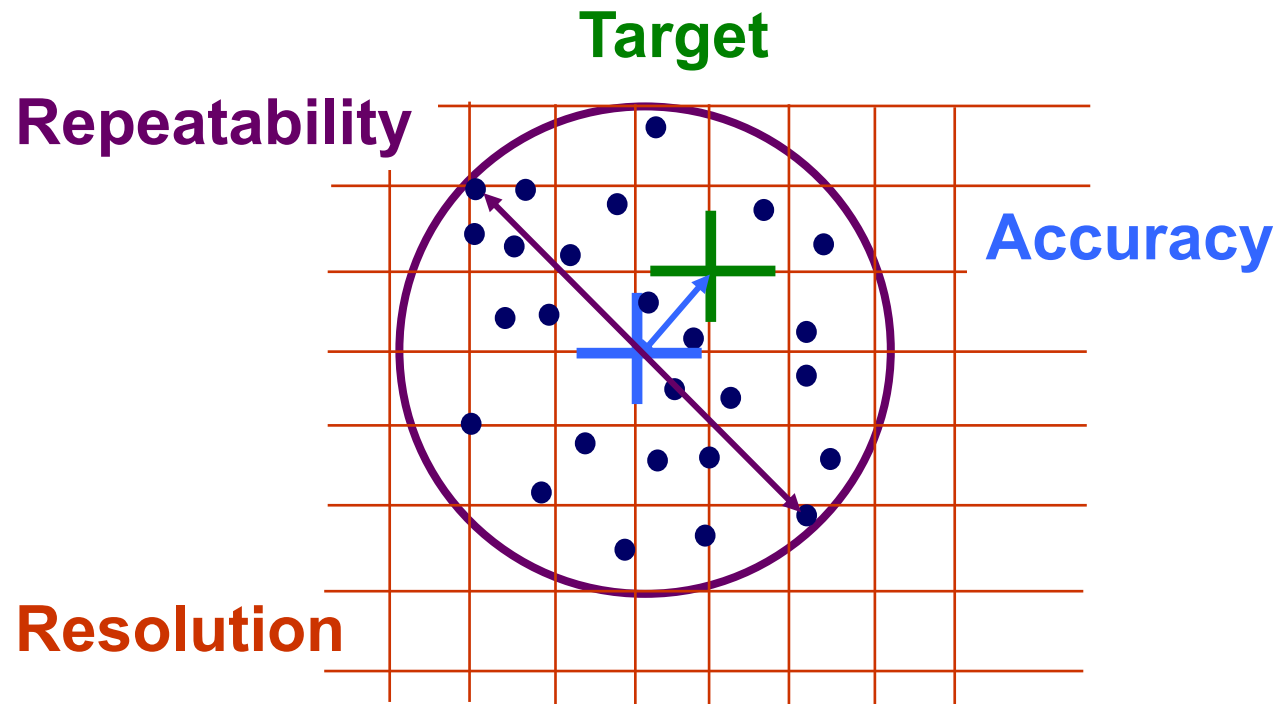


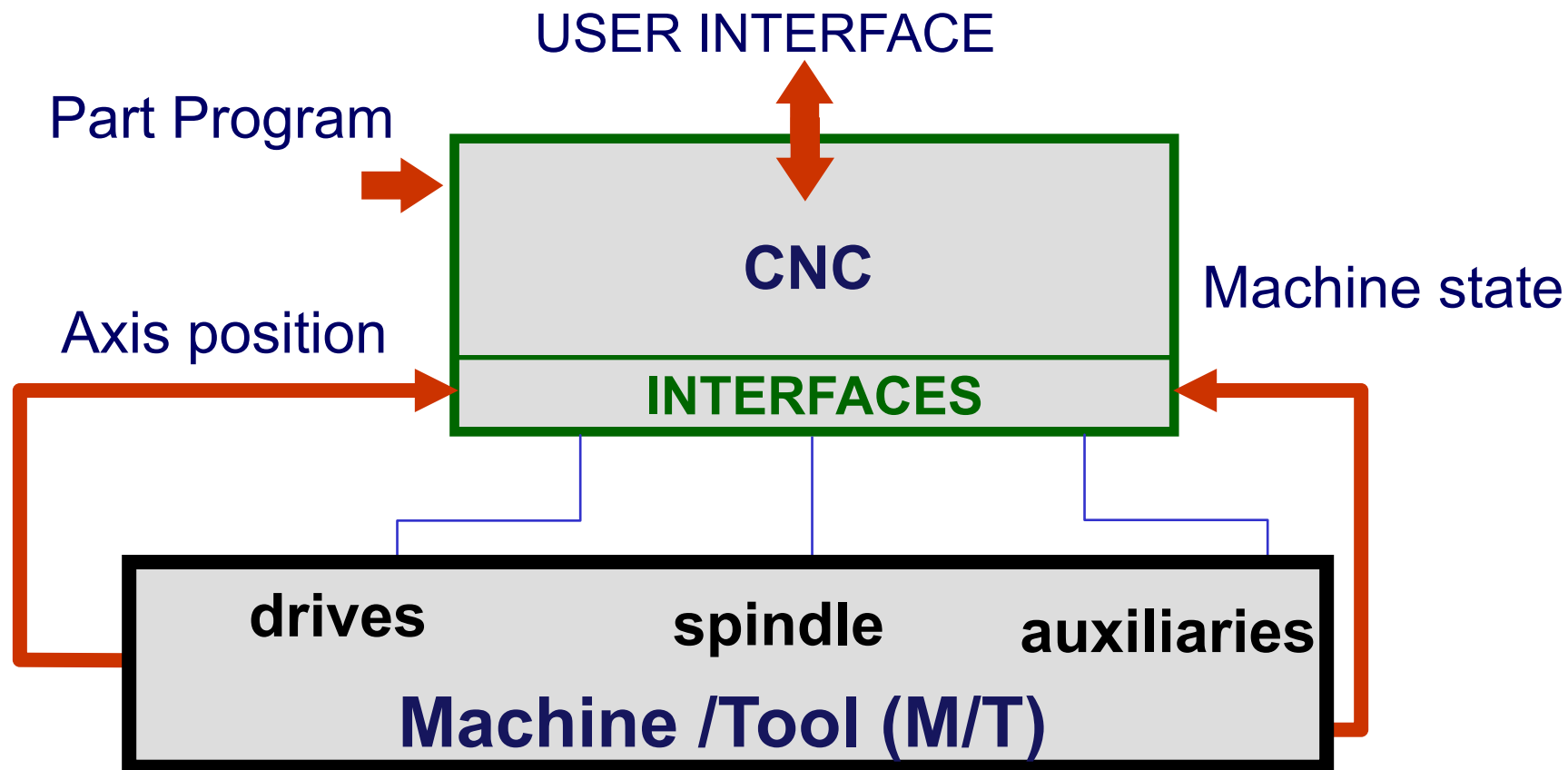
ACCURACY, REPEATABILITY, RESOLUTION

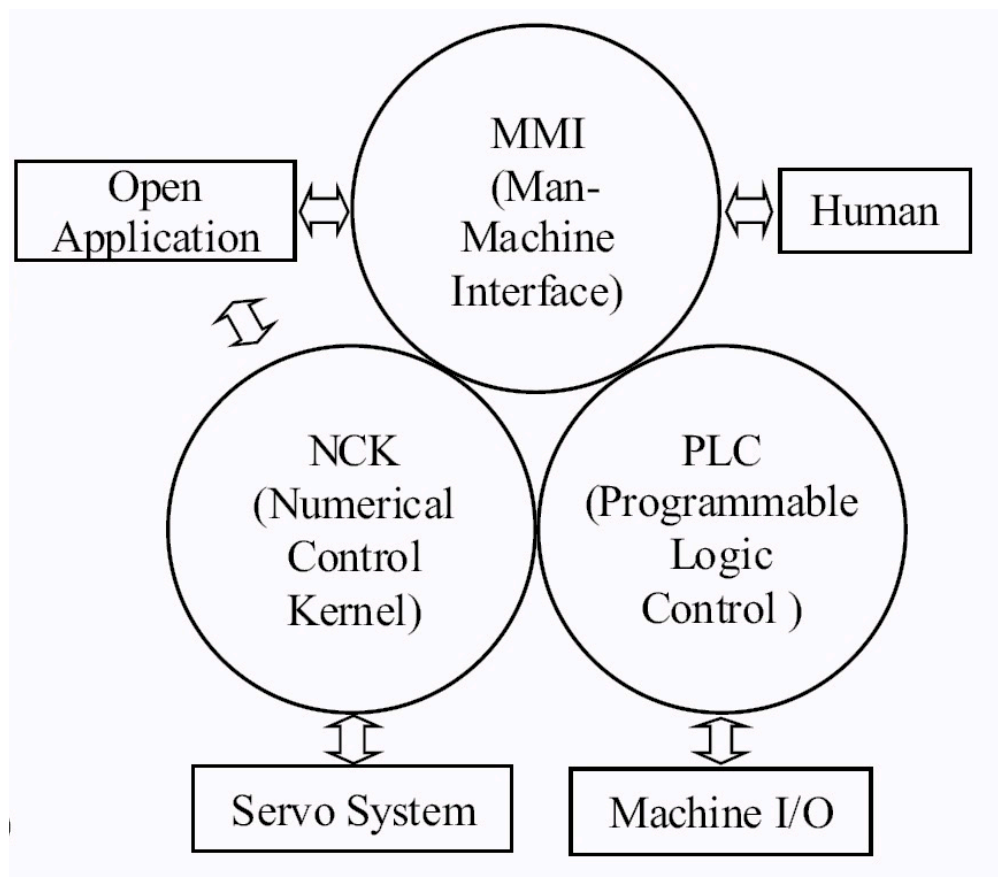
- Accuracy:** Related to the average error on the positioning of the machine.
- Repeatability:** Related to the variability of positioning of the machine.
- Resolution:** Defined as the smallest displacement of the machine axis.

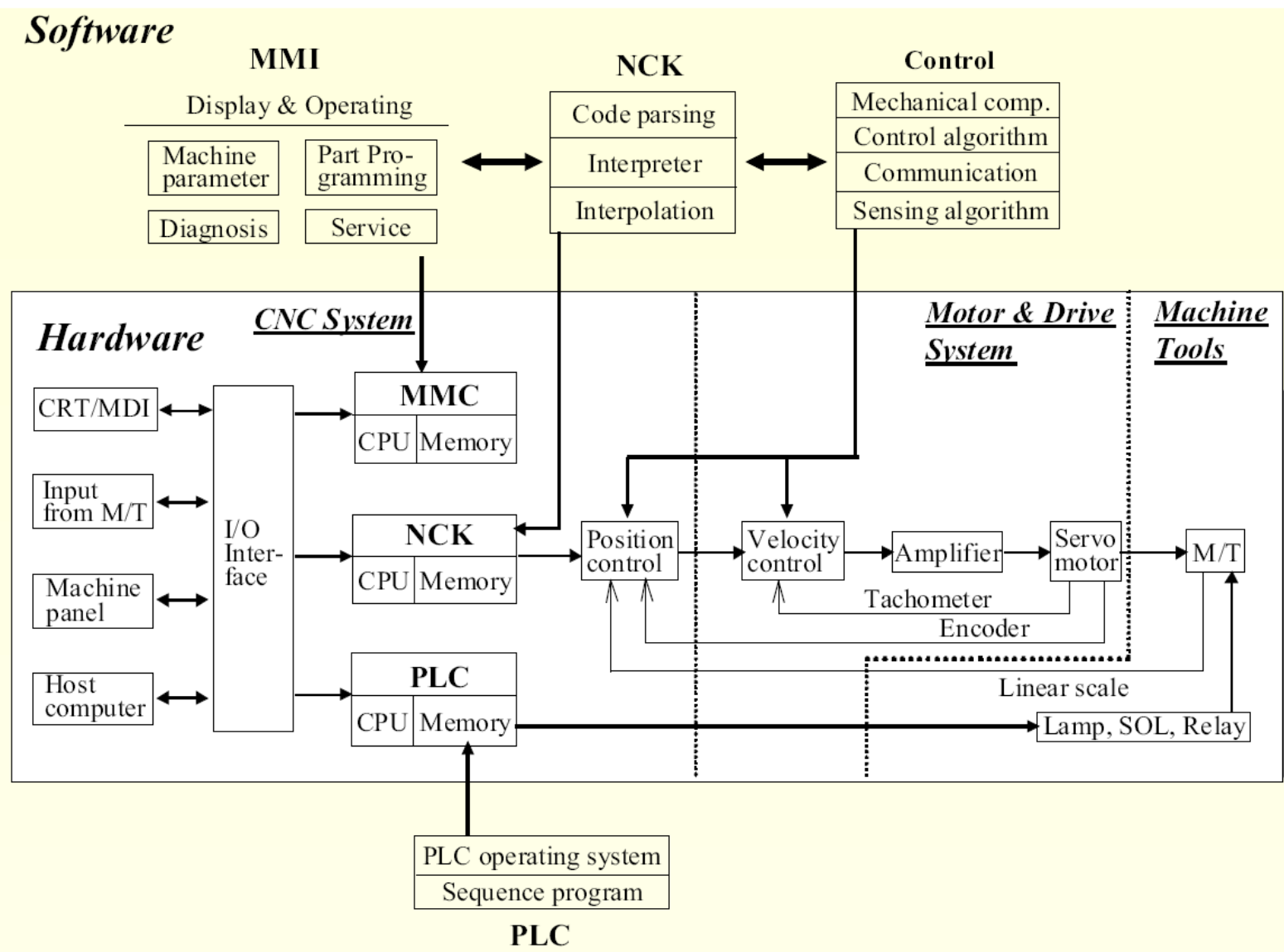


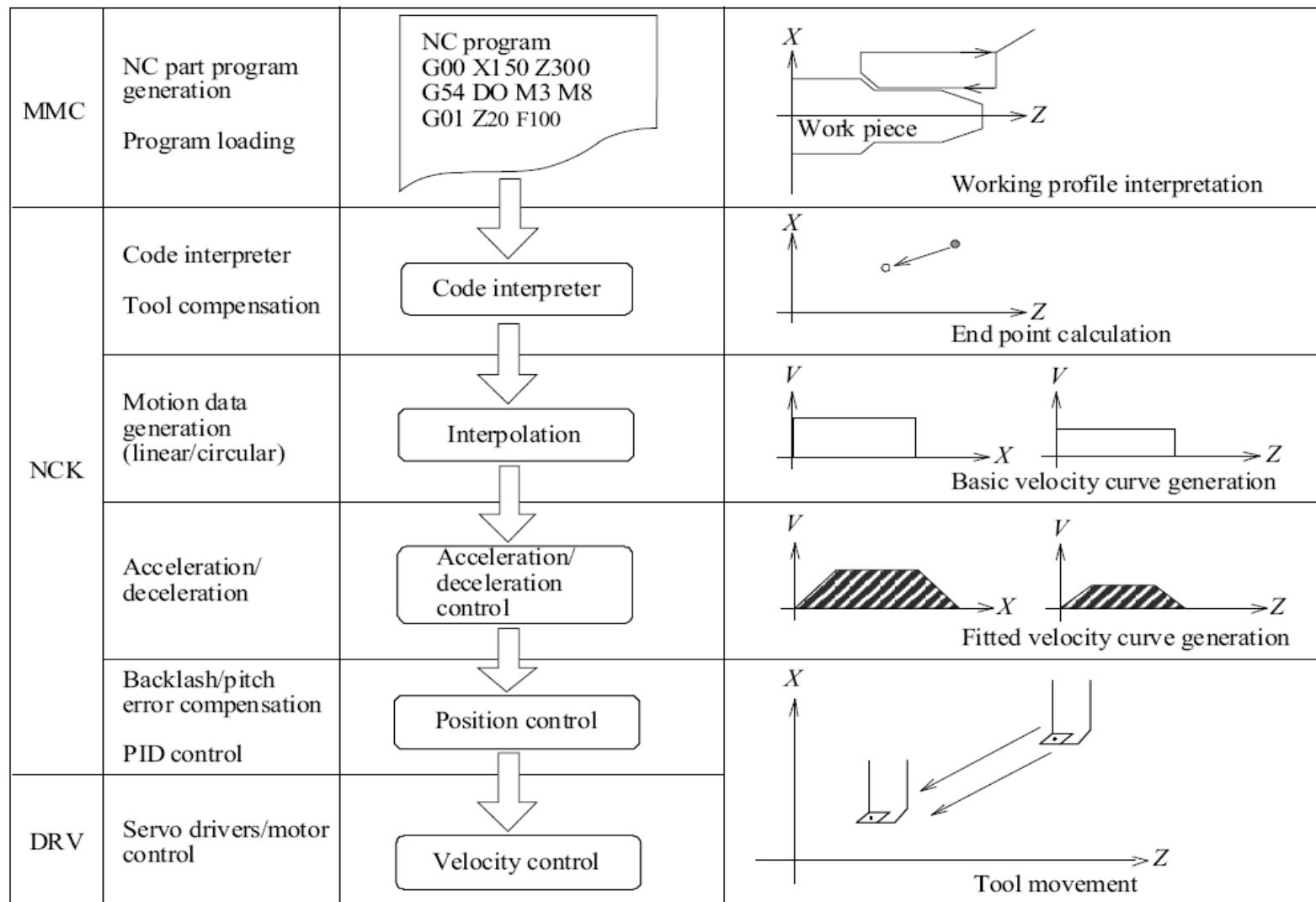
ACCURACY, REPEATABILITY, RESOLUTION











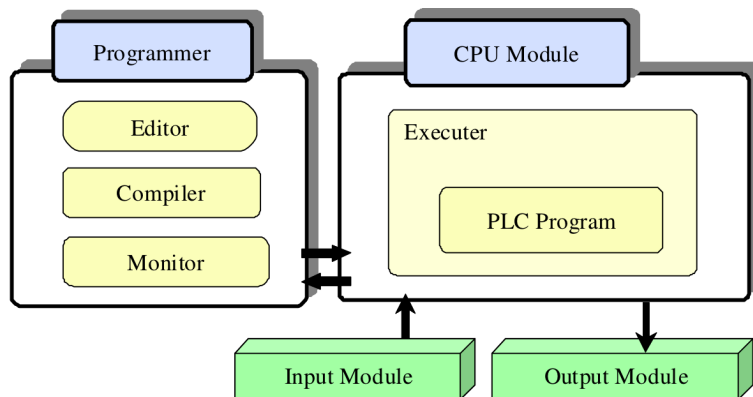


PLC – PROGRAMMABLE LOGIC CONTROL

The logic controller is used to execute sequential control in a machine and an industry. PLC systems consist of a few devices including microprocessors and memory, able to carry out logical operations, a counter function, a timer function and arithmetic operations. Therefore, a PLC system can be defined as a software-based logic controller.

It oversees logic control

- Tool change
- Spindle speed
- Workpiece change
- In/out signals



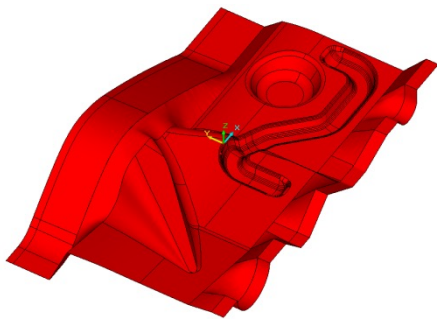


- Communications
- Editing
- Executing the machine operation command
- Displaying machine status

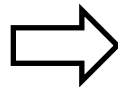




*Ordered set of instructions in a language and in a format required to cause operations to be effected under automatic control, which is either written in the form of a **machine program** on an input medium or prepared as input data for processing in a computer to obtain a machine program (ISO 2806:1994).*

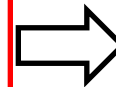


Component



```
***  
G01    X0.0    Y0.0;  
G01    X2.0    Y0.0;  
G01    X2.0    Y2.0;  
G01    X0.0    Y2.0;  
G01    X0.0    Y0.0;  
***
```

Part program



Numerical control



It oversees the servo and driving control

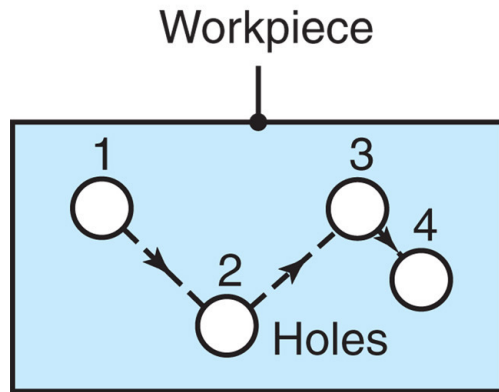
- Code parsing
- Interpreter
- Interpolator
- Error compensator



*Path described by a specified point on a cutting tool
(ISO 2806:1994).*



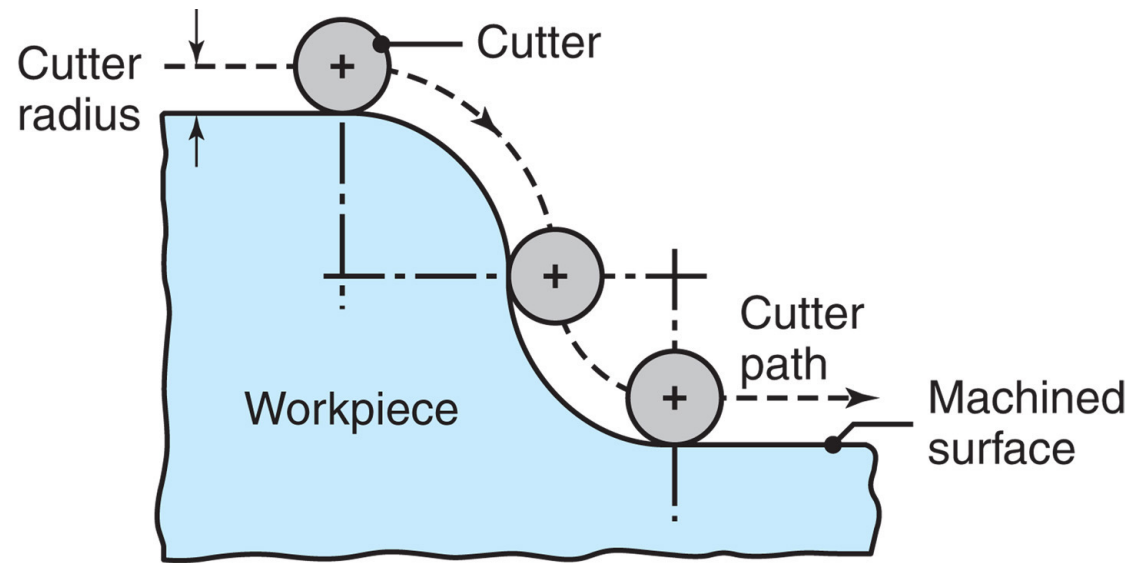
In machine tools
the *CNC controls*
the tool path during
manufacturing
operations.



(a)

Point-to-Point

- Machines axis are driven independently
- Minimize movement time
- Drilling, punching and one-axis milling



(b)

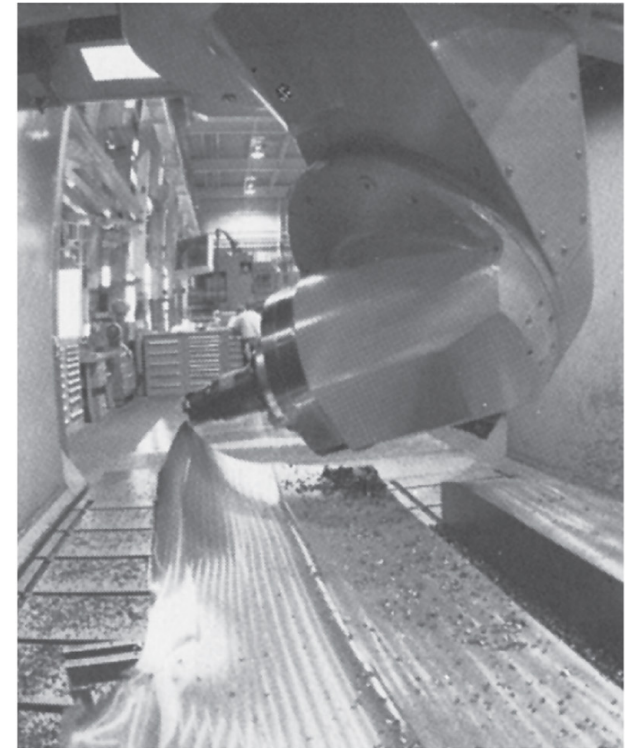
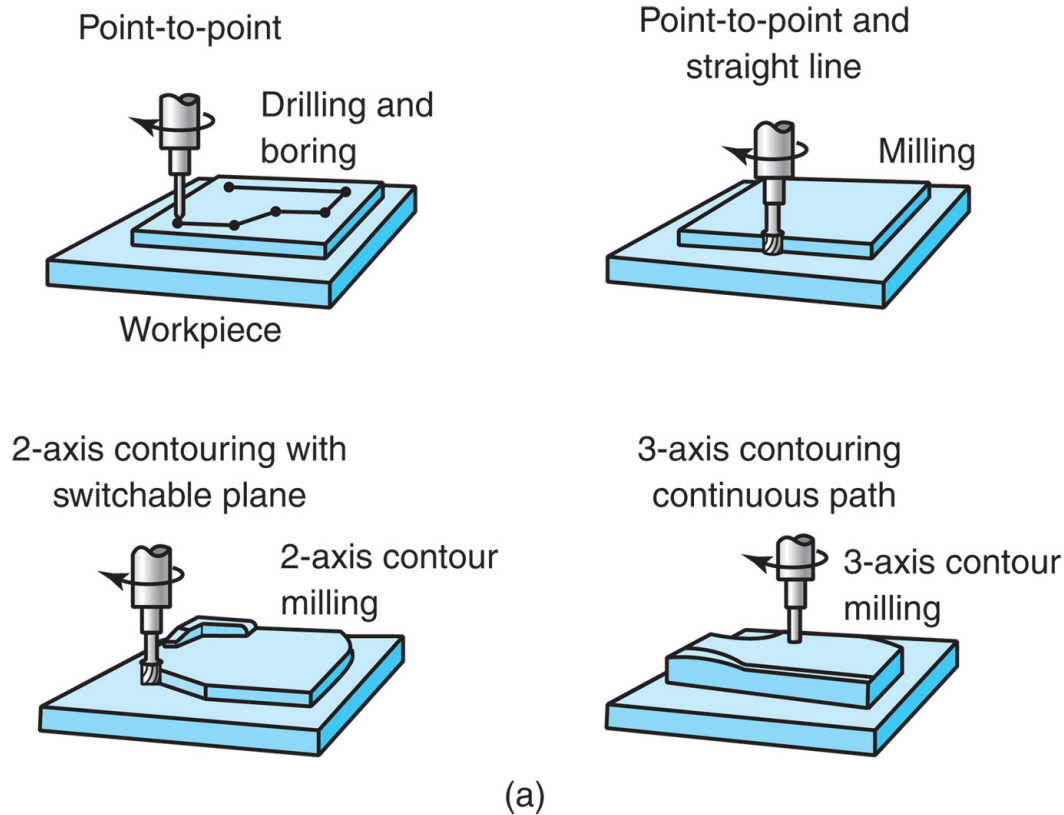
Continuous Path or Contouring

- Fully controlled path
- Complex trajectories in lathes, grinders, machining centers, welding machinery, etc.



TYPES OF TOOL PATH CONTROL

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(b)

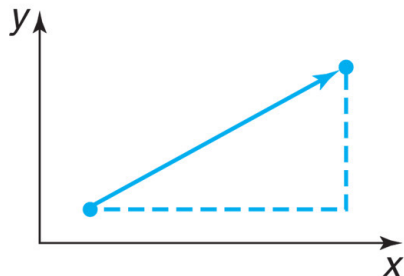
- (a) Schematic illustration of drilling, boring, and milling along various paths.
- (b) Machining a sculptured surface on a 5-axis numerical-control machine tool.



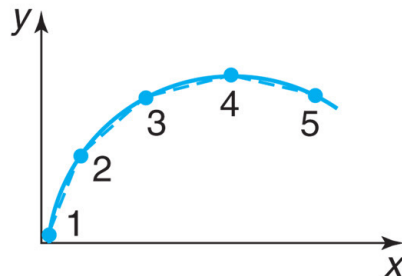
TYPES OF TOOL PATH CONTROL: INTERPOLATION⁵⁸

Determination of points intermediate between known points on a desired path or contour in accordance with a given mathematical function, for example linear, circular or higher order functions (ISO 2806:1994).

Linear

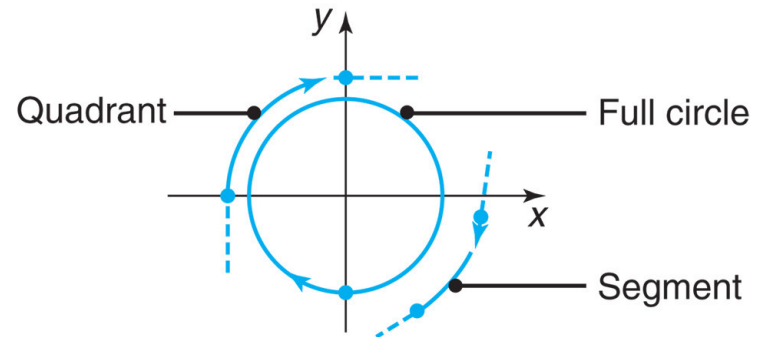


(a)



(b)

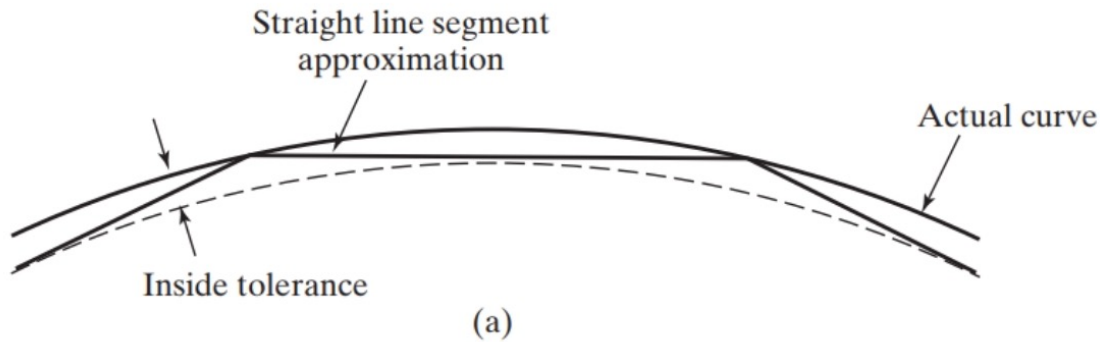
Circular



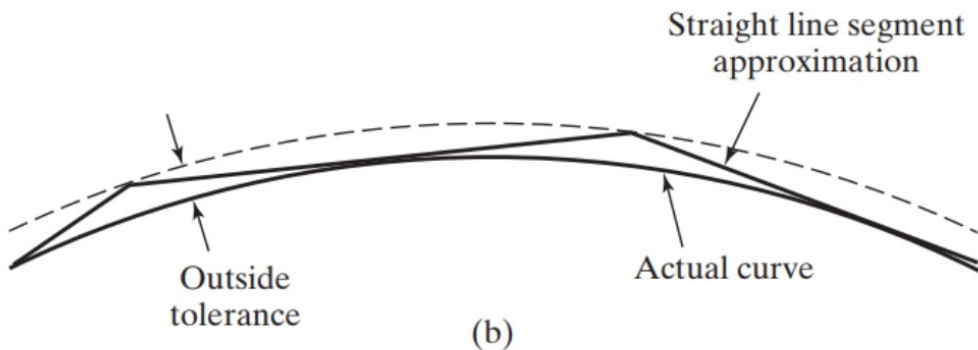
(c)



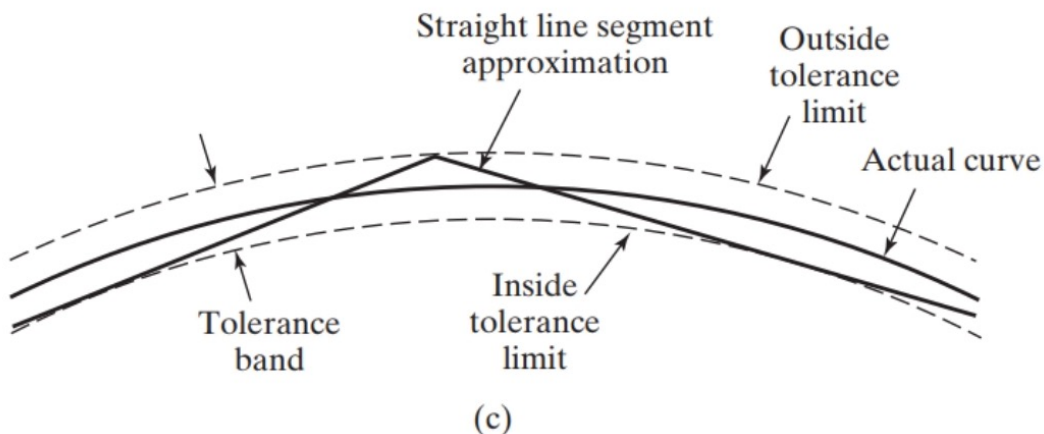
TYPES OF TOOL PATH CONTROL: INTERPOLATION⁵⁹



- a) Tolerance is defined on only the inside of the desired curve



- b) Tolerance is defined on only the outside of the desired curve



- c) Tolerance is defined on both the inside and the outside of the desired curve



TYPES OF TOOL PATH CONTROL: INTERPOLATION⁶⁰

Helical interpolation

This method combines the circular interpolation scheme for two axes with linear movement of a third axis. This permits the definition of a helical path in three-dimensional space.

Parabolic and cubic interpolations

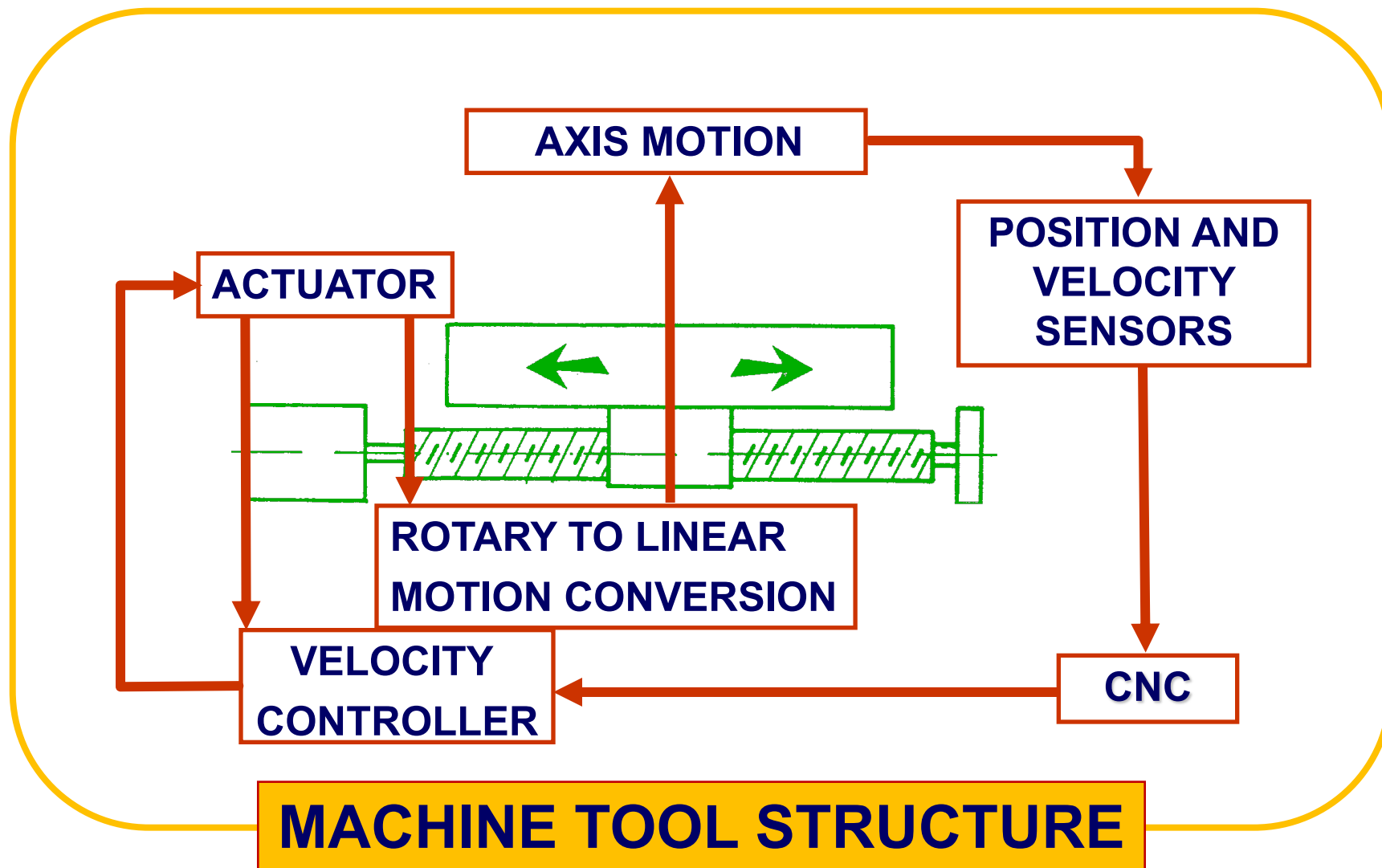
These routines provide approximations of free-form curves using higher order equations.

They generally require considerable computational power and are not as common as linear and circular interpolation.

Most applications are in the aerospace and automotive industries for free-form designs that cannot accurately and conveniently be approximated by combining linear and circular interpolations.

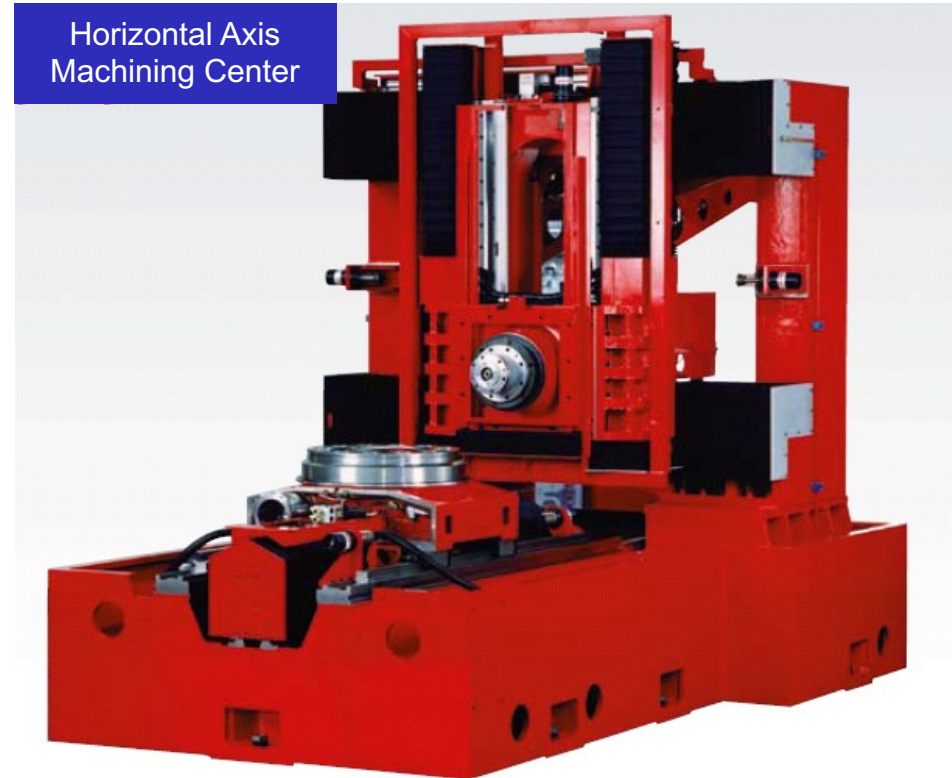


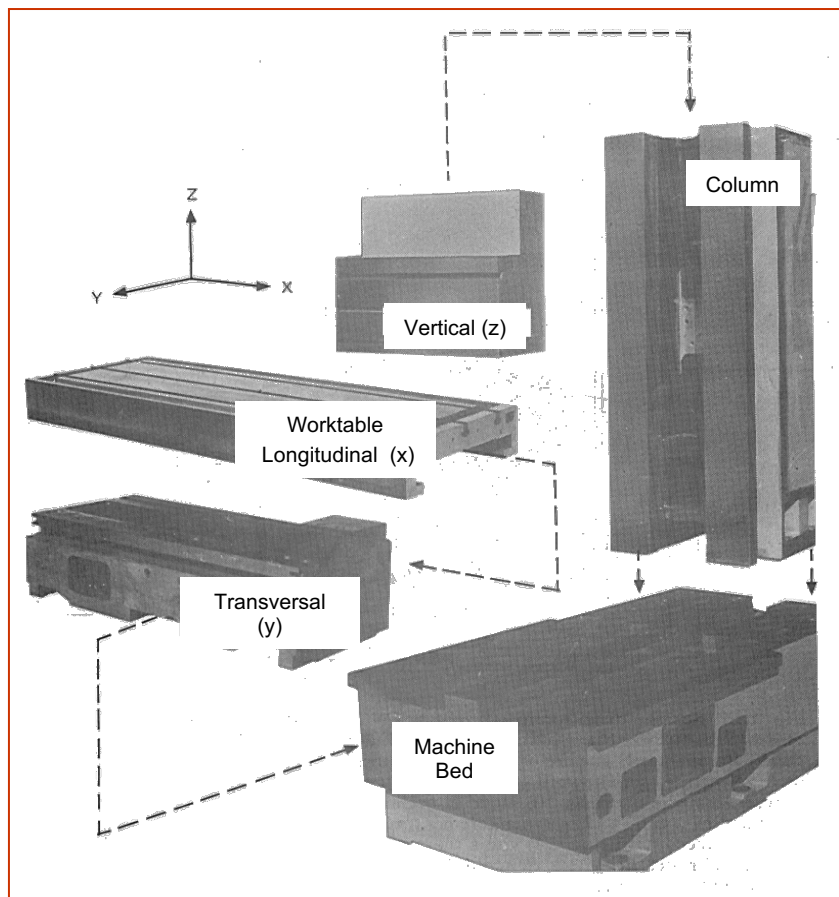
CNC – MACHINE TOOL STRUCTURE



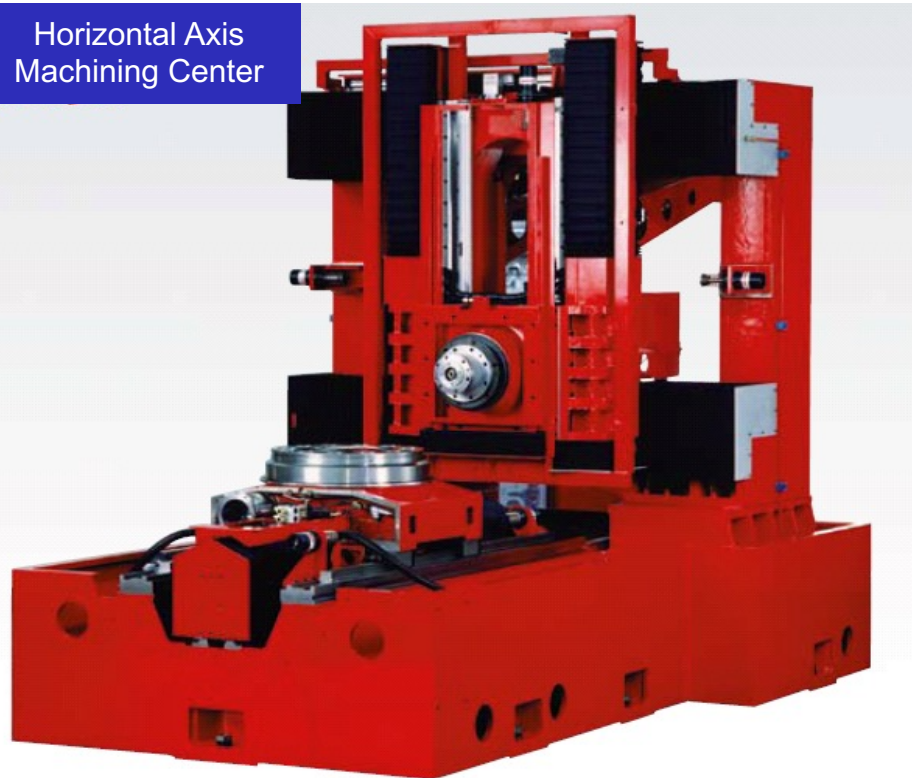


- Stiffness
- Damping
- Thermal distortion
- Assembly techniques
- Guideways
- Linear motor drives
- Machine foundations





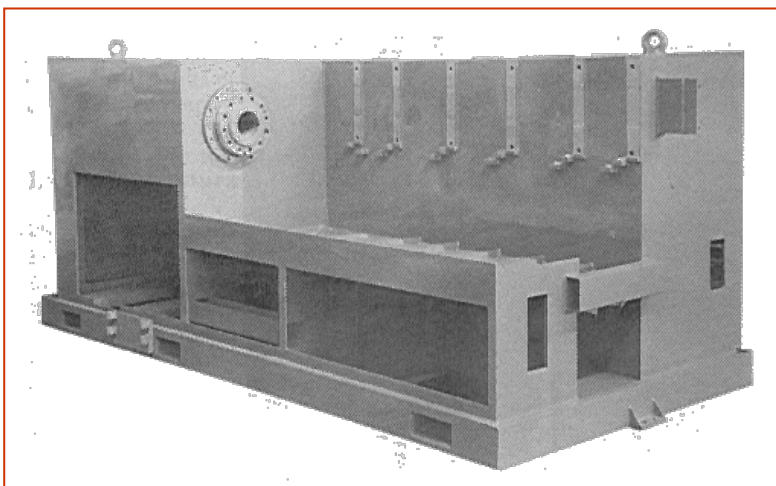
Horizontal Axis
Machining Center



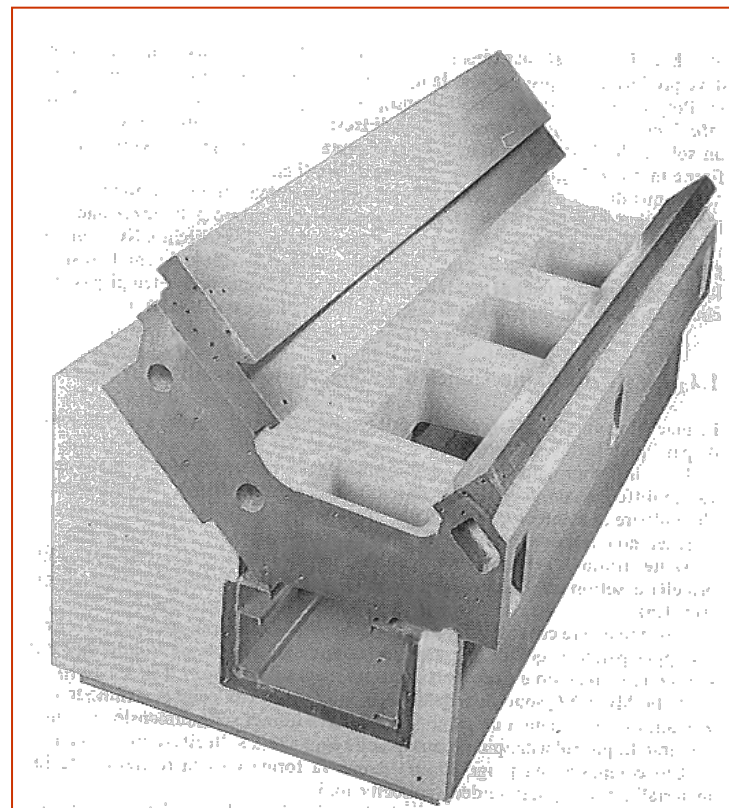


Machine Tool Bed or Base

Welded Steel



Cast Iron





Gray cast iron: Good damping capacity, low cost, heavy. Heat treatment (annealing, ageing) to remove residual stresses emerged in casting.

Welded steel: Lighter than cast-iron structures, high mechanical properties, low damping capacity. Heat treatment (annealing, ageing) to remove residual stresses emerged in welding.

Polymer concrete can be used to manufacture a structure component or can be poured into steel or cast iron structures to increase the mass and the damping capacity.



Single-purpose machines

- Evolution of the traditional machine tools
- Numerical control
- Specific machining processes

Multi-purpose machines

- Numerical control
- Very flexible (several machining processes)
- Two types: turning center and machining center



MACHINE TOOL: CHARACTERISTICS

Working envelope

Spindle: number, type, orientation, power, torque, velocity

Axis: number, type, orientation

Technical characteristics:

- Velocities, accelerations, idle times
- Accuracy, repeatability, resolution (in positioning)
- Reliability



Space in which workpieces can be machined.

- It depends on the machine axes and component spaces
- It defines in practice the size of the machine
- *Working cube* is also used as synonymous



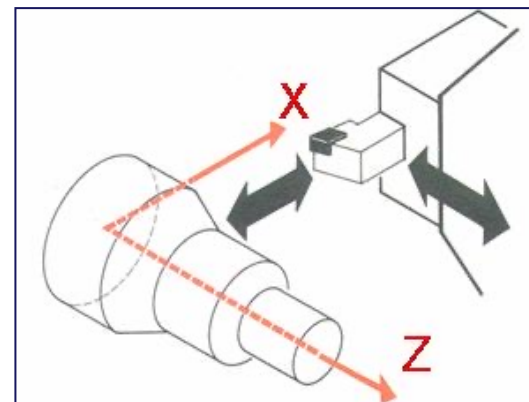
MACHINE TOOL: CHARACTERISTICS

- Working envelope
- Kinematic (axis)
- Accuracy, resolution and repeatability

**Product mix
characteristics**

- Spindle Power
- Velocity and acceleration
- Reliability

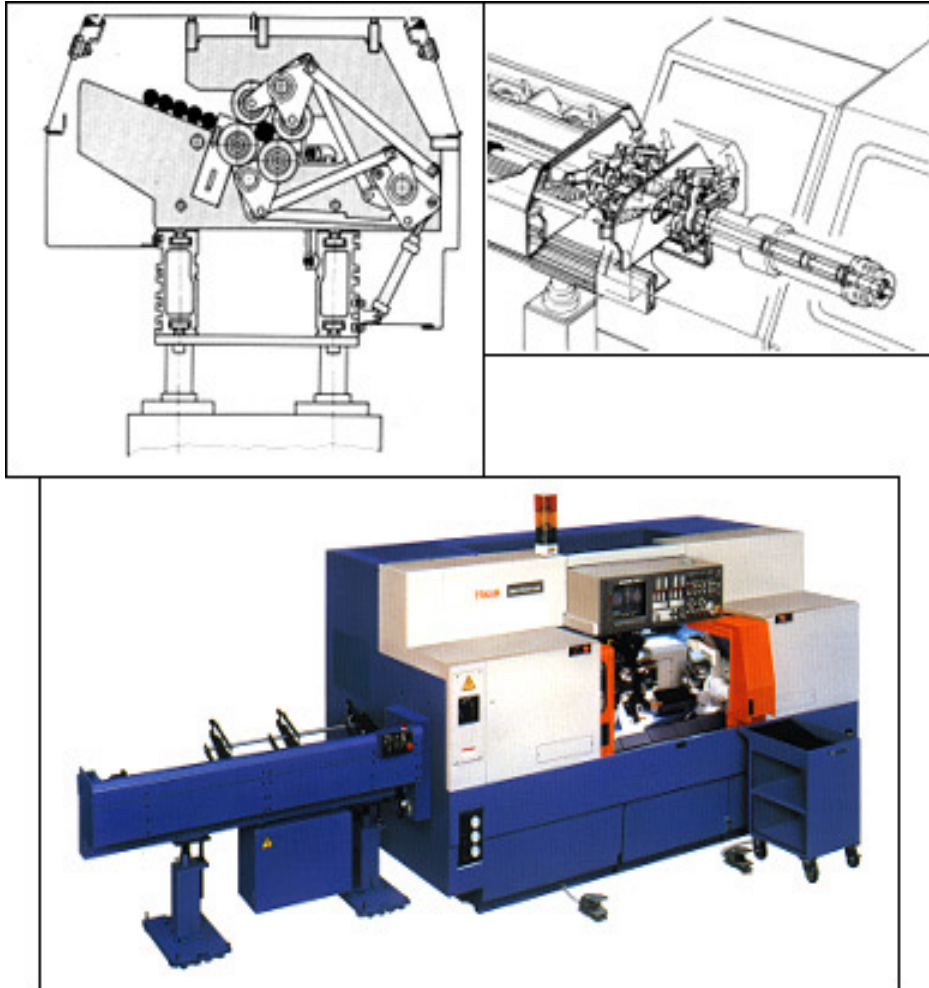
Efficiency



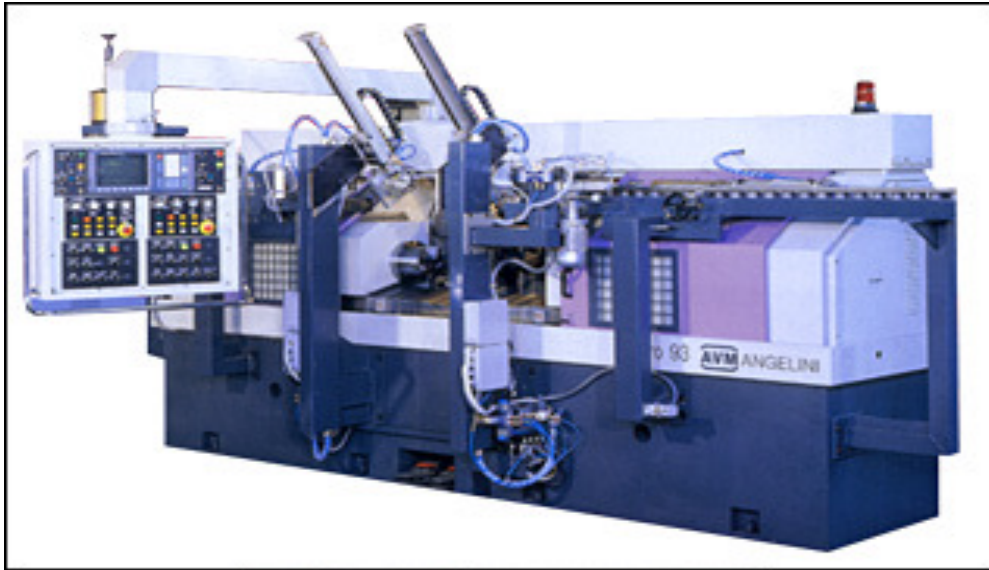
Main lathe machine axis



- Loader, feeder, unloader
- Turrets
- Automated chip removal
- Tool storage
- Tool monitoring and setting

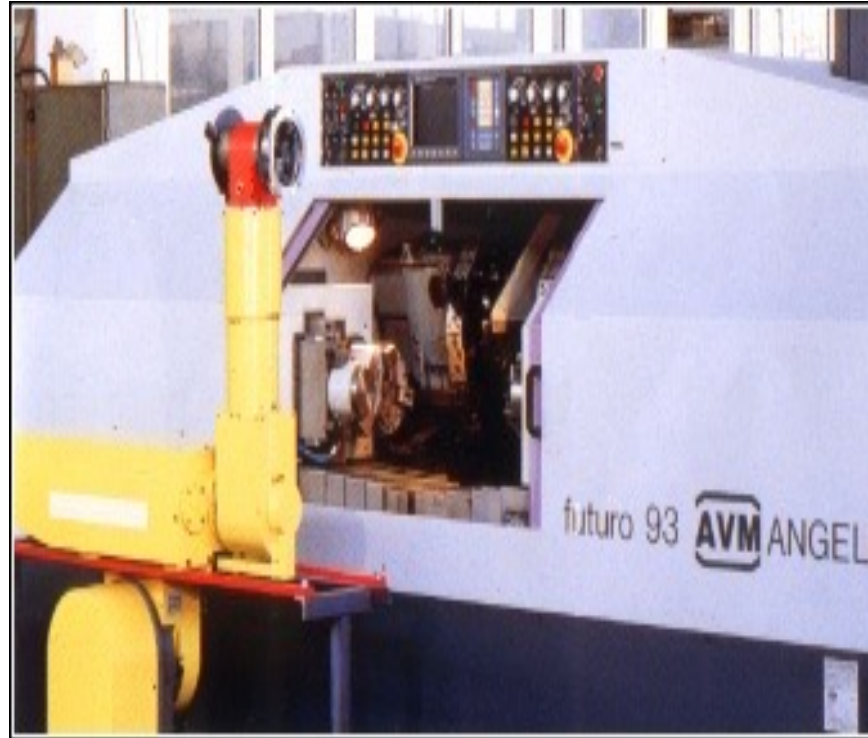


Automatic bar (with different section shapes) loader and feeder through the spindle



The unloader puts the finished part on a belt conveyor.

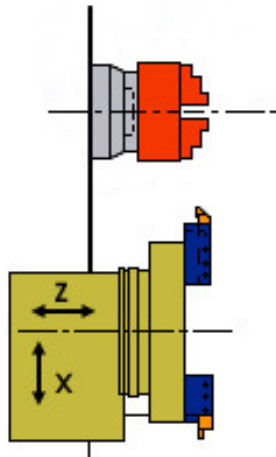
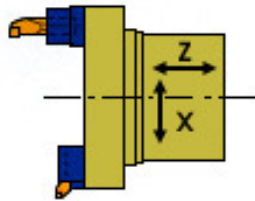




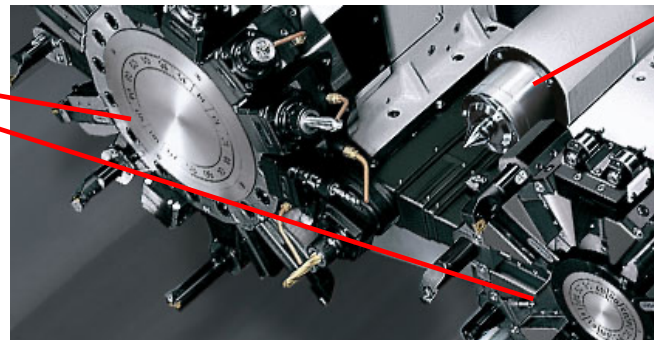
A manipulator loads and unloads the parts.



Turret



Two turrets
with
simultaneous
use of tools

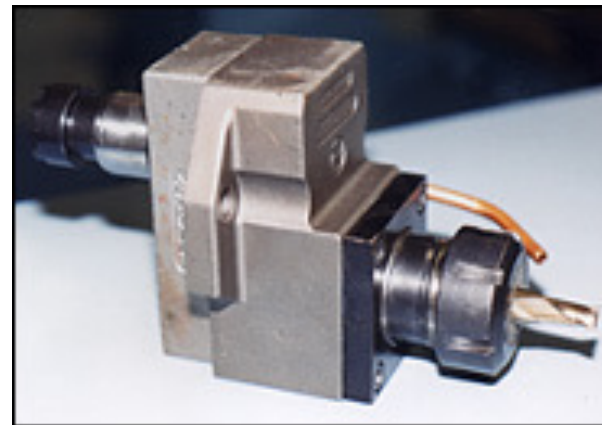
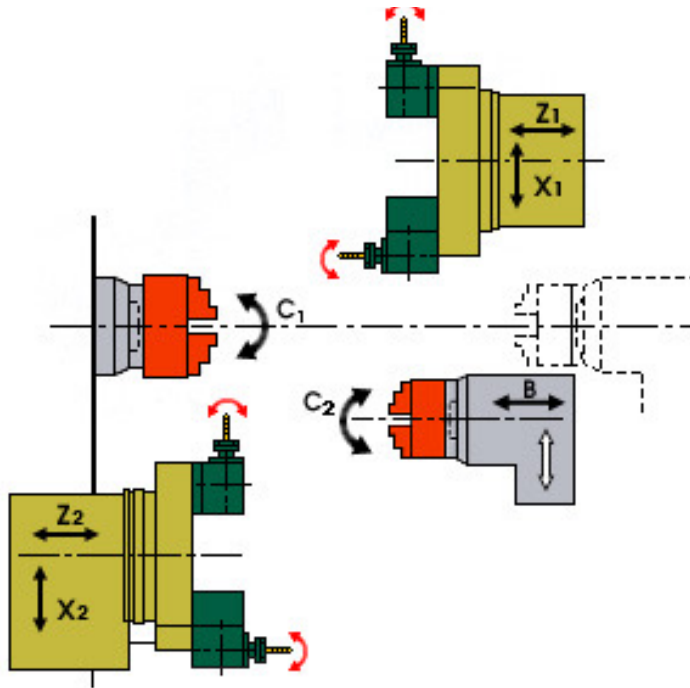


tailstock



TURNING CENTER: DRIVEN TOOLS

Turret with driven tools for drilling, milling and tapping.



drilling

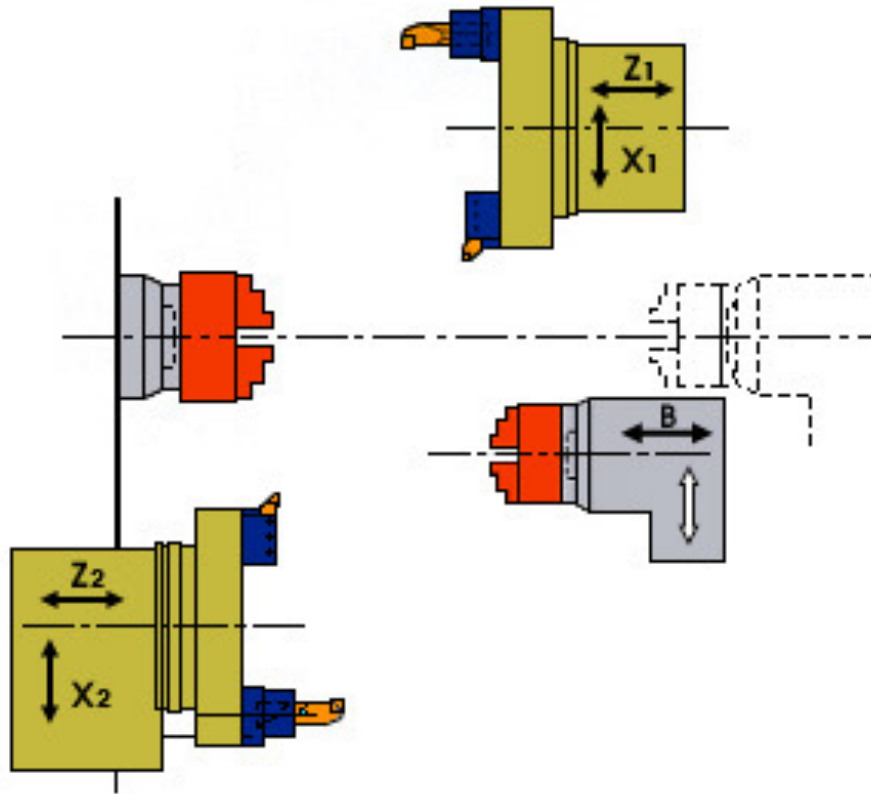


tapping

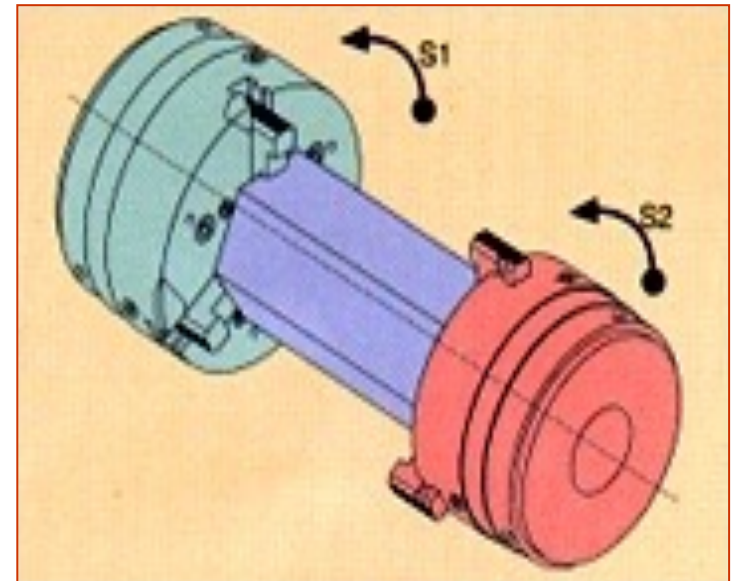


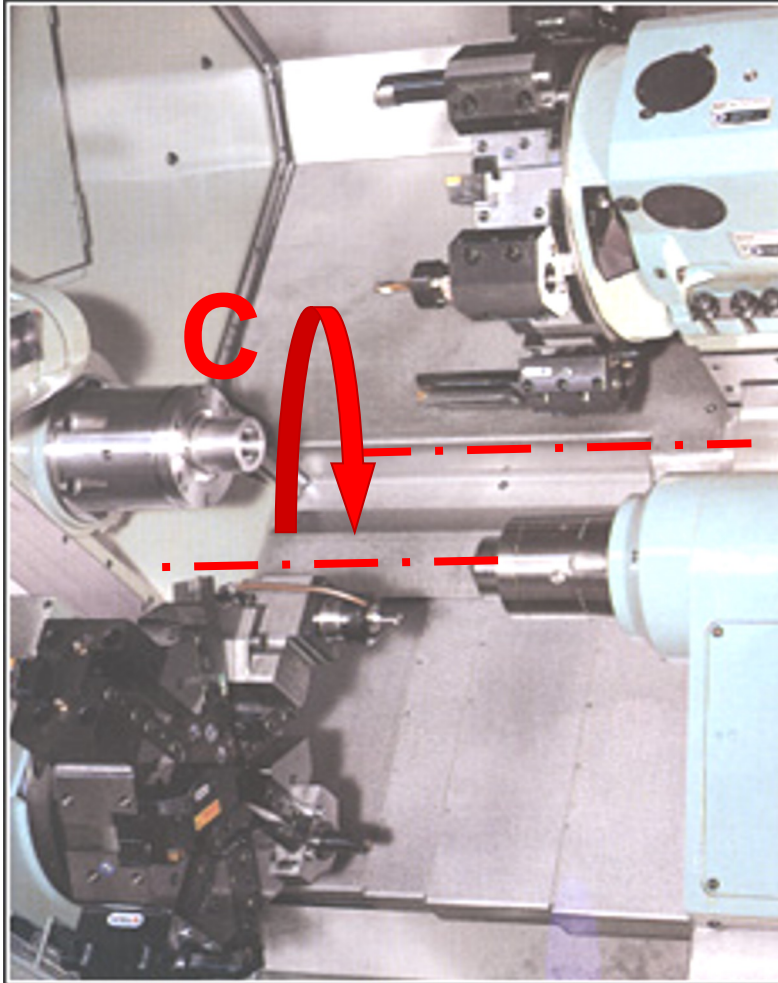
milling



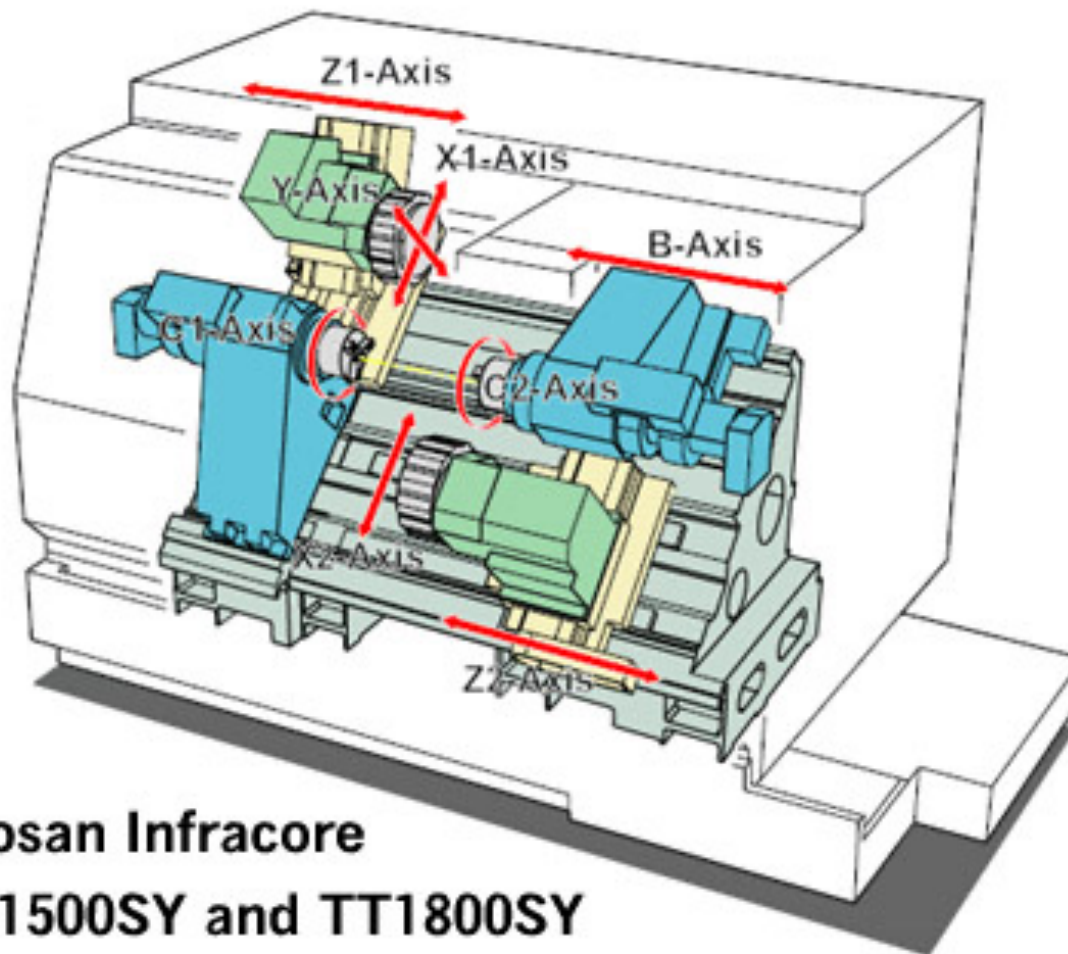


Two turrets and
two spindles

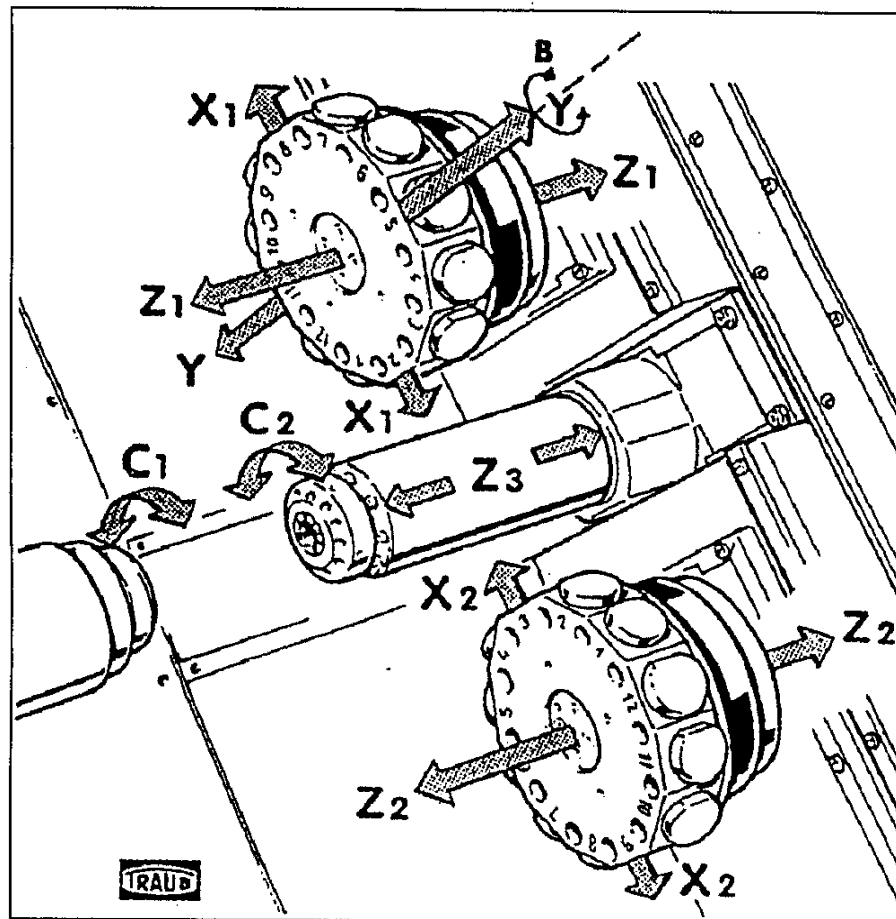




The axis C on both spindles allows the CNC the synchronization of the part movement between the two blockages.



**Doosan Infracore
TT1500SY and TT1800SY**

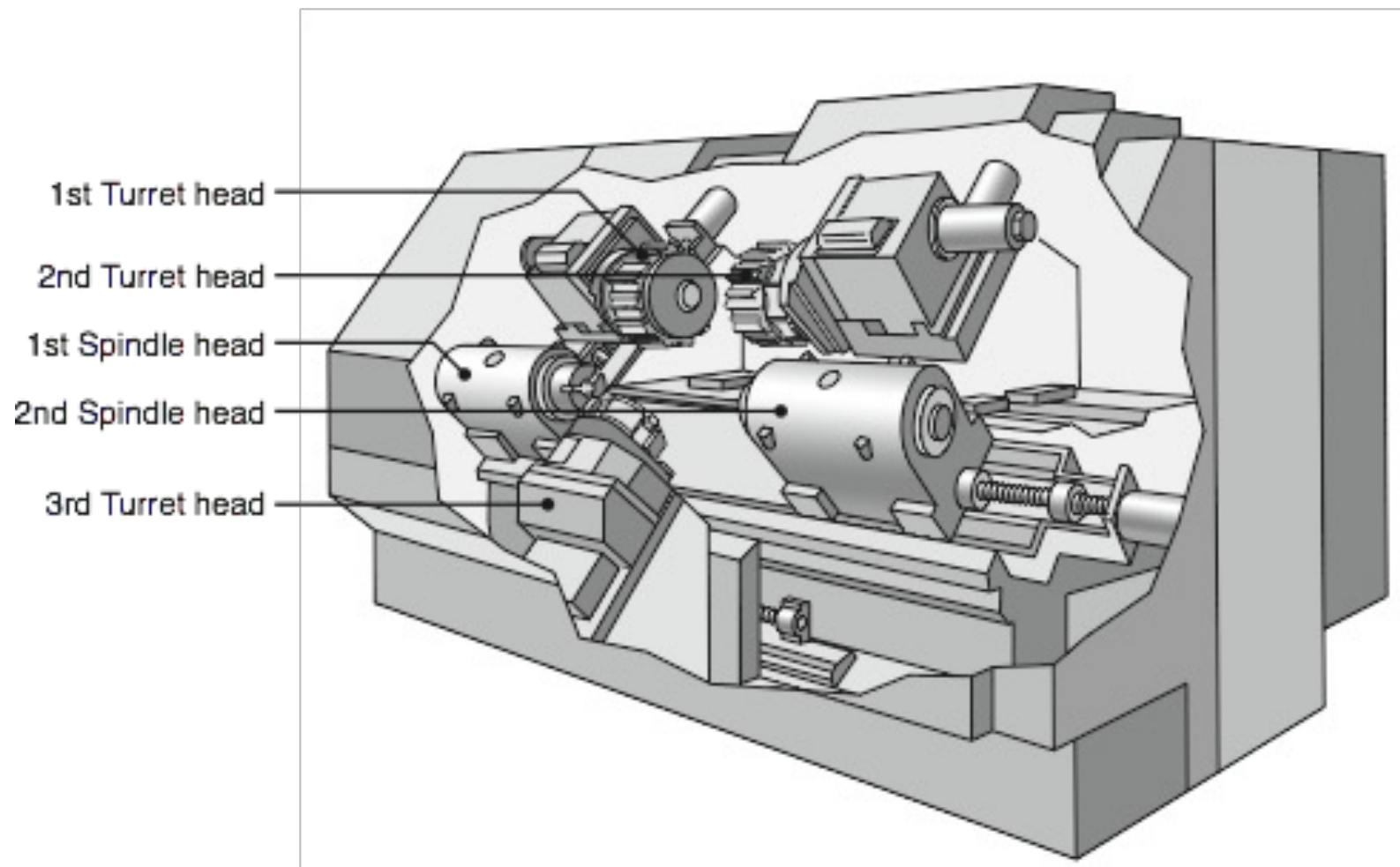


9-axis machine



TURNING CENTER

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Technical characteristics

- *Turning diameter*
- *Turning length with chuck*
- *Spindle nose*
- *Bar passage*
- *Spindle power*
- *Driven tools power*
- *Spindle speed*
- *Driven tools speed*
- *Tailstock travel*
- *“Z” axis travel*
- *“Z” axis rapid*
- *Ball screw diameter x pitch*
- *“X” axis travel*
- *“X” axis rapid*
- *Number of positions turret*
- *α indexing time*
- *Tool size*
- *Overall dimension*
- *Net weight*
- *Chip conveyor*
- *Tool setter*
- *Tool loader and unloader*



MORI SEIKI
THE MACHINE TOOL COMPANY

<https://youtu.be/zhkv-YjNdbw>

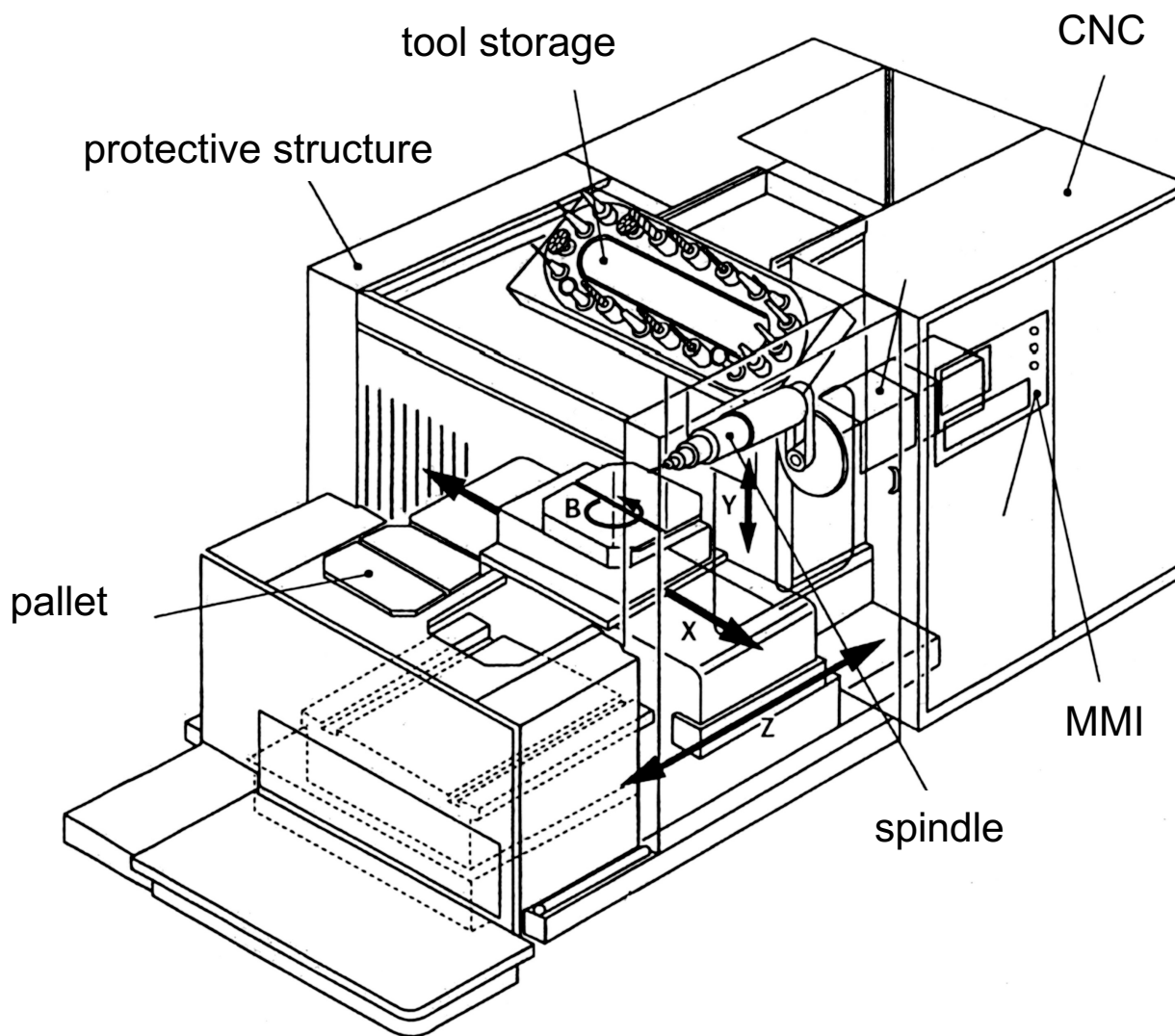


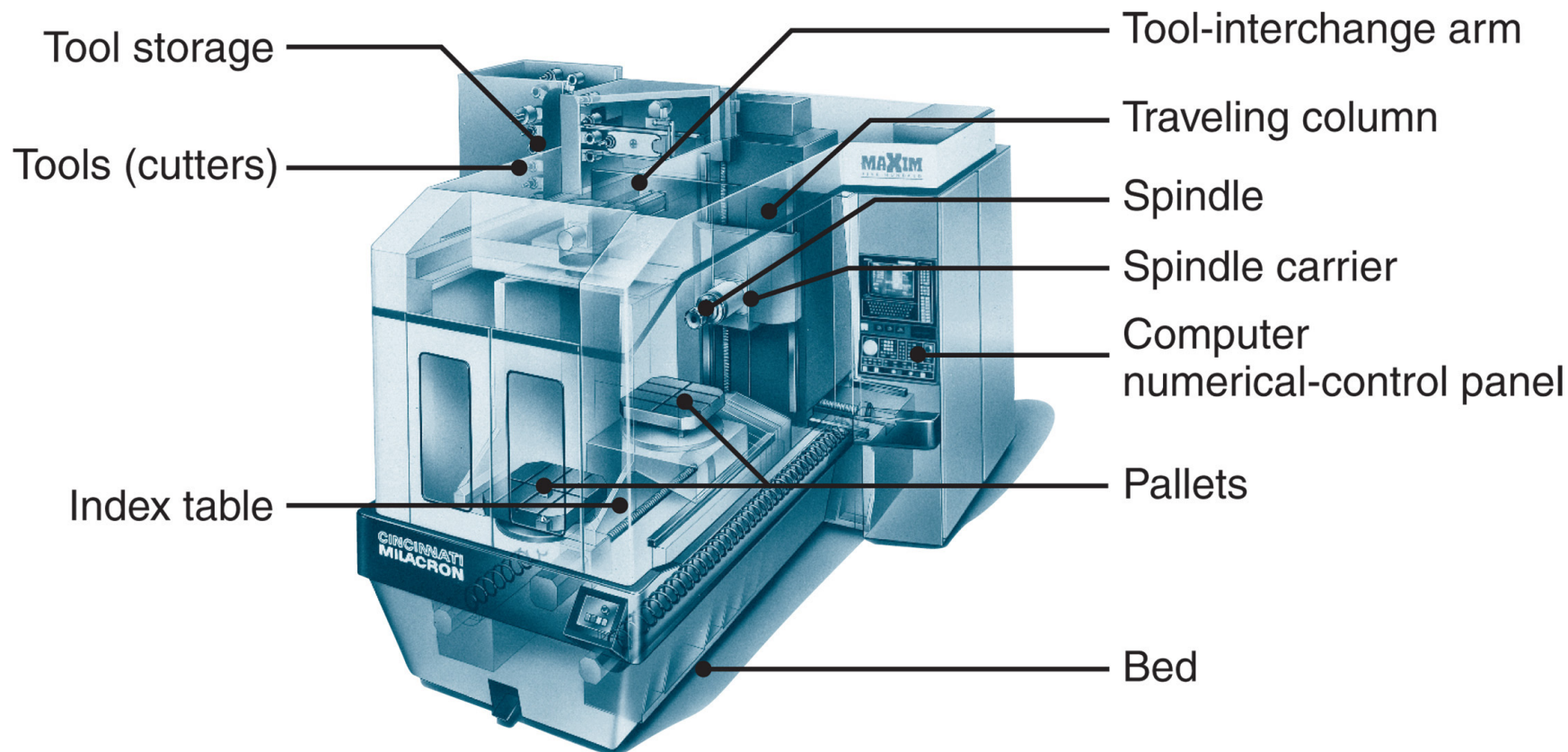


<https://youtu.be/6XTjyYVtZyc>



<https://youtu.be/MyPcOHLBW7w>





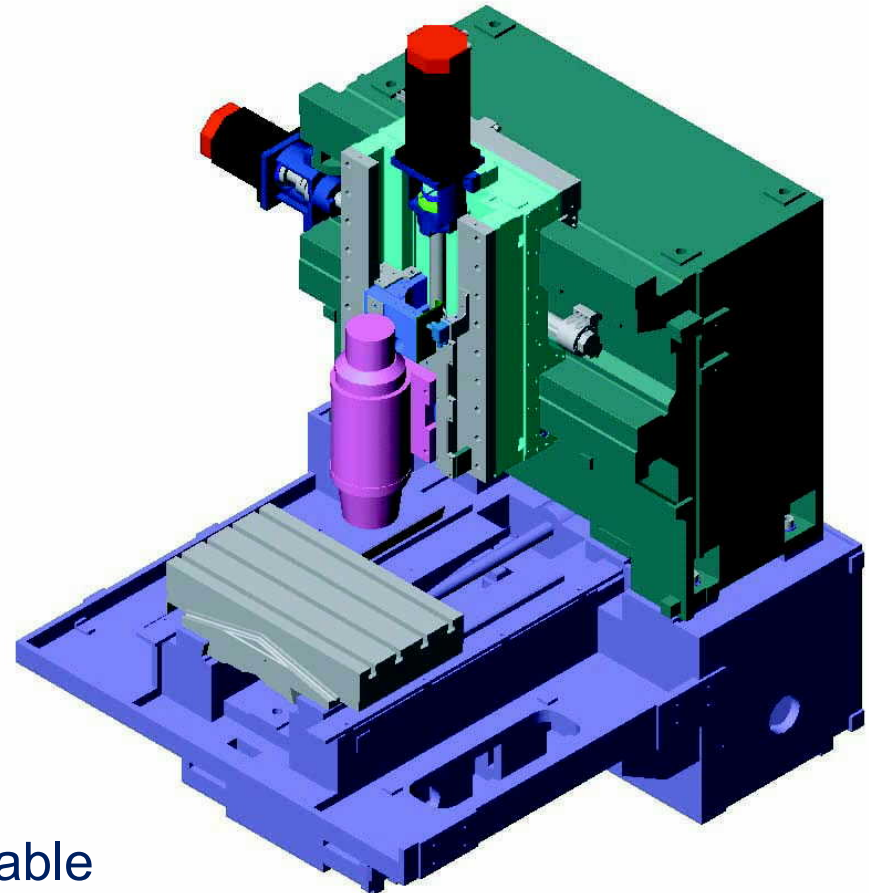
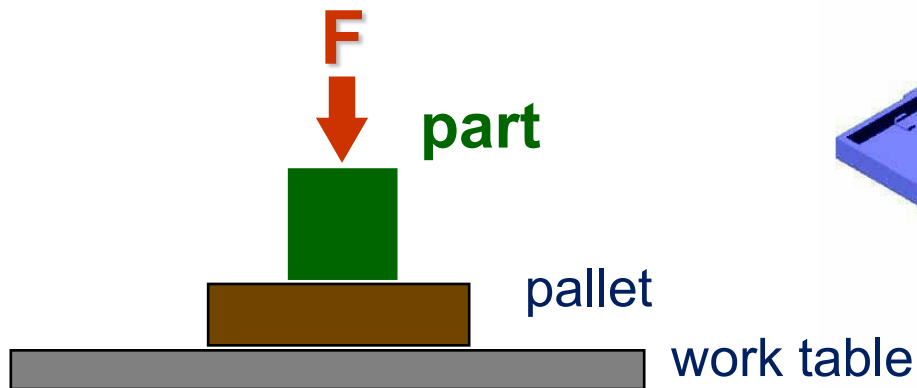
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- Spindle orientation
 - horizontal
 - vertical
- Motion:
 - Tool motion
 - Part motion (working table)

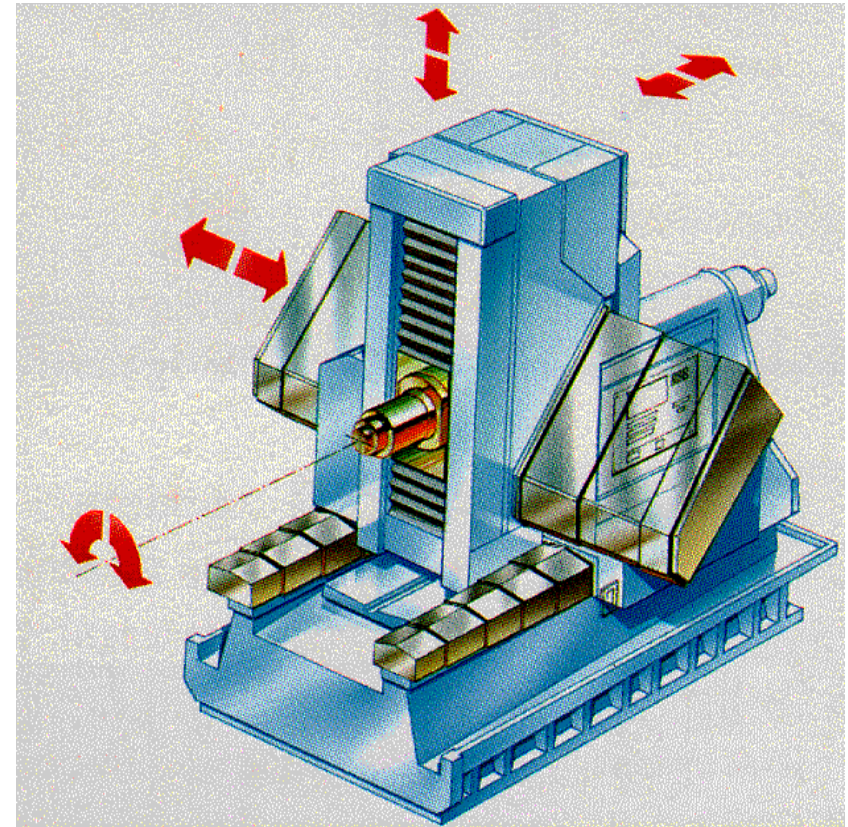
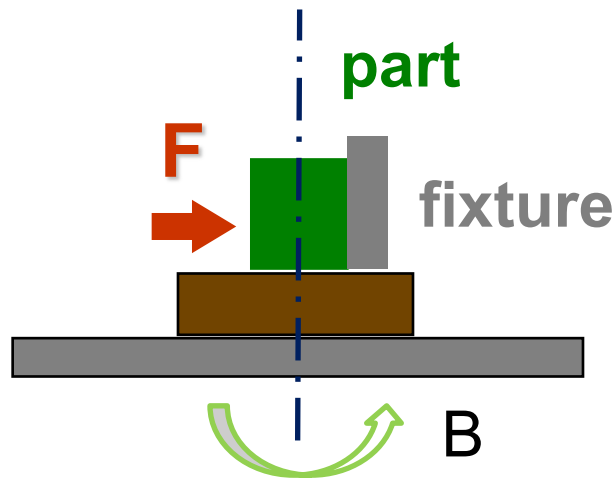


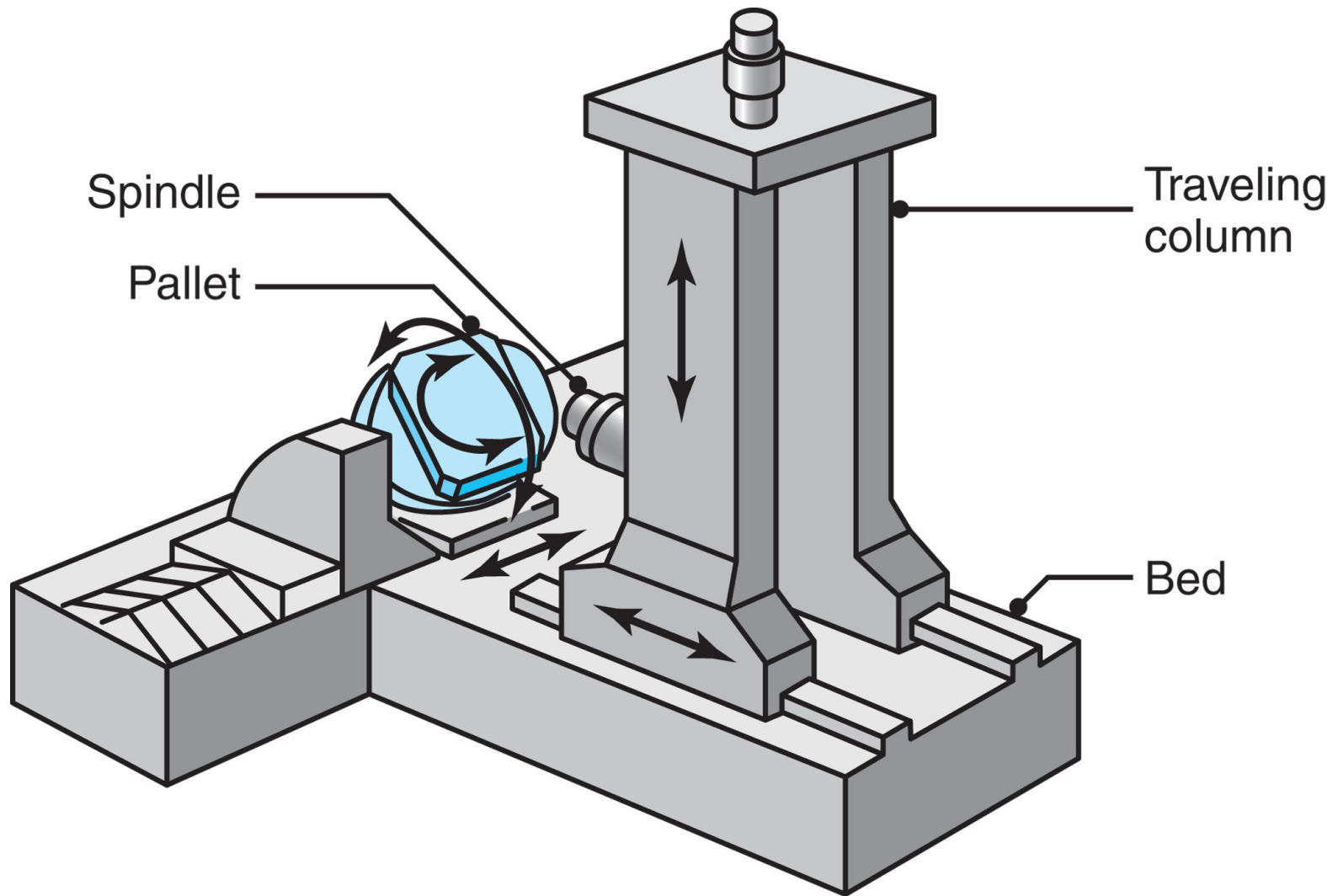
- Forces are sustained by the table
- Problems with chip removal

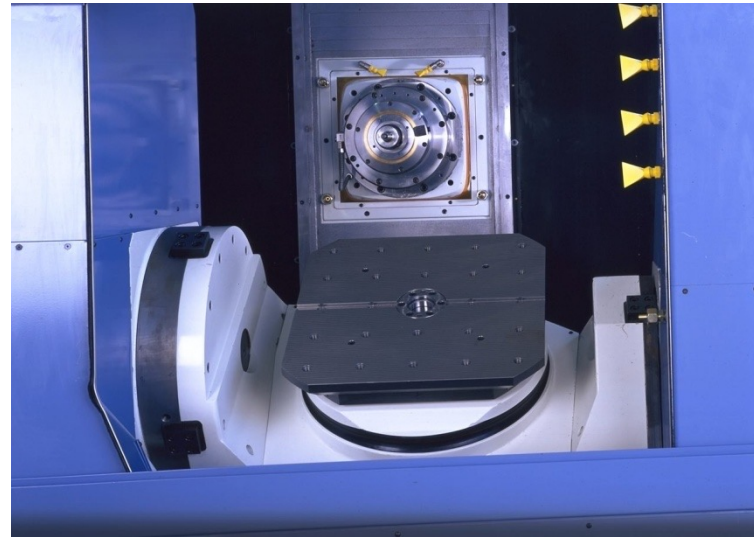
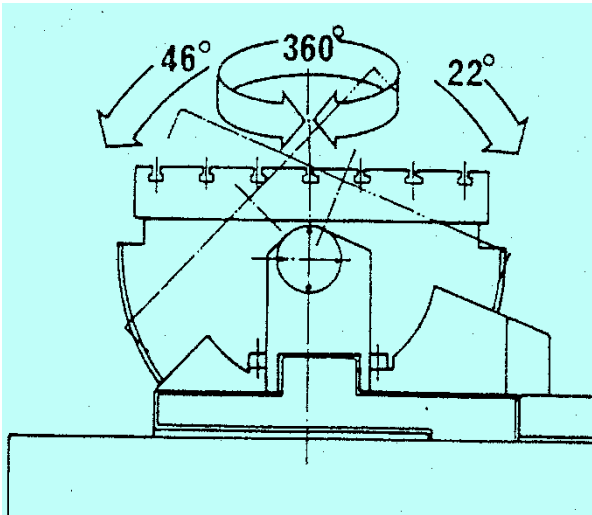




- Higher surface accessibility by moving axis B
- Forces are sustained by the fixture





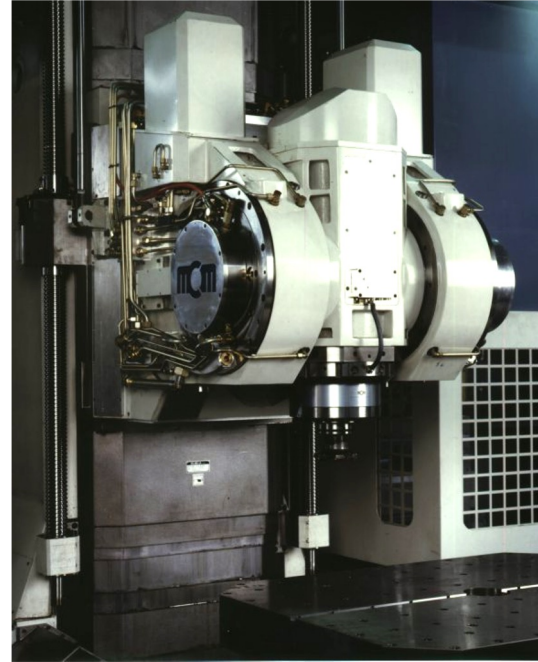
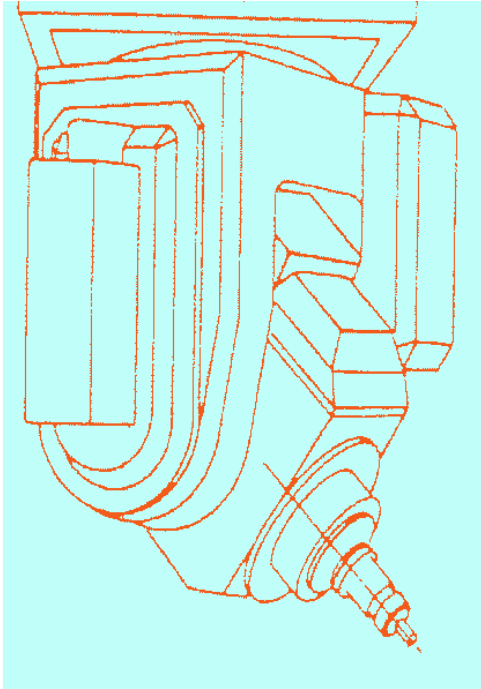


Rotary table

- 4th axis in horizontal spindle machines (B axis)

Tilting table

- 5th axis in horizontal spindle machines (A axis)
- 4th or 5th axis in vertical spindle machines



Tilting spindle head

- 4th or 5th axis in vertical spindle machines
- 5th axis in horizontal spindle machines



Two types of rotational axis:

Continuous: rotational motions can be interpolated by the CNC with linear ones.

Indexed: it is necessary to fix the orientation angle before machining (no interpolation possible).

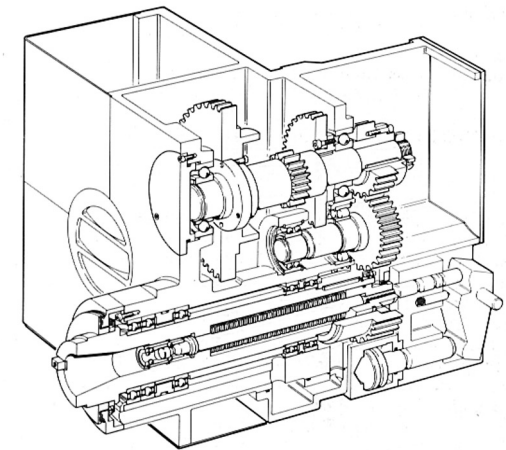


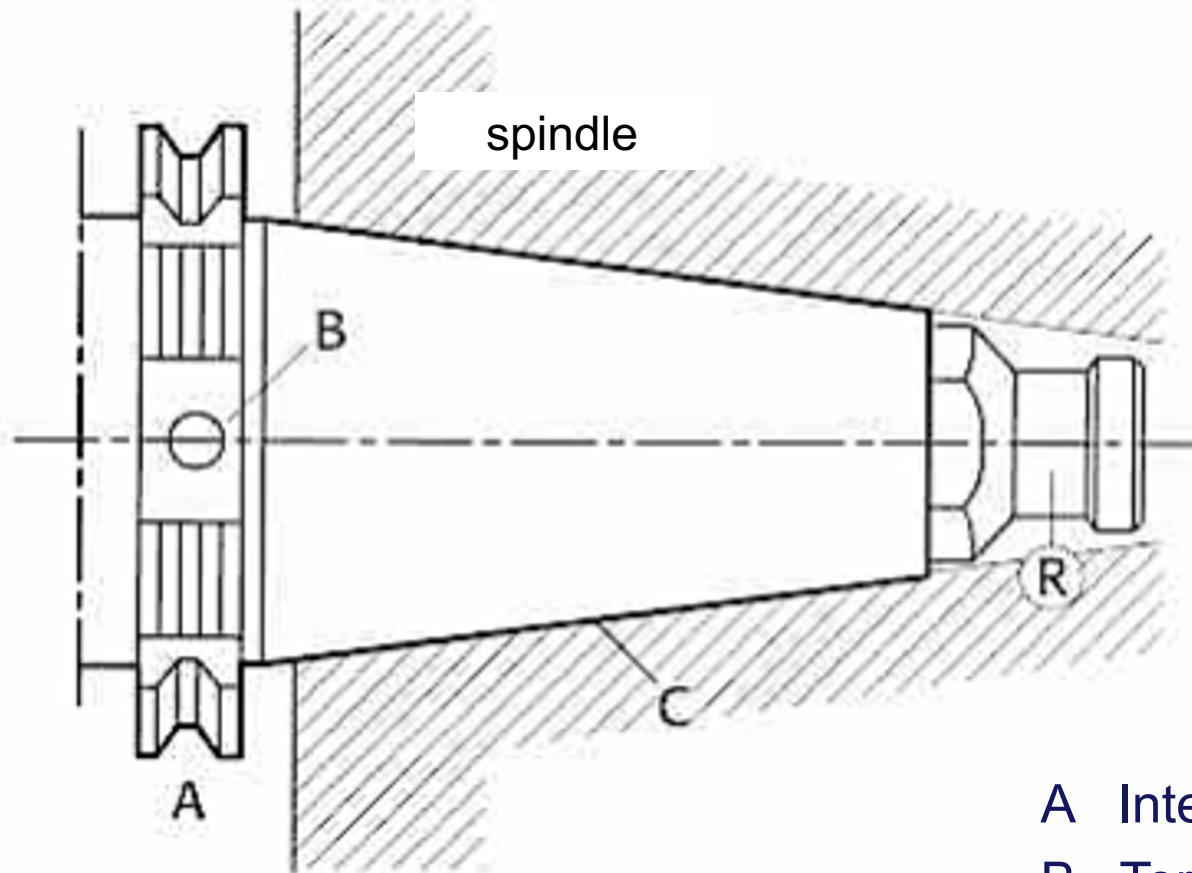
- Automatic Tool Changer (ATC)
- Tool storage
- Tool monitoring and setting
- Automatic Pallet Changer (APC)
- Automatic chip removal



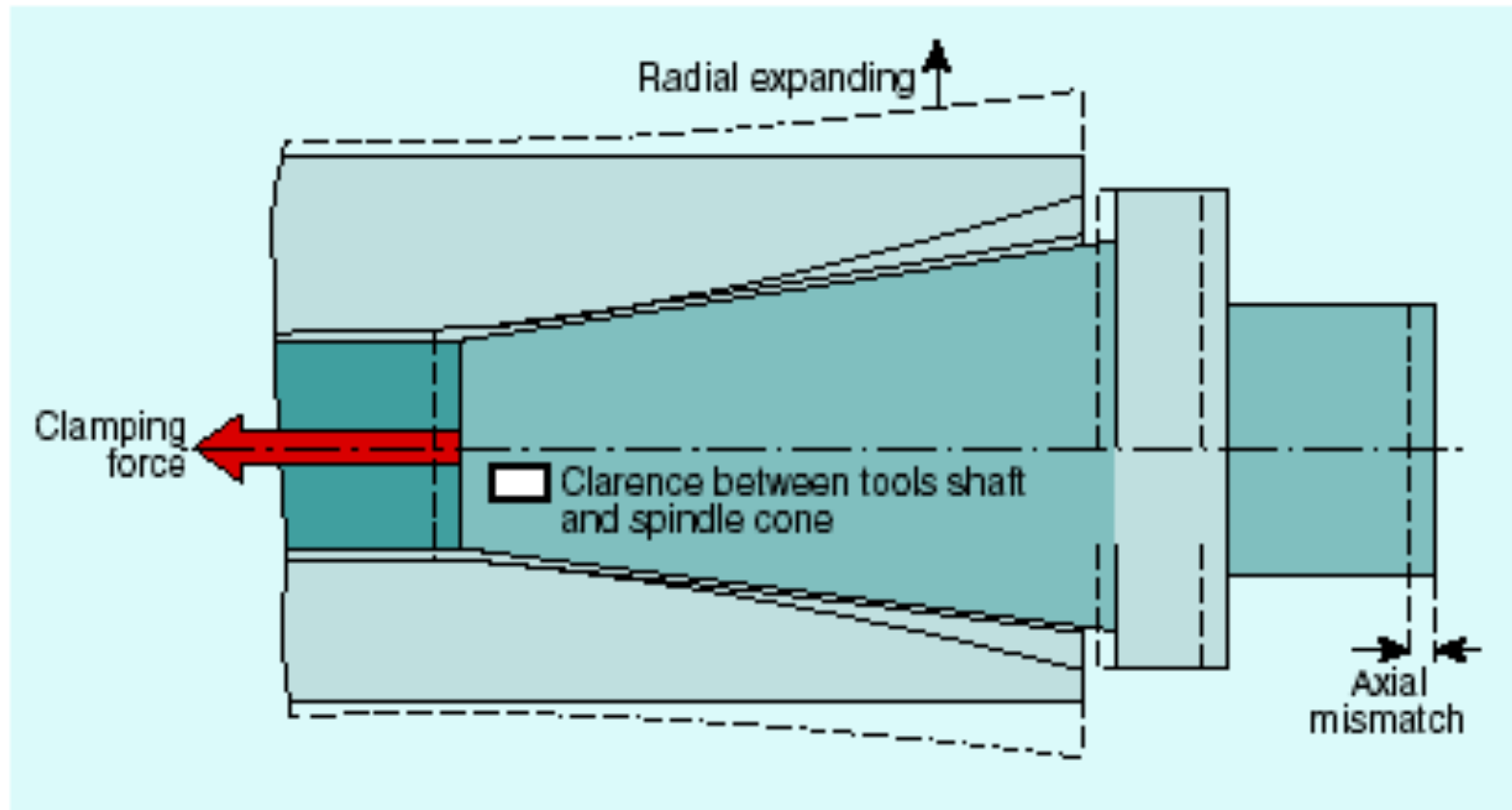
Functionalities

- Hardware interface between the tool and the machine
- Centering
- Depth reference
- Blockage
- Torque transmission
- Liquid for lubrication and cooling
- Interface for manipulation
- Data saving

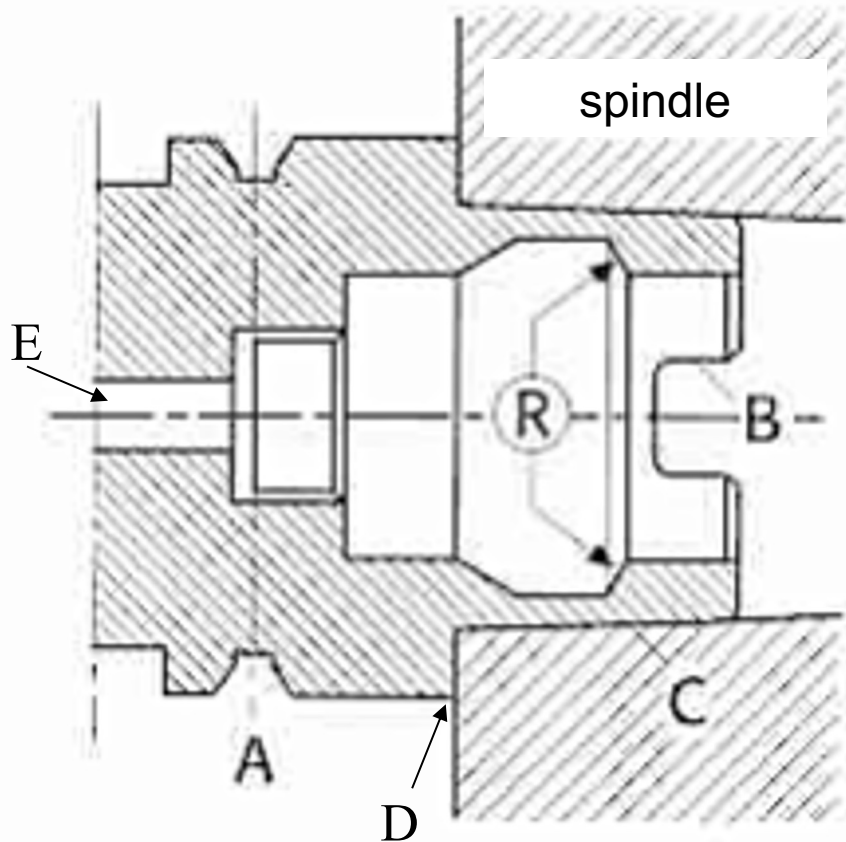




- A Interface for manipulation
- B Torque transmission
- C Centering
- R Blockage



Effect of the deformation caused by the high speed on ISO toolholder.

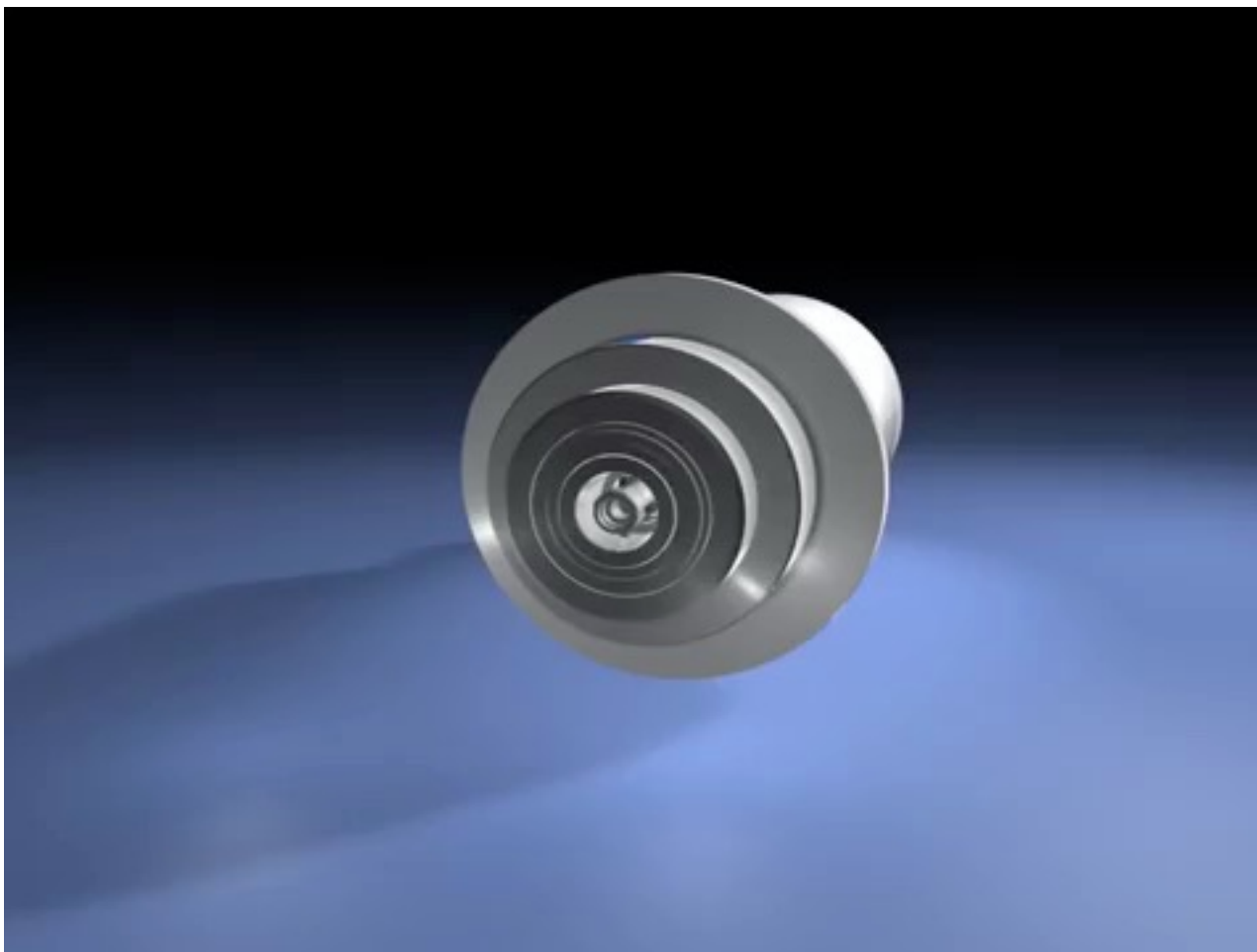


- A Interface for manipulation
- B Torque transmission
- C Centering
- D Depth reference
- E Liquid
- R Blockage



TOOLHOLDER: HSK – HOLLOW SHAFT

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<https://youtu.be/p-8uM0qEAfc>

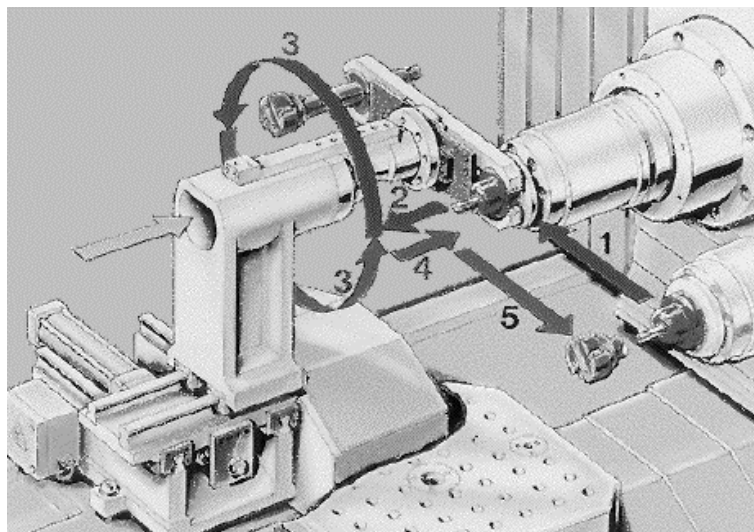


AUTOMATIC TOOL CHANGER (ATC)

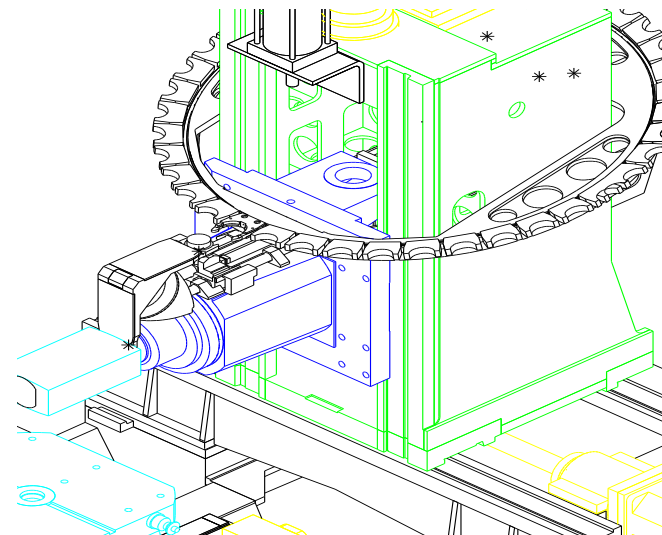
It replaces from the spindle the used tool with the next one to use.

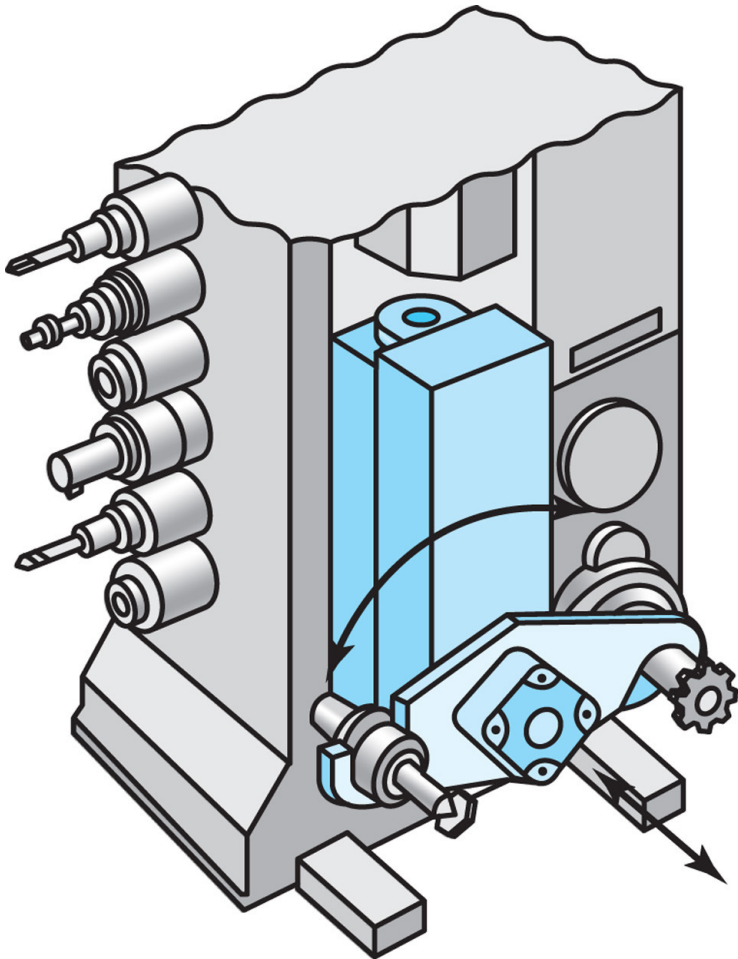
Main characteristics

- Motions and logic
- Types of tools
- Travel time from the tool storage to the change position
- Tool change time (from chip to chip)

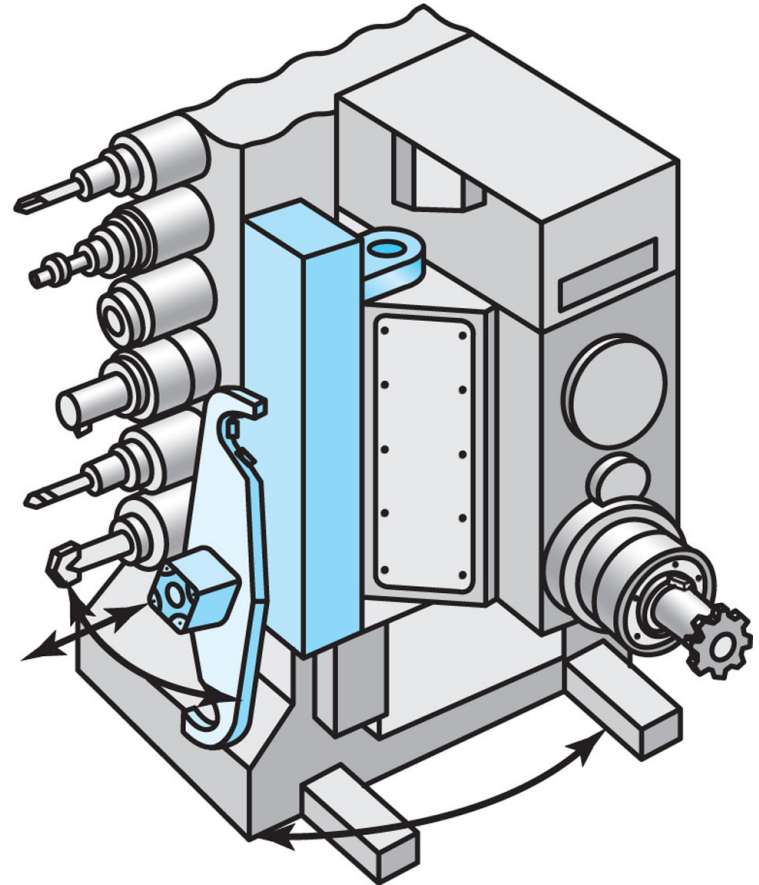


One rotational axis



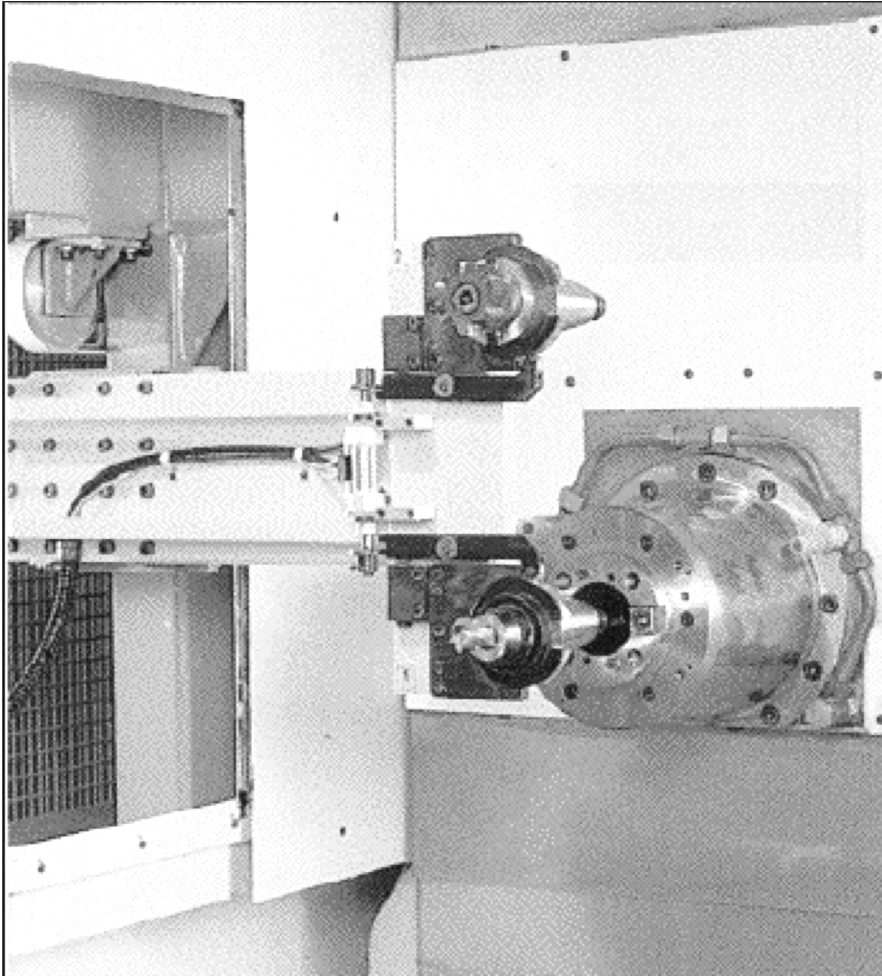


(a)



(b)

Two rotational axis



Linear axis



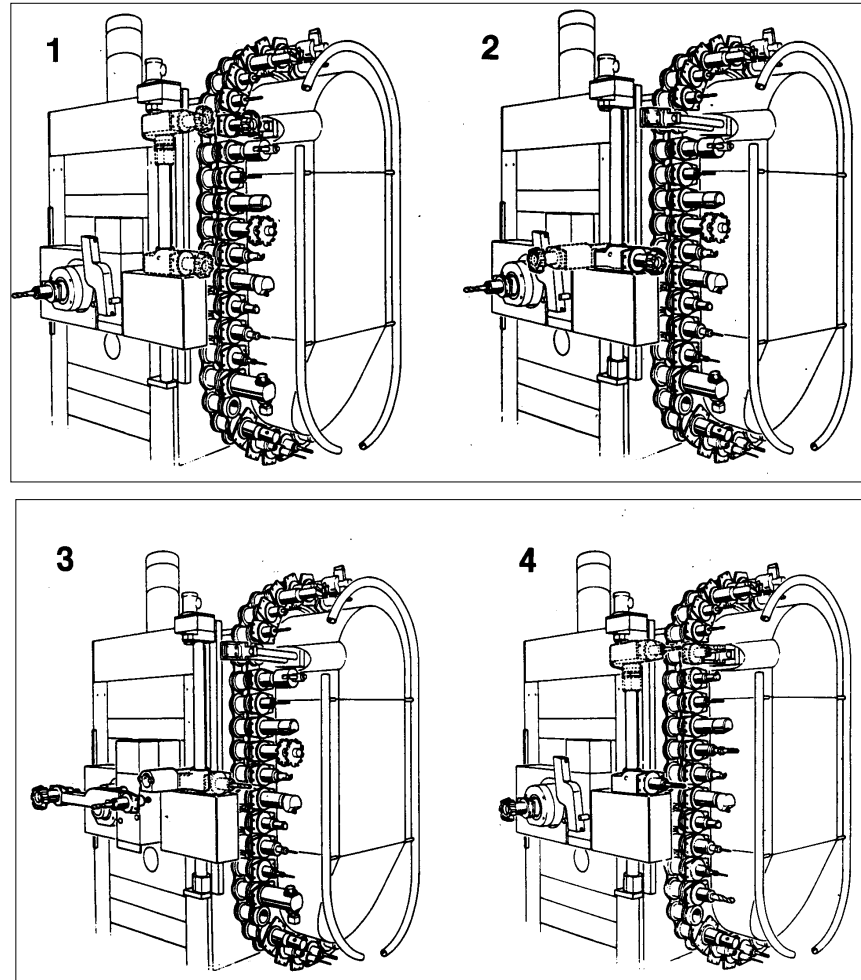
Tool storage



Tool changer



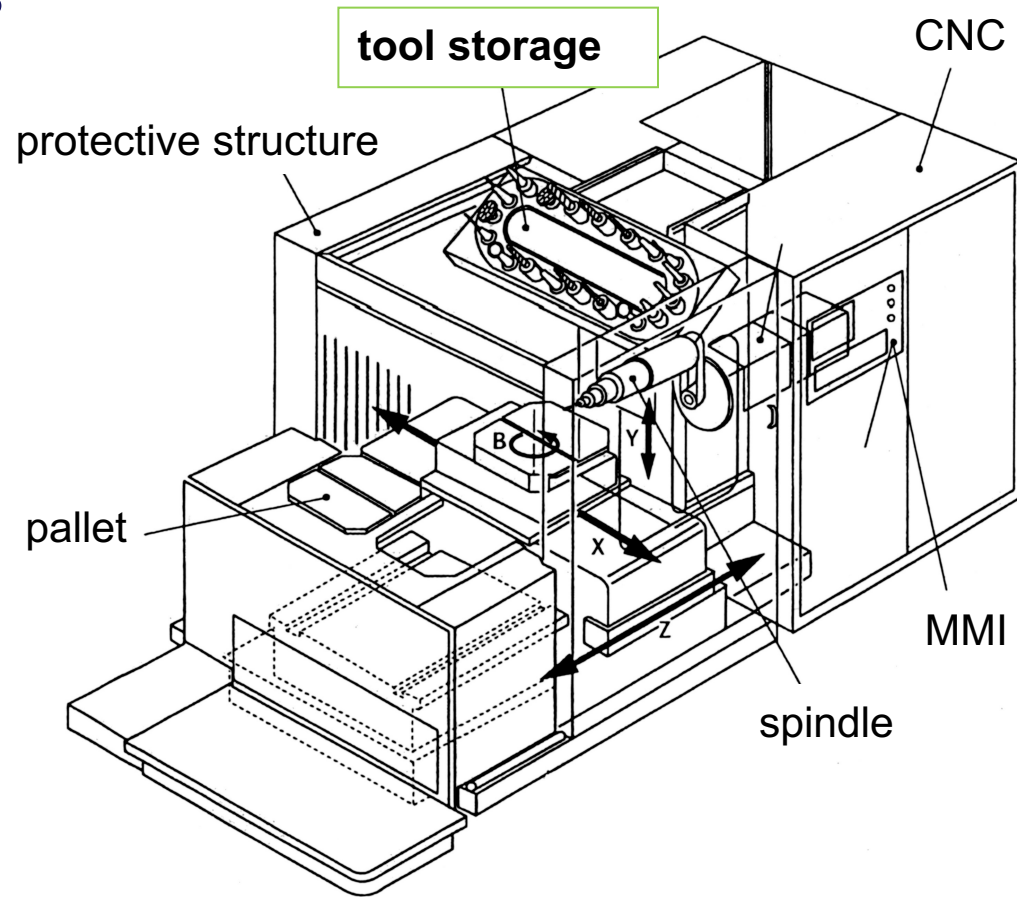
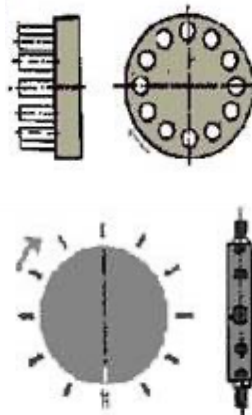
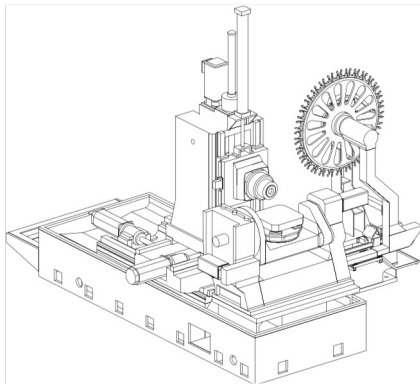
Spindle





Possible Configurations

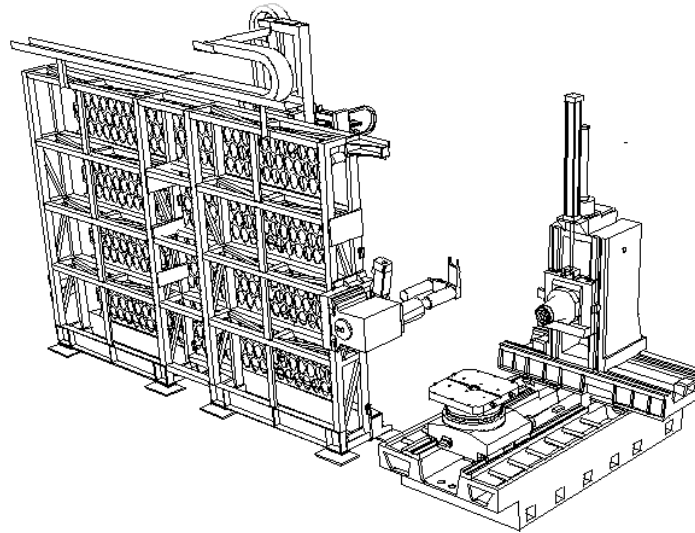
- Disk
- Drum
- Turret
- Chain
- Planar



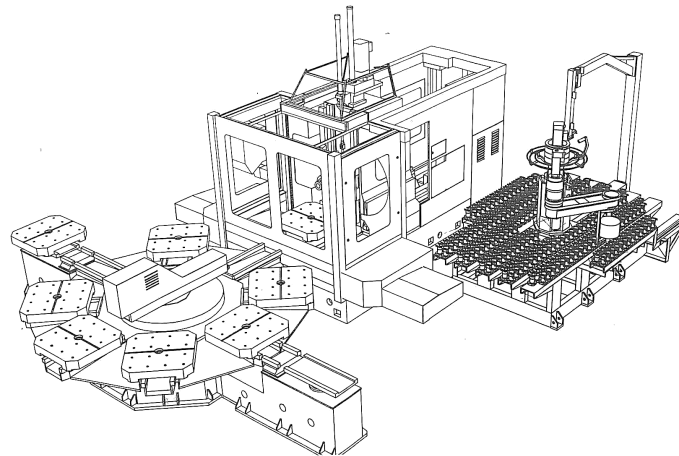


MACHINING CENTER: PLANAR TOOL STORAGE

Vertical



Horizontal





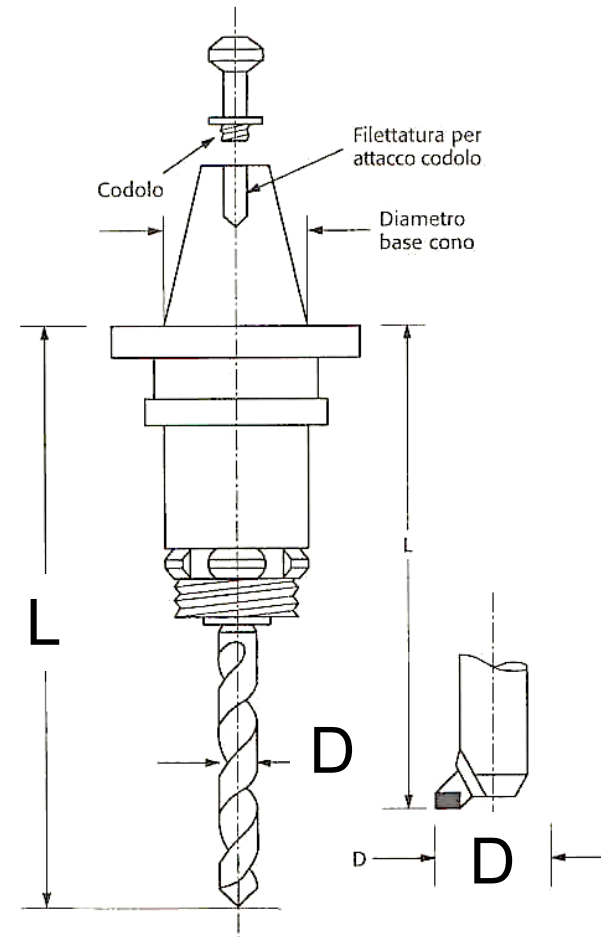
Main characteristics

- Shape and logic
- Types of tools storable (toolholder)
- Number of available slots
- Mean seeking time
- Possibility to interface with a tool transporter
- Floor space



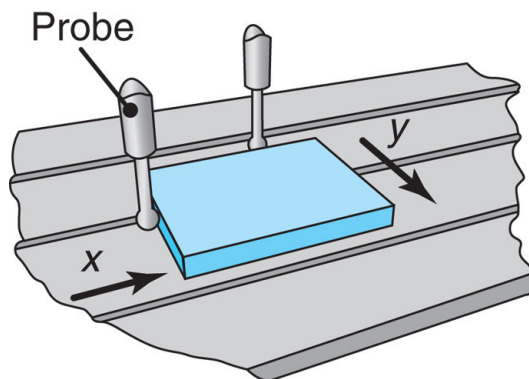
Tool data:

- Tool type
- Copy number
- Tool life
- Residual tool life
- Correctors “L” and “D”
- Manufacturer
- Supplier
- Price
- Delivery lead time

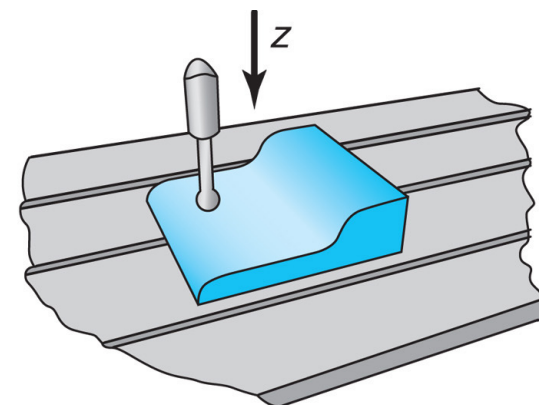




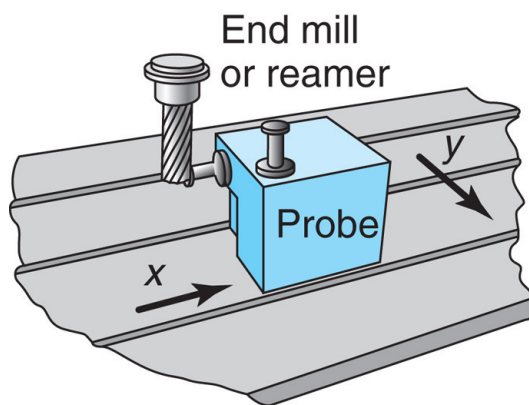
- Touch probe determining
- (a) the X - Y (horizontal) position of a workpiece;
 - (b) the height of a horizontal surface;
 - (c) the planar position of the surface of a cutter (cutter-diameter compensation);
 - (d) the length of a tool for tool-length offset



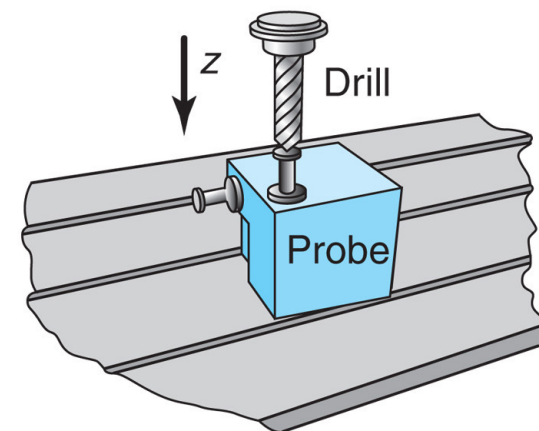
(a) x - y direction



(b) z direction



(c) Diameter compensation



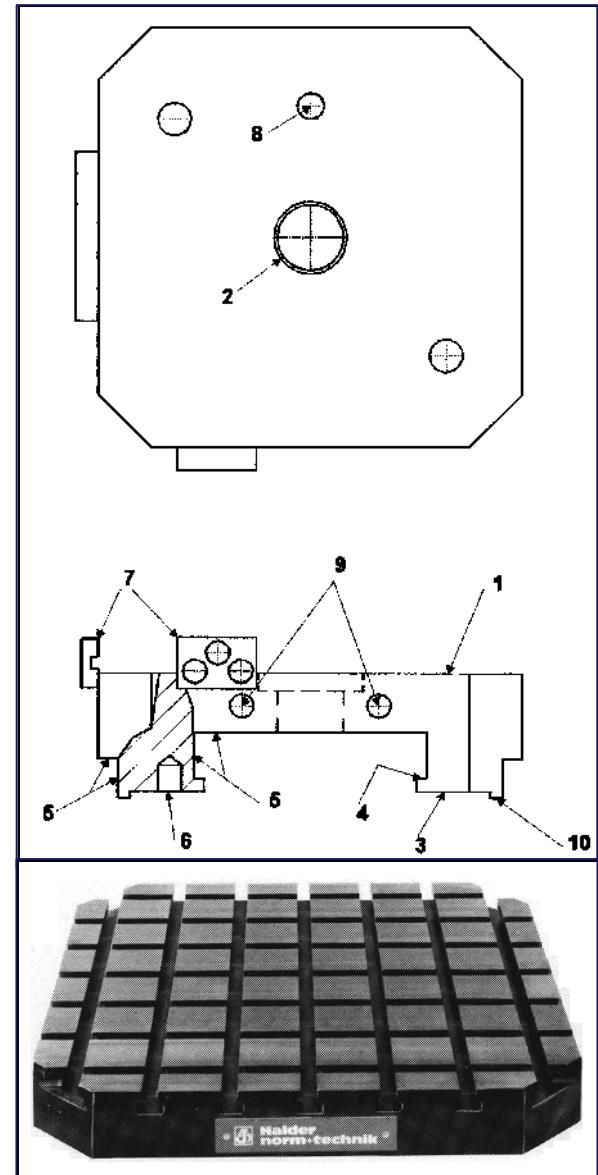
(d) Tool-length offset



Functionalities

- Reference for the fixture
- Fixture blockage
- To allow the positioning on the machine
- To allow the blockage on the machine
- To allow manipulation

Standard pallets follow the ISO8526

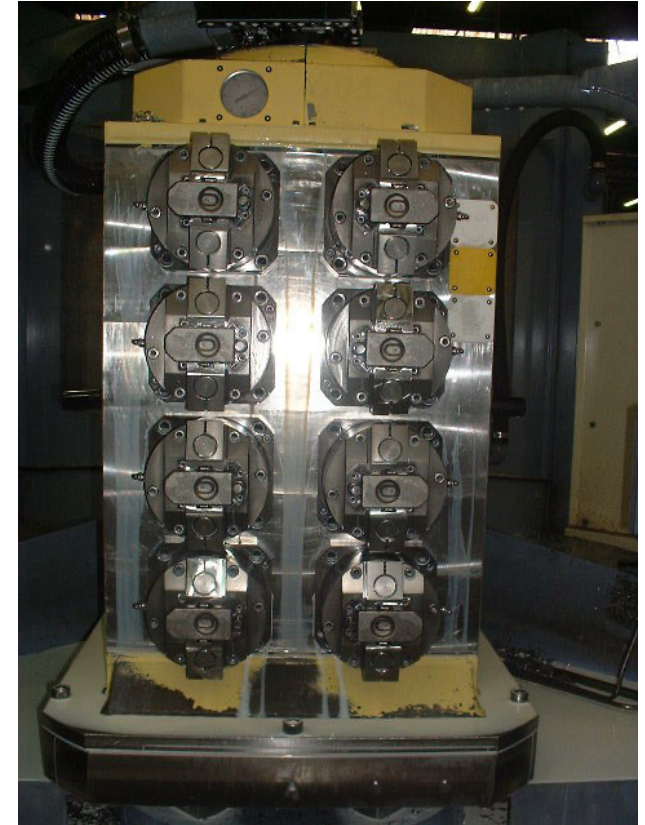


Functionalities

- Reference for the workpiece
- Workpiece blockage
- To support the workpiece during machining and movements
- To allow chip removal

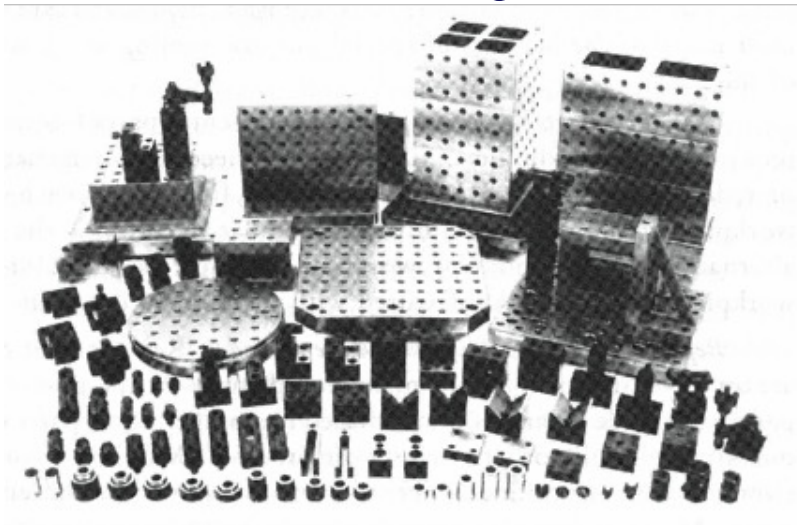
Components

- Basic element or structure
- Positioning elements
- Blocking elements (mechanical, hydraulic)





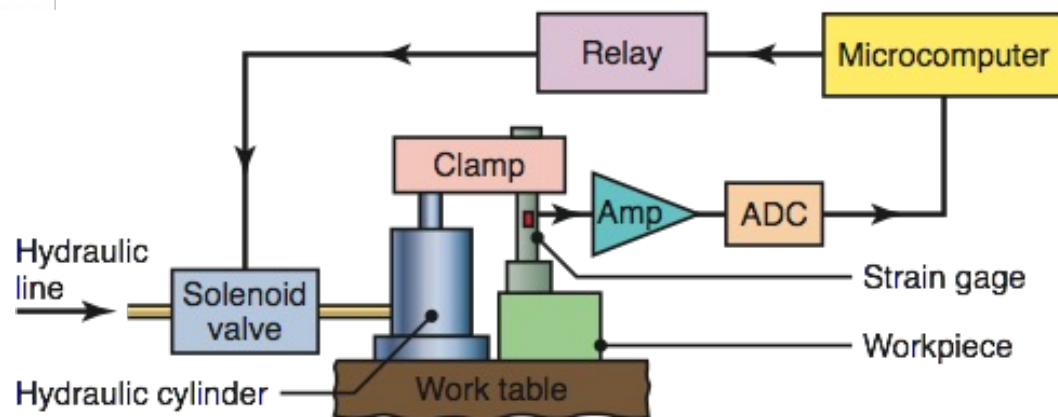
Elements in fixturing



Clamping



Work holding system

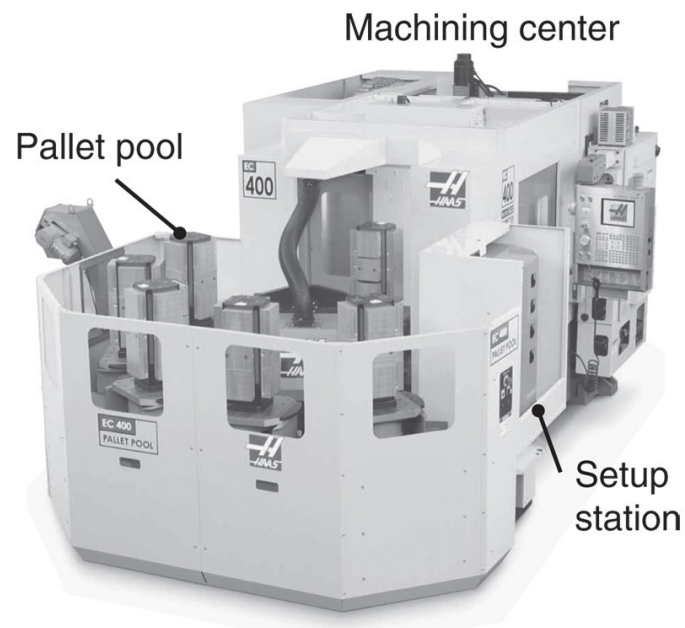
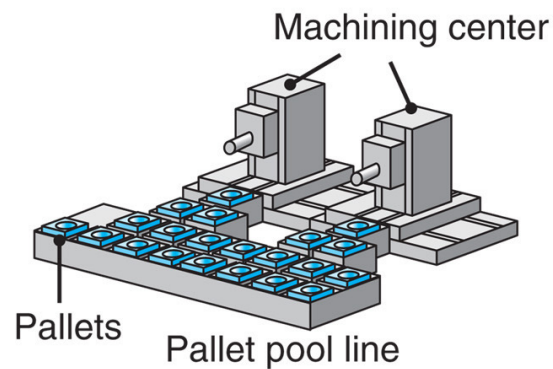
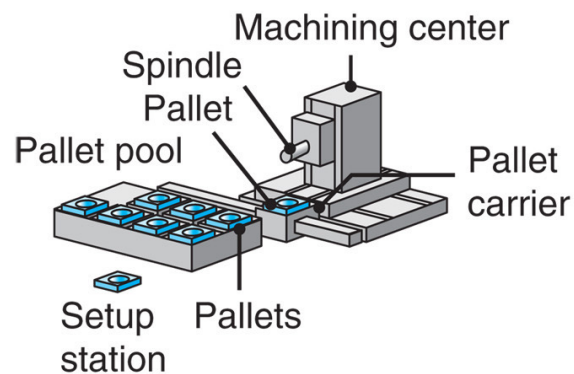


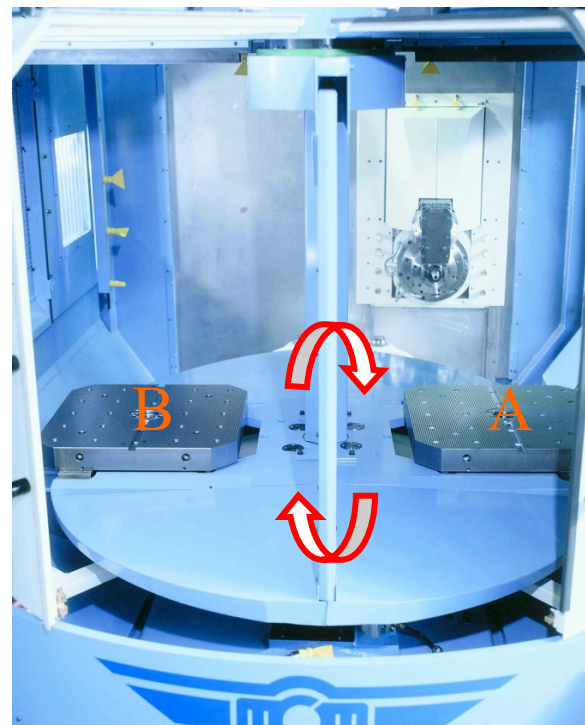
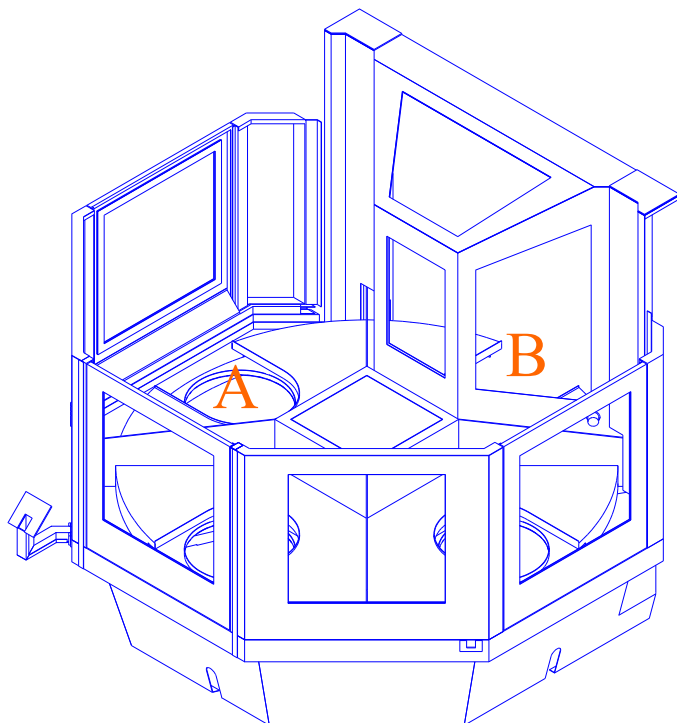
Adjustable-force clamping system



AUTOMATIC PALLET CHANGER (APC)

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1. *Pallet to process in position A*
2. *Machined pallet in position B*
3. *Rotation $A \Leftrightarrow B$*



高速・高精度5軸複合加工機 High speed, High accuracy

<https://youtu.be/CqePrbeAQoM>